

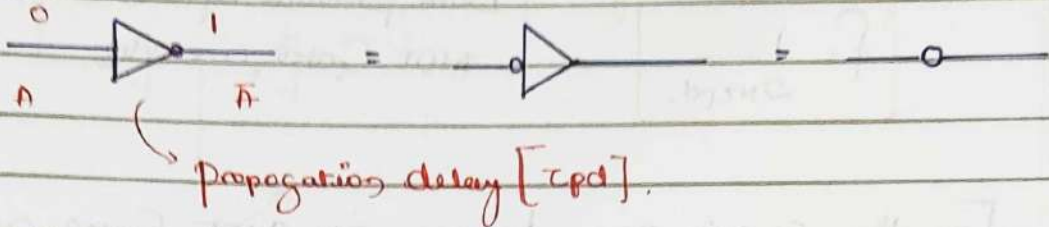
Digital Logic [Till Date]

[Vivekanand Vernekar](#) Parakram C Weekday

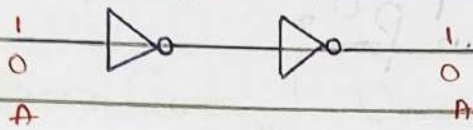
Mentor: Chandan Sir

LOGIC GATES

1. NOT Gate [Inverter, Negation, Complement Logic].

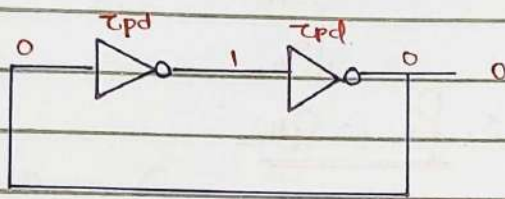


"Buffer" (for even)



No. of not gates are even - "Buffer".
No. of not gates odd - "Inverter".

[High.]



2τpd 2τpd 2τpd 2τpd

1 → 1 → 1 → 1 →

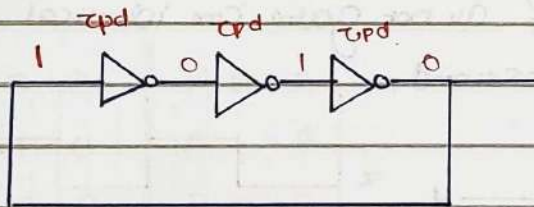
2τpd 2τpd 2τpd

0 → 0 → 0 →

[Low]

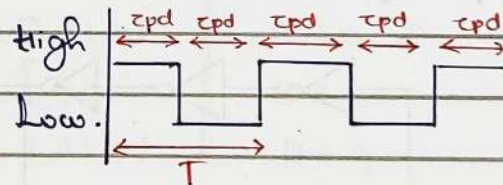
Whenever even numbers of 'not' gate in circuit (loop).

1. Basic memory element
2. Bistable multivibrator
3. DC Generator ($F = 0 + 1$).



τpd τpd τpd τpd τpd

1 → 0 → 1 → 0 → 1



When Odd no. of NOT Gate:

1. Astable multivibrator
2. Square wave generator
3. Clock generator
4. Free running circuit
5. Ring oscillator.

$$T = 2\tau_{pd}$$

$$f = \frac{1}{2\tau_{pd}}$$

$$f = \frac{1}{T}$$

$$T = 2 \times N \times \tau_{pd}$$

N = no. of not gate in loop.

$$f = \frac{1}{2N\tau_{pd}}$$

* Both formulae applicable when number of NOT Gate are odd.

Q1. For the circuit given below, all NOT Gate are identical to each other and having Propagation delay 10 ps . Find the frequency of generated wave form? 10^{12} .

$$N = 5, \quad \tau_{pd} = 10 \times 10^{-12}$$

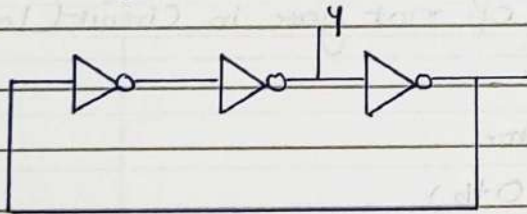
$$f = \frac{1}{2N\tau_{pd}} = \frac{1}{2 \times 5 \times 10 \times 10^{-12}} = \frac{10^{12}}{10 \times 10} \text{ Hz}$$

(odd)

$$= 10^9 \text{ Hz}$$

$$\therefore f = 10 \text{ GHz}$$

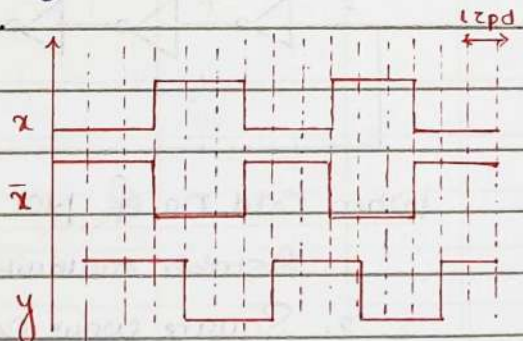
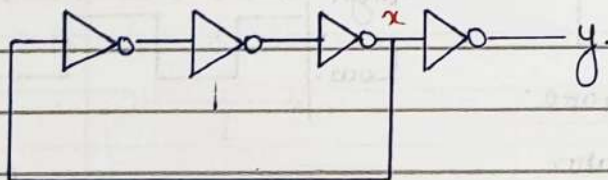
Q2. Circuit given below are called. Astable Multivibrator.



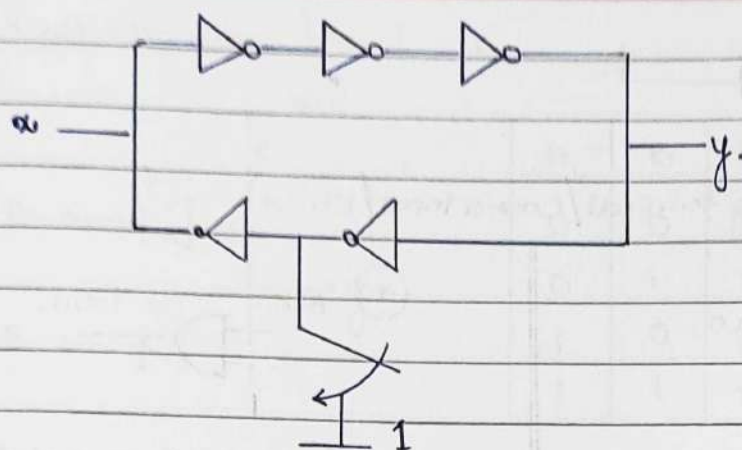
Note: always, For calculation of frequency consider the no. of NOT gate in loop.

HW

Q3. Sketch the waveform of y if all not gate are identical and having Propagation delay of 1 microsecond.



Q4. For the circuit given below x and y condition will be - x and y both toggle.



Odd no. of NOT gate in the loop
 $\therefore x$ and y will toggle.

Basic Gate.

NOT

AND

OR

Universal Gate

NAND

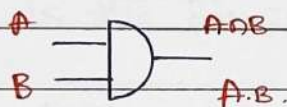
NOR

Exclusive Gate.

X-OR

X-NOR

AND Gate

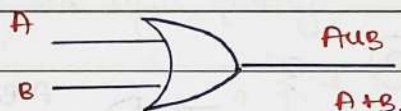


$$A = \{1, 2, 3, 4\} \quad B = \{2, 3, 4\}$$

$$A \text{ AND } B = \{2, 3, 4\}$$

$$\begin{array}{l|l} 0 \cdot 0 = 0 & A \cdot A = A \\ 0 \cdot 1 = 0 & A \cdot 1 = A \\ 1 \cdot 0 = 0 & A \cdot 0 = 0 \\ 1 \cdot 1 = 1 & A \cdot \bar{A} = 0 \end{array}$$

OR Gate

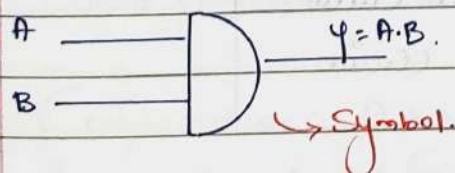


$$A = \{1, 2\} \quad B = \{3, 4\}$$

$$A \text{ OR } B = \{1, 2, 3, 4\}$$

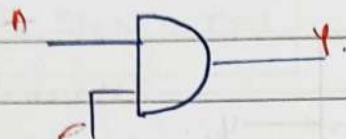
$$\begin{array}{l|l} 0 + 0 = 0 & A + A = A \\ 0 + 1 = 1 & A + 0 = A \\ 1 + 0 = 1 & A + 1 = 1 \\ 1 + 1 = 1 & A + \bar{A} = 1 \end{array}$$

AND Gate [Intersection]

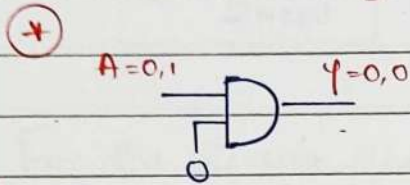


A	B	$Y = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

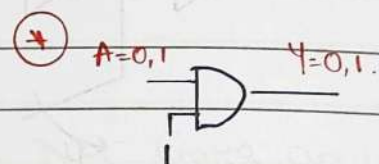
Truth Table



floating terminal / Control Input / Strobe



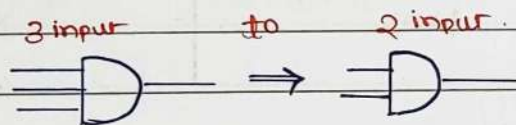
Control '0' disabled.



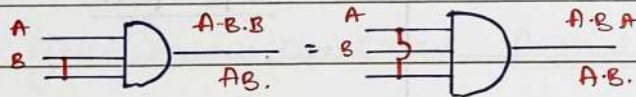
Control '1' enabled.

* Commutative Law : $A \cdot B = B \cdot A$ * Associative Law : $A \cdot B \cdot C = (A \cdot B) \cdot C$

} AND is both commutative and associative.

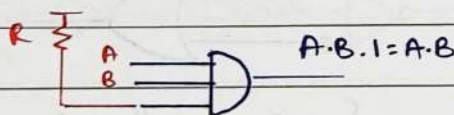
TTL = Transistor Transistor Logic.

①.



Favour ↓

②.

 $A \cdot B \cdot 1 = A \cdot B$

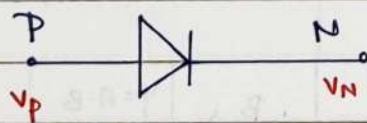
pull up arrangement.

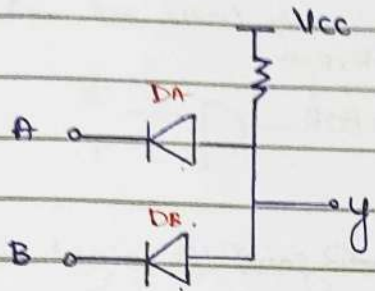
Costly

③.



Noise margin ↓

"Diode" $V_p > V_n \rightarrow$ Forward bias $\rightarrow$ Short circuit $V_p < V_n \rightarrow$ Reverse bias $\rightarrow$ Open circuit

Circuit diagram: [AND]

A	B	DA	DB	y
0	0	FB	FB	0
0	1	FB	Re	0
1	0	Re	FB	0
1	1	Re	Re	1

Note:

- * Whenever logic are designed by TTL the floating terminal always connect as high.
- * Whenever logic are designed by ECL (Emitter Coupled Logic) then floating terminal always connect as low.
- * Noise Margin: Maximum noise added to the input which will not affect the output are called Noise margin.
- * Fan out: The number of logic driven by the logic are called fan out.
- * Fan in: Number of input of a logic are called fan in.
- * ECL is the fastest logic among all the logic family.

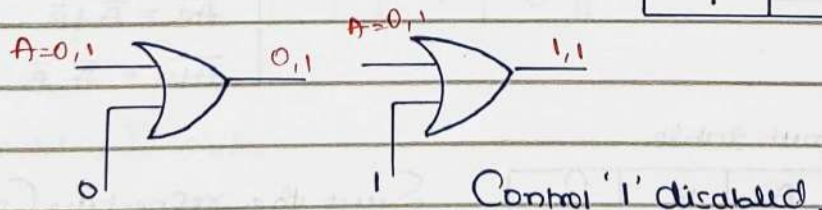
OR Gate.

2. Truth Table

1. Symbol.

A	B	y
0	0	0
0	1	1
1	0	1
1	1	1

3. Enable/Disable

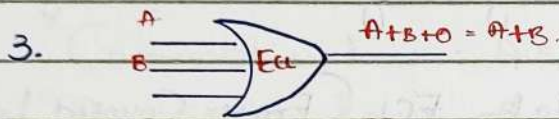
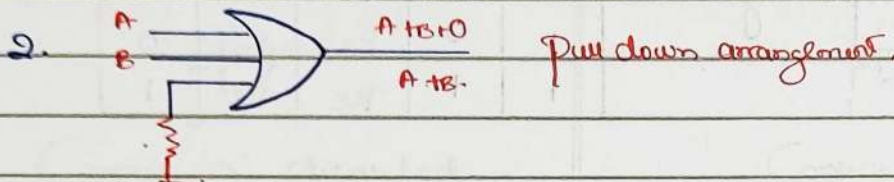
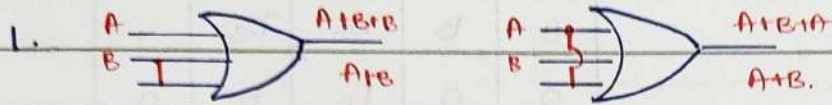


Control '0' enabled.

4. Commutative: $A + B = B + A$
5. Associative: $A + (B + C) = (A + B) + C$

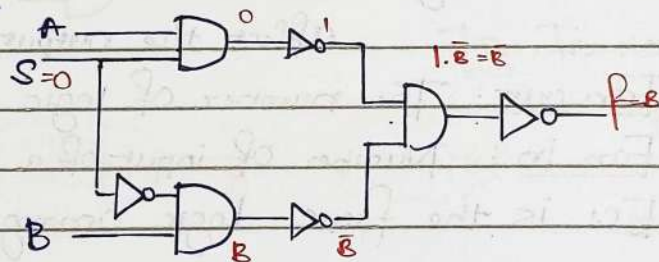


ECL - Emitter Coupled Logic.

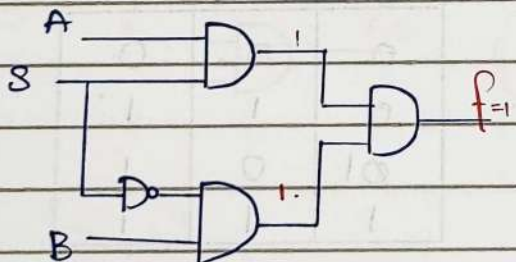


Q1. Output at $S=0$ is B

$\rightarrow f=B$



Q2. $f=1$ for Combination of A, B, S.



[None of the options are correct.]

De Morgan's Law

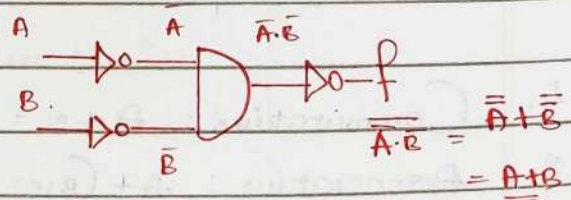
$$\overline{AB} = \overline{A} + \overline{B}$$

$$\overline{A+B} = \overline{A} \cdot \overline{B}$$

Q3. For the given truth table.

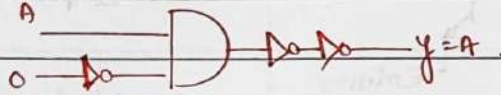
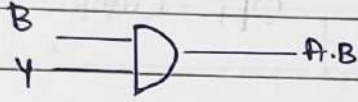
A	B	f
0	0	0
0	1	1
1	0	1
1	1	1

Select the respective Combinat. diagram

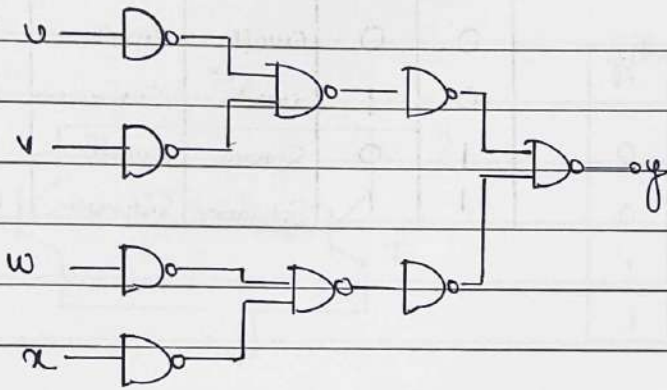


11/11

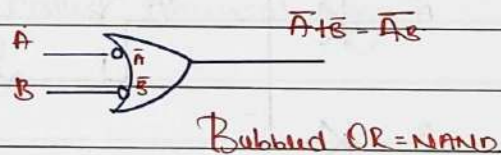
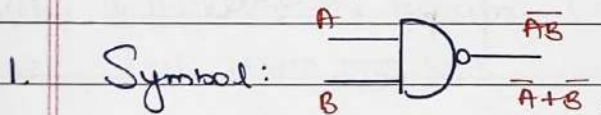
Q4. A logical circuit is as shown below, which of the following circuit can be used to get the desired expression.



Q5. The logic circuit shown below, is equivalent to.



NAND Gate.

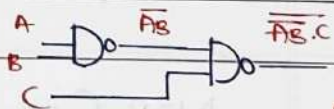


2. Truth Table.

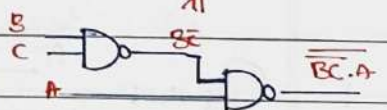
A	B	y
0	0	1
0	1	1
1	0	1
1	1	0

4. Commutative law. ✓

5. Associative law. ✗

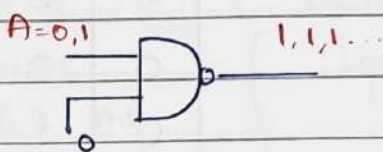


Associative ✗

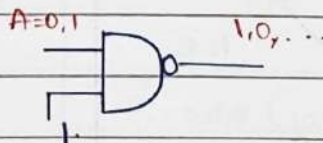


NAND follows commutative law but does not follow associative law.

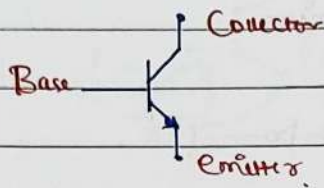
3. Enable/Disable.



Control '0' disabled.



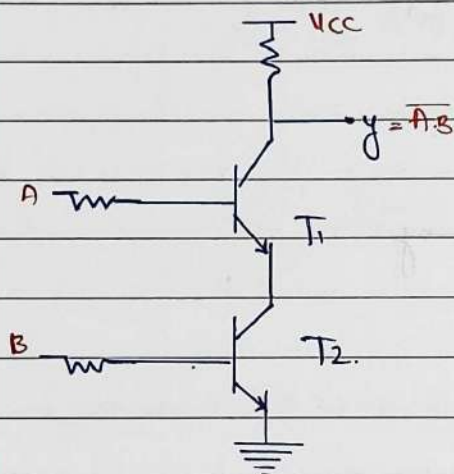
Control '1' enabled

Transistor

Input = 1

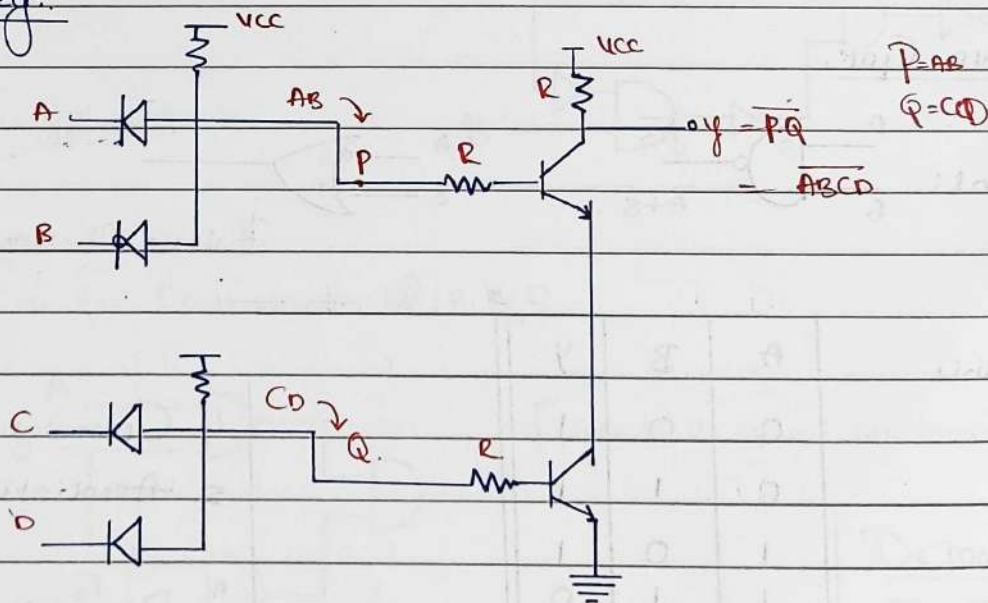
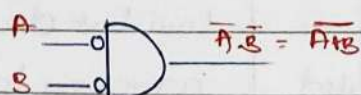
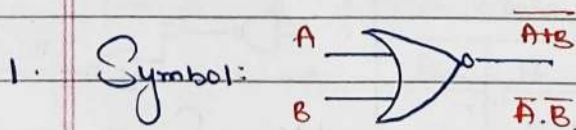
ON = Saturation $\rightarrow$ Short Circuit

Input = 0

OFF = Cutoff $\rightarrow$ Open Circuit

A	B	T_1	T_2	y
0	0	Cutoff	Cutoff	1
0	1	Cutoff	Saturation	1
1	0	Saturation	Cutoff	1
1	1	Saturation	Saturation	0

eg.:

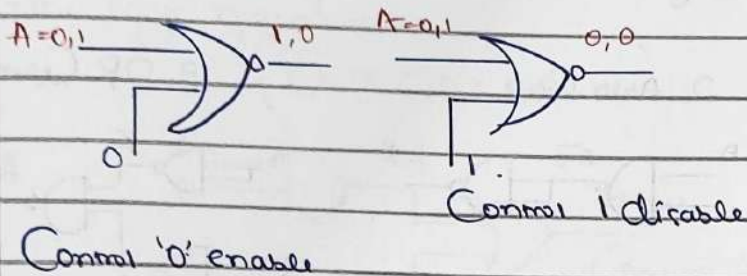
NOR Gate

Bubbled AND = NOR

2. Truth Table:

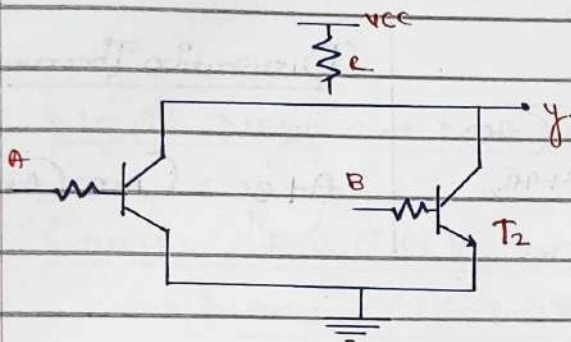
A	B	y
0	0	1
0	1	0
1	0	0
1	1	0

3. Enable/Disable



4. Commutative Law ✓

5. Associative Law. ✗



A	B	T ₁	T ₂	Y
0	0	Cutoff	Cutoff	1
0	1	Cutoff	Saturation	0
1	0	Saturation	Cutoff	0
1	1	Saturation	Saturation	0

Note: NAND, NOR are called "Universal logic".

Q1. Which of the following option is called universal logic?

Ans. Both NAND and NOR. (option given)

Q2. Which of the following is called Universal logic? [MCQ]

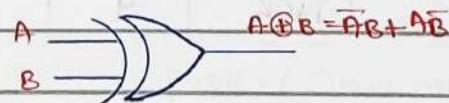
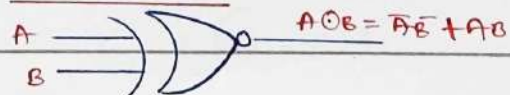
Ans. NAND (if both are given choose NAND)

Q3. Which of the following Option(s) is/are called universal logic? [MCQ]

Ans. $(A+B)$, $(\bar{A} \cdot \bar{B})$, $(A+B)$, $A \cdot \bar{B}$. $\bar{A} \cdot \bar{B}$ $\bar{A} + \bar{B}$ $\bar{A} + B$ $A + \bar{B}$ $\bar{A} \cdot B$, $A \cdot \bar{B}$

Multiplexer

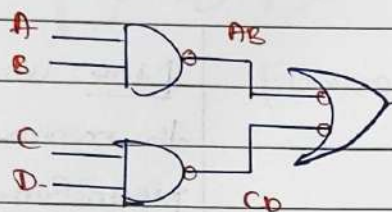
Decoder + OR logic

Universal
Logic.X-OR Gate:X-NOR Gate

$\therefore 1+2+2 = \underline{5}$ required

Q2 Find the minimum no. of two input NAND Gate required to implement the logic given below.

$$f(A, B, C, D) = AB + CD$$



AB + CD

(AND, OR, Inverter)

approach...

Create using AOT approach and make it with NAND by adding inverter ('o'), to balance out!

∴ 3 NAND Gate required

No. of NAND and NOR Gate Required:

Case (1): $A.B.\bar{C}.D.E...$

$$\rightarrow \text{NAND} = (2n - 2) + k$$

n = total no. of variables

k = no. of Complement Variables

$$\rightarrow \text{NOR} = (3n - 3) + k$$

n = no. of variables

k = no. of Complement Variables

eg: $f = A.B.C$

$n=3, k=1$

$$\text{NAND} = (2n - 2) + k$$

$$= (2 \times 3 - 2) + 1 = 5$$

$$\text{NOR} = (3n - 3) + k$$

$$= (3 \times 3 - 3) + 1$$

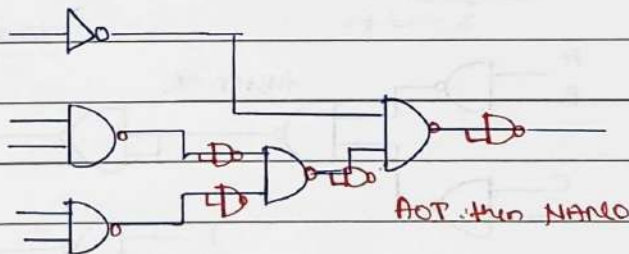
$$= 6 - 1 = 5$$

eg: $f = \bar{A}BCDE$

$n=5, k=1$

$$\text{NAND} = (2 \times 5 - 2) + 1$$

$$= 9$$



Case (2): $A + B + \bar{C} + D...$

$$\text{NAND} = (3n - 3) - k$$

$$\text{NOR} = (2n - 2) + k$$

eg: $f = \bar{A} + B$, $n=2, k=1$

$$\text{NAND} = (3n - 3) - k$$

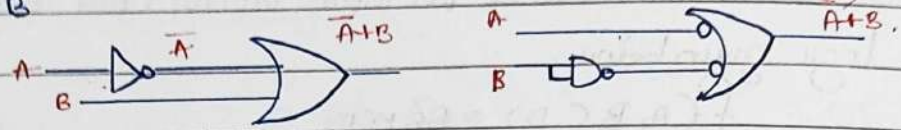
$$= (3 \times 2 - 3) - 1 = 2$$

$$\text{NOR} = (2n - 2) + k$$

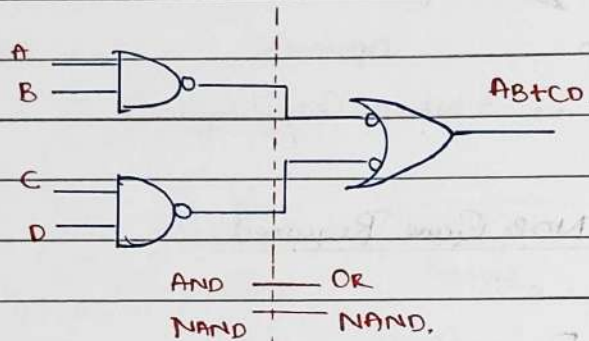
$$= (2 \times 2 - 2) + 1$$

$$= 3$$

eg: $\bar{A} + B$

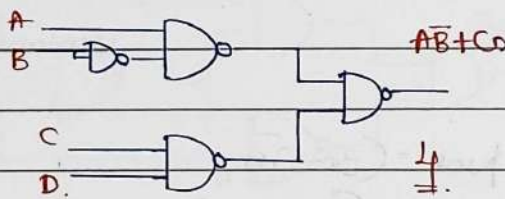


Case (3) $f = AB + CD$ [Sum of product]

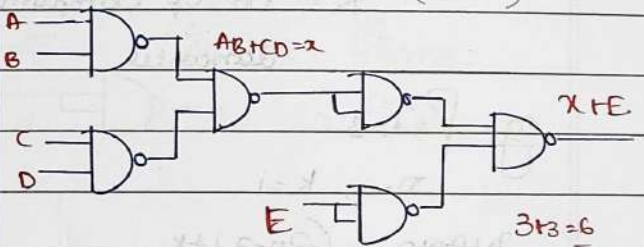


Note: Whenever in the problem the minimum number of NAND Gate are asked then write in SOP form and implement it by AOI and this is exactly equal to NAND NAND implementation.

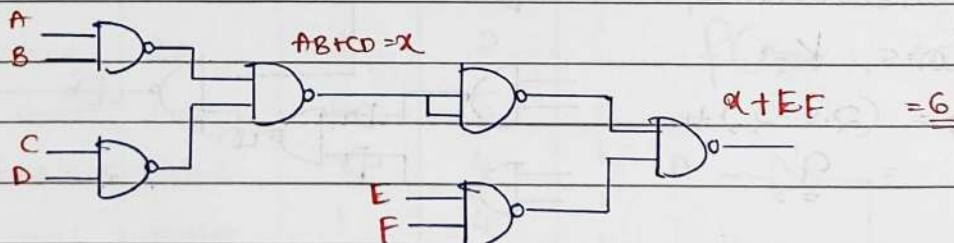
Q. $f = A\bar{B} + CD$



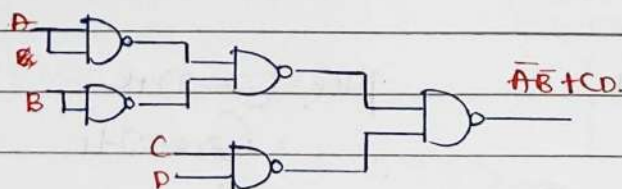
$f = AB + CD + E$ $f = x + E$ $(AB + CD)$



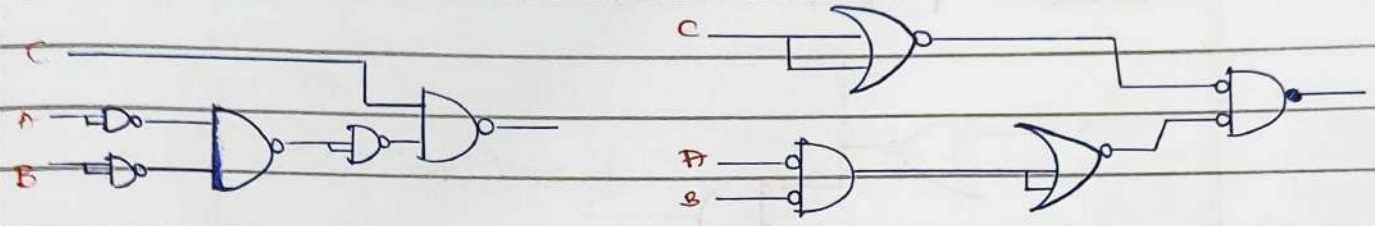
Q. $f = AB + CD + EF$
 $x \rightarrow 3 + 3$



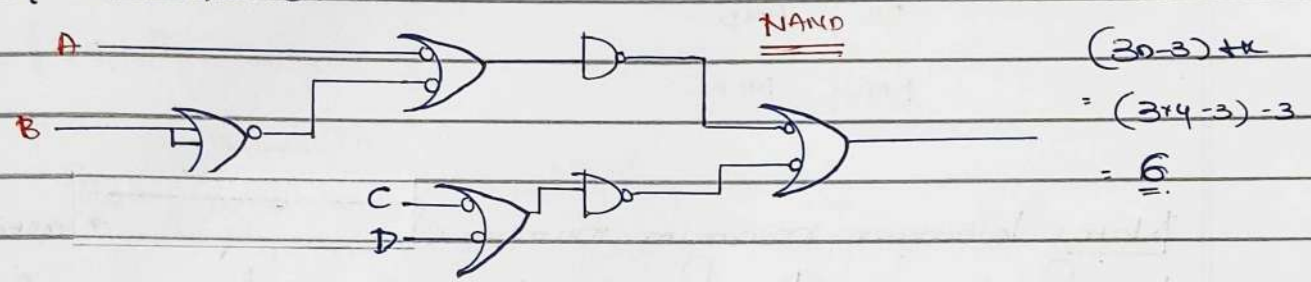
Q. $\bar{A}\bar{B} + CD$



Q1. $f = \bar{A}\bar{B}C$

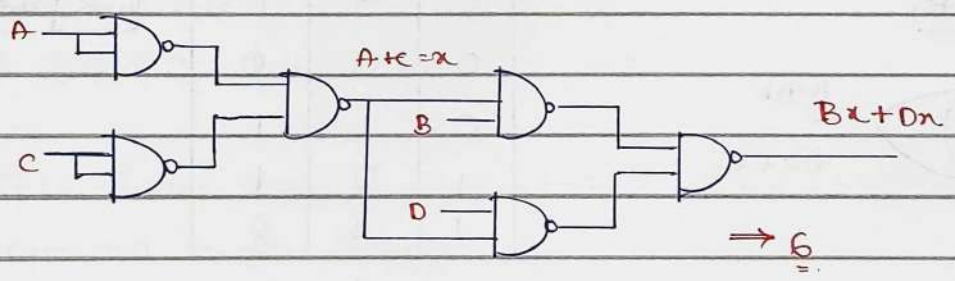


Q2. $f = \bar{A} + B + \bar{C} + \bar{D}$



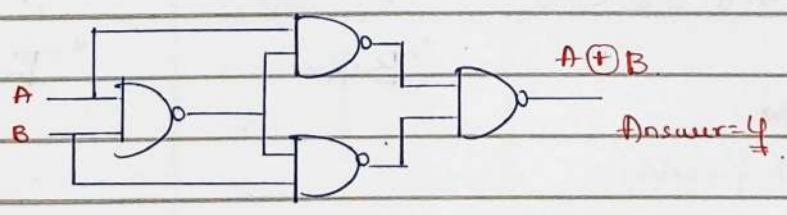
Q3. $f = AB + BC + CD + DA$

$f = B(A+C) + D(A+C)$
 $f = Bx + Dx, \quad A+C = x$

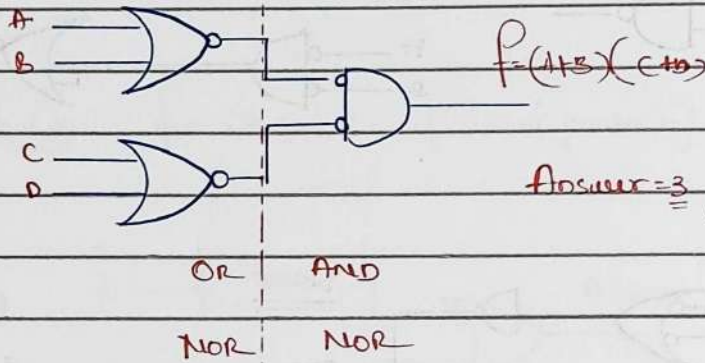


Case (4). Most Important.

Q1. $\bar{A}B + A\bar{B} \rightarrow \text{XOR}$



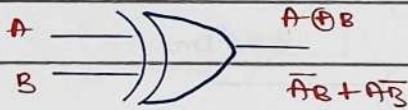
Case (5) $F = (A+B) \cdot (C+D) \rightarrow$ Product of Sum Form.



Answer = 3

Note: Whenever minimum number of NOR Gate are asked, write the function in Pos form and implement by using AND which is AND-OR Implementation which is exactly equal to NOR implementation.

XOR Gate.



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

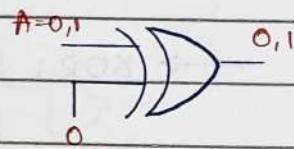
Truth Table

$A \oplus A = 0$
 $A \oplus 0 = A$
 $A \oplus 1 = \bar{A}$
 $A \oplus \bar{A} = 1$

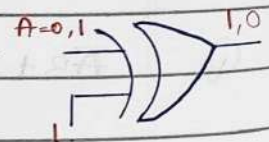
eg::

$A \oplus B = C$
 $A \oplus C = B$
 $B \oplus C = A$
 $A \oplus B \oplus C = 0$

Enable/Disable



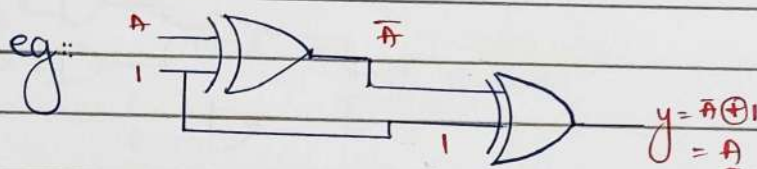
"Buffer"



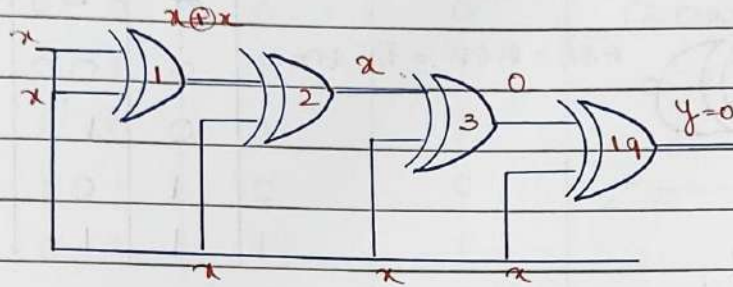
"Inverter"

$\rightarrow A \oplus A = 0$ (for even)

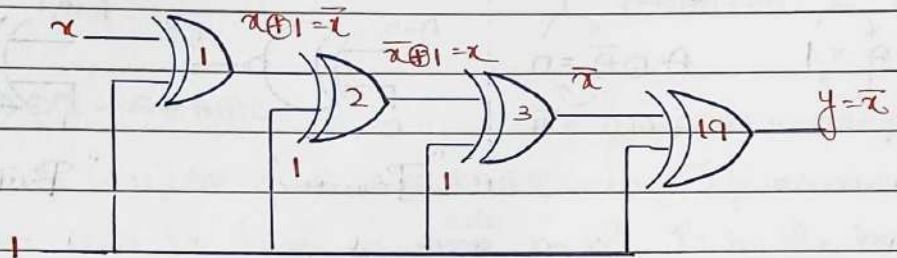
$A \oplus A \oplus A = A$ (for odd)



Q1. Find the output y.



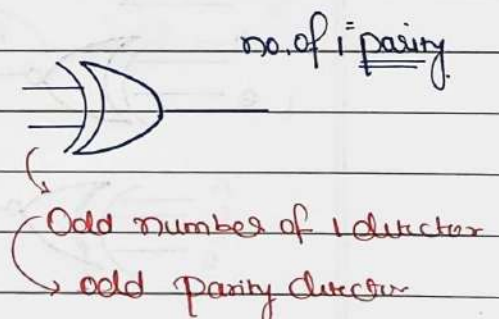
Q2. Find the output y.



- XOR Gate follows Commutative as well as associative law
 $A \oplus B = B \oplus A \rightarrow$ Commutative law
 $A \oplus B \oplus C = (A \oplus B) \oplus C \rightarrow$ associative law

- X-OR Gate output will be high when odd number of 1's are connected in the inputs.

ABC	$A \oplus B \oplus C$	$(A \oplus B) \oplus C$
000	0	0
001	1	1
010	1	1
011	0	0
100	1	1
101	0	0
110	0	0
111	1	1



X-NOR Gate:

Symbol.



$$A \oplus B = A \odot B = \overline{A}B + A\overline{B}$$

A	B	$A \odot B$
0	0	1
0	1	0
1	0	0
1	1	1

Truth Table

$$A = B \quad y = 1$$

$$A \neq B \quad y = 0$$

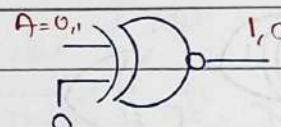
It is also called as "even parity detector / Equivalence logic / Equal detector / Coincidence logic".

$$A \odot A = 1$$

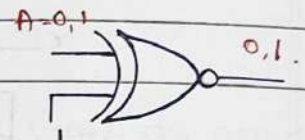
$$A \odot 1 = A$$

$$A \odot \overline{A} = 0$$

$$A \odot 0 = \overline{A}$$

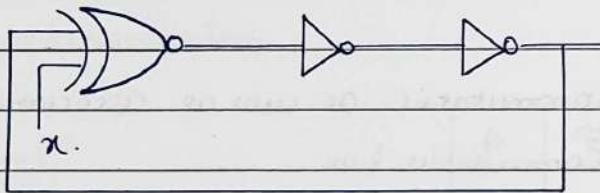


"Inverter".



"Buffer".

eg.:

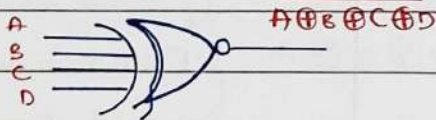
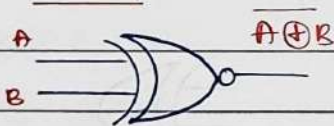

 $x = 0$ - Astable multivibrator

 $x = 1$ - Bistable multivibrator

* X-NOR Gate follows commutative as well as associative law.

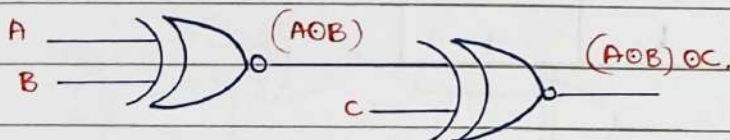
* For even number of variable and even number of inputs.

Case 1:



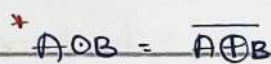
Output will be high for even no. of inputs are high.

Case 2:



$$(A \circ B) \circ C = A \circ (B \circ C)$$

→ Associative Law

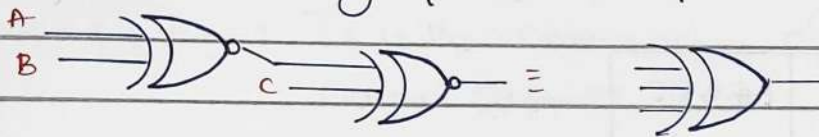


$$A \oplus B \oplus C \oplus D = \overline{A \oplus B \oplus C \oplus D}$$

Variable

$$(((A \oplus B) \odot C) \odot D) \odot E = A \oplus B \oplus C \oplus D \oplus E$$

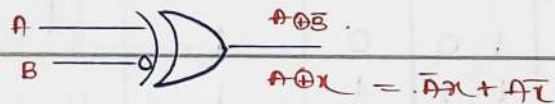
* Output will be high for ~~even~~ odd no of 1's in the input.



✓ Output will be high for even no. of 1's in input.



Diagram of an XOR gate with inputs A and B. The output is labeled $A \oplus B$ and $\overline{A}B + A\overline{B}$.

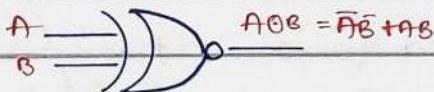


$$A \oplus x = \bar{A}x + A\bar{x}$$

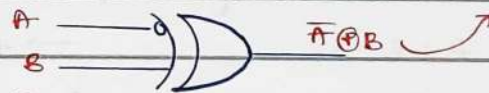
$$\overline{A}B + A\overline{B}$$

$$\bar{A}\bar{B} + AB = A \odot B$$

X-NOR



$$A \odot B = \overline{A} \overline{B} + AB$$



$A \oplus B$

$$\overline{A \cdot B} = \overline{A} \overline{B} + \overline{A} B$$

$$= AB + \overline{A}\overline{B}$$

$$= \overline{A} \overline{B} + AB$$

Sample

$$\# A \odot B = \overline{A} \oplus B = \overline{A} \oplus \overline{B} = \overline{A \odot B} = \overline{A \oplus B}$$

$$\# \quad A \oplus B = \overline{A} \odot B = A \odot \overline{B} = \overline{\overline{A} \oplus B} = \overline{A \odot B}$$

Q1. The boolean function given: $f(A, B) = A \oplus B \oplus AB$
Which Statement is/are Correct?

$$\begin{aligned} f(A, B) &= (A \oplus B) \oplus AB \\ &= 1 \oplus AB & A \oplus B = 1 &= \bar{A}B + A\bar{B} \\ &= \bar{1} \cdot AB + 1 \cdot \bar{AB} & \bar{A \oplus B} = \bar{1} &= A \oplus B = \bar{A}\bar{B} + AB \\ &= [\bar{A}\bar{B} + AB] \cdot AB + [\bar{A}B + A\bar{B}] \cdot [\bar{A} + \bar{B}] \\ &= \bar{A}\bar{B} \cdot AB + AB \cdot AB + \bar{A}B \cdot \bar{A} + A\bar{B} \cdot \bar{B} + \bar{A}\bar{B} \cdot \bar{B} + AB \cdot B \\ &= AB + \bar{A}\bar{B} + AB \\ &= AB + \bar{A}\bar{B} + \bar{A}B \\ &= A[\bar{B} + \bar{A}] + \bar{A}B \\ &= A \cdot 1 + \bar{A}B \Rightarrow A + \bar{A}B = (A + \bar{A})(A + B) = A + B \text{ (OR)} \end{aligned}$$

Ans: $\rightarrow$ It is a OR Gate
 $\rightarrow$ It requires 3 NAND Gate to implement the function.

A B	AB	$A \oplus B \oplus C$
0 0	0	0
0 1	0	1
1 0	0	1
1 1	1	1

$f = A + B$

Q2. Minimized expression will be $Y = A \oplus (A + B)$

$A + B = 1$

$A \oplus 1$

$\bar{A} \cdot 1 + A \cdot \bar{1}$

$\bar{A}[A + B] + A[\bar{A} + \bar{B}]$

$\bar{A} \cdot A + \bar{A} \cdot B + A \cdot \bar{A} + A \cdot \bar{B}$

$= \bar{A}B$

$\rightarrow$ Associative law

does not follow

by χ -OR Gate out

OR Gate.

$A \oplus (A + B)$

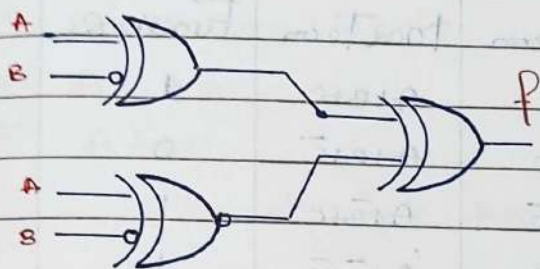
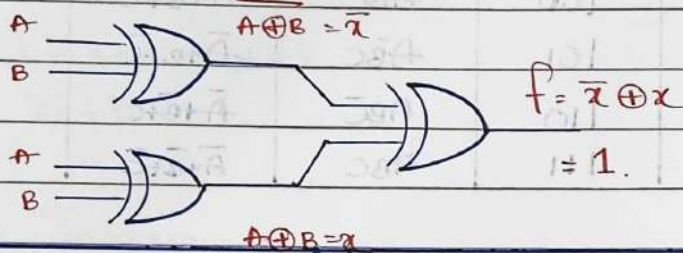
$(A \oplus A) + (A \oplus B)$

$0 + A \oplus B$

$A \oplus B$

A B	A	$(A+B)$ χ	$A \oplus \chi$	$A \oplus (A+B)$ $A \oplus \chi$
0 0	0	0	0	
0 1	0	1	1	
1 0	1	1	0	
1 1	1	1	0	

eg:

f will be 1. $A \oplus B = \bar{x}$

$$f = \bar{x} \oplus x$$

$$= 1$$

Laws of Boolean Algebra

Boolean function: It is the combination of inputs on which output depends.

eg: $f(A, B) = \underline{AB} + \underline{\bar{A}\bar{B}}$

No. of variables = 2

No. of terms = 2, Literals = 4

Function.

SOP

POS

Sum of Product
(Minterm)Product of Sum
(Maxterm)

Standard Canonical Form

Each term should contain all the variables

eg: $f(A, B) = \bar{A}B + A\bar{B}$ ✓
 $f(A, B) = A + \bar{A}B$

Decimal	ABC	Min Term	Max Term	Functions
0	000	$\bar{A}\bar{B}\bar{C}$	$A+B+C$	1
1	001	$\bar{A}\bar{B}C$	$A+B+\bar{C}$	0
2	010	$\bar{A}B\bar{C}$	$A+\bar{B}+C$	1
3	100	$A\bar{B}\bar{C}$	$A+\bar{B}+\bar{C}$	1
4	101	$A\bar{B}C$	$\bar{A}+B+C$	0
5	110	$A\bar{B}\bar{C}$	$\bar{A}+B+\bar{C}$	0
6	111	ABC	$\bar{A}+\bar{B}+C$	0
7	111	ABC	$\bar{A}+\bar{B}+\bar{C}$	1

Standard Canonical SOP form:

$$f(A,B,C) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + ABC$$

$$= m_0 + m_2 + m_3 + m_7$$

$$= \sum m(0, 2, 3, 7)$$

$$= \sum (0, 2, 3, 7)$$

Standard Canonical Pos' form:

$$F(A,B,C) = (A+B+C)(\bar{A}+B+C)(\bar{A}+B+\bar{C})(\bar{A}+\bar{B}+C)$$

$$= m_1 \cdot m_9 \cdot m_2 \cdot m_6$$

$$= \pi M(1, 4, 5, 6)$$

$$= \pi (1, 4, 5, 6)$$

Q1. No. of terms present in Standard Canonical form (SOP)?

Ans. $f(A,B,C) = A+BC$

$$= A(\bar{B}+B)(\bar{C}+C) + (\bar{A}+A)BC$$

$$= A\bar{B}\bar{C} + A\bar{B}C + A\bar{B}C + ABC + \bar{A}BC$$

$\therefore$ 5 terms are present.

① Distribution Theorem

$$(A+B)(A+C)$$

$$= A.A + A.C + AB + BC$$

$$= A + AC + AB + BC$$

$$= A[1+C+B] + BC$$

$$= A \cdot 1 + BC$$

$$= A + BC$$

$$\text{eg: } A + \overline{A}B$$

$$(A + \overline{A})(A + B)$$

$$= 1(A + B)$$

$$= A + B$$

$$A + BC = (A + B)(A + C)$$

$$Q2. f(A, B) = \overline{A} + \overline{A}B$$

$$= (\overline{A} + A)(\overline{A} + B)$$

$$= \overline{A} + B$$

2. Consensus Theorem

$$f(A, B, C) = AB + \overline{A}C + BC$$

$$= AB + \overline{A}C + (\overline{A}B + A)BC$$

$$= AB + \overline{A}C + \overline{A}BC + ABC$$

$$= AB[1+C] + \overline{A}C[1+B]$$

$$= AB + \overline{A}C$$

$BC \rightarrow$ Redundant term

* 3 variable function

* each term consist of 2 variable.

* each variable repeated twice except one.

* One variable repeated in form of complement.

$$\text{eg: } AB + \overline{B}C + AC$$

$$\Rightarrow AB + \overline{B}C$$

$AC \rightarrow$ redundant.

3. Transpose Theorem

$$(A+B)(\overline{A}+C)$$

$$= A\overline{A} + AC + \overline{A}B + BC$$

$$= AC + \overline{A}B + BC$$

$$= AC + \overline{A}B$$

$$(A+B)(\overline{A}+C)$$

$$= AC + \overline{A}B$$

4. De Morgan's Law

$$\overline{ABC} = \overline{A} + \overline{B} + \overline{C}$$

$$\overline{A+B+C} = \overline{A} \cdot \overline{B} \cdot \overline{C}$$

Q1. Find min. no of NAND Gate required to implement $f(A, B, C) = A + ABC + \overline{A}\overline{B}\overline{C}$.

$$A[1+BC+\overline{B}\overline{C}] = A$$

$\therefore$ 0 NAND Gate required.

Q2. Find the minimum no. of NAND gate required to implement the boolean function

Ans
$$f(A,B,C) = A + ABC + ABC$$

$$= A [1 + BC + BC]$$

$$= A$$

$$\therefore 0 \text{ NAND Gate required}$$

Q3. Minimize the expression $f(A,B) = A + AB$

Ans
$$= A [1 + B]$$

$$= A$$

Q4. Minimize the expression $f(A,B) = \bar{A}\bar{B} + \bar{A}B + AB$

Ans
$$= \bar{A} [\bar{B} + B] + AB$$

$$= \bar{A} + AB = (\bar{A} + A) (\bar{A} + B)$$

$$= \underline{\underline{\bar{A} + B}}$$

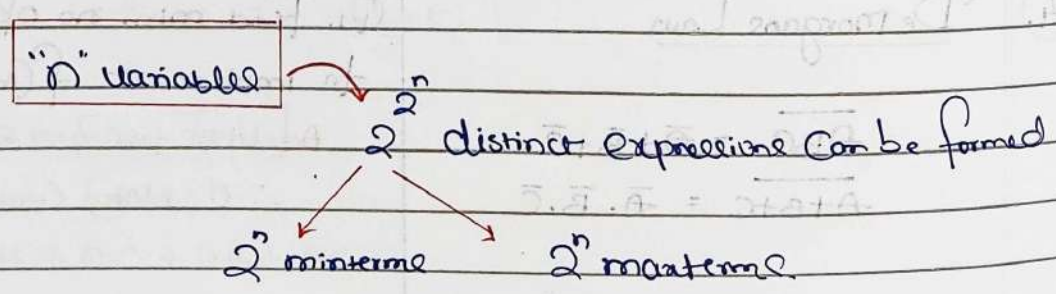
Q5. Minimize the expression $f(A,B) = \bar{A}\bar{B} + \bar{A}B + AB + AB$

Ans
$$= \bar{A} [\bar{B} + B] + A [\bar{B} + B]$$

$$= \bar{A} + A \Rightarrow \underline{\underline{1}}$$

$n=1$		$n=2$			
A	}	$\bar{A}\bar{B}$	$\bar{A} + \bar{B}$	A	0
$\bar{A}$		$\bar{A}B$	$\bar{A} + B$	$\bar{A}$	1
1		$A\bar{B}$	$A + \bar{B}$	B	$\bar{A}B + A\bar{B}$
0		AB	$A + B$	$\bar{B}$	$\bar{A}\bar{B} + AB$
		minterms maxterms			

16 different terms



Q6. Minimize the expression $f(A, B) = \bar{A}B + A\bar{B} = A \oplus B$.
 Ans. Already minimized

Q7. Minimize the expression: $f(A, B) = AB + \bar{A}C + BC$
 ↪ function written in wrong format.

$$\begin{aligned} \text{eg: } f(A, B, C) &= \overset{001}{\bar{A}\bar{B}C} + \overset{010}{\bar{A}B\bar{C}} + \overset{100}{A\bar{B}\bar{C}} + \overset{111}{ABC} \\ &= \sum (1, 2, 4, 7) \\ &= A \oplus B \oplus C \quad (\text{Already Minimized}) \end{aligned}$$

$$\begin{aligned} \text{eg: } f(B, A) &= \overset{01}{\bar{A}B} + \overset{10}{A\bar{B}} + \overset{11}{AB} = \sum m(1, 2, 3) \times \\ &\quad \uparrow \uparrow \\ &= B\bar{A} + \bar{B}A + BA = \sum m(1, 2, 3) \checkmark \\ &\quad \text{note the order, function should be written in same order.} \end{aligned}$$

Q8. Minimize the expression: $f(A, B, C) = \bar{A}\bar{B} + \bar{A}C + \bar{B}C$
 Ans. $= \bar{A}C + \bar{B}C$

Q9. Minimize the expression: $f(A, B, C) = (A+B)(A+C)(\bar{B}+C)$
 Ans. $= (A+B)(\bar{B}+C)$

Q10. Write the function for truth table and minimize it.

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	1

SOP form:

$$Y = \bar{A}\bar{B} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + AB\bar{C}$$

$$Y = \bar{A}\bar{B} + A\bar{B} + AB$$

$$Y = \bar{A}\bar{B} + A[\bar{B}+B]$$

$$Y = \bar{A} + A\bar{B} = (A+\bar{A})(A+\bar{B})$$

$$Y = A + \bar{B}$$

Pos:

$$Y = (A+B+1) \cdot (A+B\bar{C}) \cdot (\bar{A}+B+1) \cdot (\bar{A}+\bar{B}+1)$$

$$= 1(A+\bar{B}) \cdot 1 \cdot 1$$

$$= \underline{A + \bar{B}}$$

Q11. Write the function for truth table and minimize it.

Ans.

A	B	Y
0	0	C
0	1	$\bar{C}$
1	0	1
1	1	1

Sol:

$$Y: \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B} + AB$$

Q12. Two way switch is an example of which logic?

Ans. X-OR

Q13. If we have 4 variable, then total different expressions will be?

Ans.

$$2^{2^n} = 2^{2^4} = 2^{16} = 2^6 \times 2^{10} = 64K$$

Q14. $A + BC + \bar{A}C$ is equal to :

$$A + BC + \bar{A}C$$

$$= (A + \bar{A})(A + C) + BC$$

$$= A + C + BC$$

$$= A + C [1 + B]$$

$$= \underline{A + C} \text{ (not in option!)}$$

eg: $f = AB + \bar{A}\bar{B} + ABC + \bar{A}\bar{B}\bar{C}$

$$= AB[1 + C] + \bar{A}\bar{B}[1 + \bar{C}]$$

$$= AB + \bar{A}\bar{B} \text{ — — — (Semi minimised) } \rightarrow AB + \bar{A}\bar{B} + \bar{A}\bar{B}C$$

Modified Venn Diagram

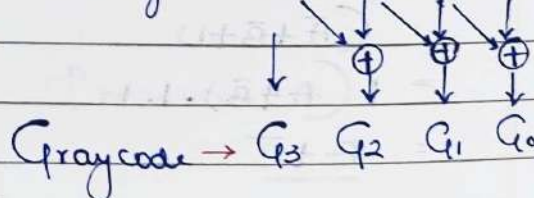
→ It is also known as "K-map".

Minimization by K-map

* Based on graycode

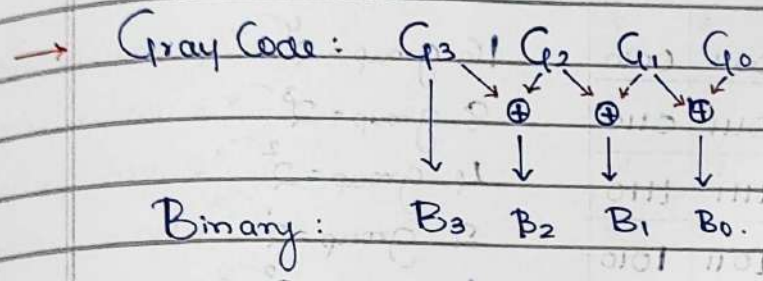
* Graycode:

→ Binary → B₃ B₂ B₁ B₀



eg: Binary: 0110101
Gray code: 0101111

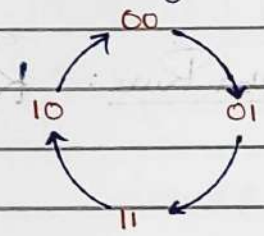
eg: Binary: 111001011
Gray code: 100101110



eg: Gray Code: 1101011
Binary: 1001101

Gray Code: Gray code is the code in which successive numbers differ by 1 bit.

Decimal	Binary	Gray Code
0	00	00
1	01	01
2	10	11
3	11	10



1. Unity Hamming distance code
2. Cyclic code
3. Reflecting code

Decimal	Binary	Gray Code
0	000	000
1	001	001
2	010	011
3	011	010
4	100	110
5	101	111
6	110	101
7	111	100

* eg: $f(A, B)$

MSB (Most Significant bit) LSB (Least Significant bit)

	$\bar{B}$	B
$\bar{A}$ 0	$\bar{A}\bar{B}$ 00 ②	$\bar{A}B$ 01 ①
A 1	$A\bar{B}$ 10 ③	AB 11 ④

→ $f(A, B, C)$

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$ 0	000 0	001 1	011 3	010 2
A 1	100 4	101 5	111 7	110 6

$f(A, B, C, D)$

* Group can be formed in order of 2^n .

		CD			
		$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
AB	$\bar{A}\bar{B}$	00	01	11	10
$\bar{A}\bar{B}$	00	0000	0001	0011	0010
$\bar{A}B$	01	0100	0101	0111	0110
AB	11	1100	1101	1111	1110
$\bar{A}B$	10	1000	1001	1011	1010

16 group = $2^4 \rightarrow$ 4 variable reduce

8 group = $2^3 \rightarrow$ 3 variable reduce

4 group = $2^2 \rightarrow$ 2 variable reduce

2 group = $2^1 \rightarrow$ 1 variable reduce

1 group = $2^0 \rightarrow$ 0 variable reduce

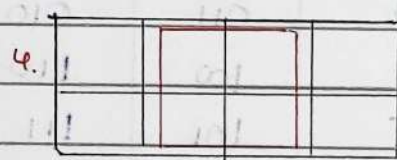
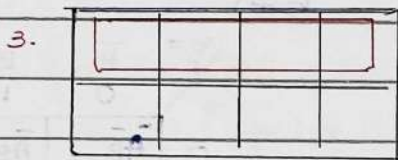
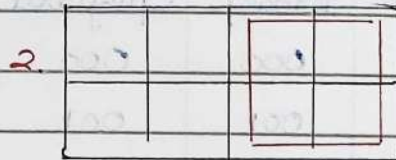
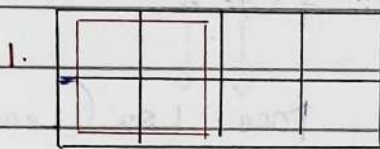
Rule of Minimization:

Termes cahi minimise

Cy Baba Rule: Kam se kam group banana hai and bade se bade group banana hai.

Variable cahi minimize.

Quad (making group)



And many similar combinations possible

etc.

Q1. $f(A, B) = \bar{A}\bar{B} + \bar{A}B + AB = \sum m(0, 1, 3)$

		B	
		$\bar{B}$	B
A	$\bar{A}$	1	1
A	$\bar{A}$	0	1

$f(A, B) = \bar{A}\bar{B} \cdot 1 + \bar{A}B \cdot 1 + AB \cdot 0 + AB \cdot 1$

$\Rightarrow \bar{A}B$

Q2. $f(A,B) = \bar{A}\bar{B} + \bar{A}B + A\bar{B} + AB = \sum m(0,1,2,3)$

	A \ B	0	1
	$\bar{A}$	1	1
	A	1	1

 $\Rightarrow 1$

Q3. $f(A,B,C) = \sum m(0,1,3,6,7)$

A \ BC				
	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0	1	1	0
A	1	0	1	1

Groupings: $\bar{A}\bar{B}$ (cells 0,1), $\bar{A}C$ (cells 1,3), AB (cells 3,7), $\bar{A}B$ (cells 1,5)

Expression: $\bar{A}\bar{B} + \bar{A}C + AB + \bar{A}B$

A \ BC				
	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0	1	1	0
A	1	0	1	1

Groupings: $\bar{A}\bar{B}$ (cells 0,1), $\bar{A}C$ (cells 1,3), AB (cells 3,7), $\bar{A}B$ (cells 1,5)

Expression: $\bar{A}\bar{B} + \bar{A}C + AB + \bar{A}B$

A \ BC				
	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0	1	1	0
A	1	0	1	1

Groupings: $\bar{A}\bar{B}$ (cells 0,1), $\bar{A}C$ (cells 1,3), AB (cells 3,7), $\bar{A}B$ (cells 1,5)

Expression: $\bar{A}\bar{B} + \bar{A}C + AB + \bar{A}B$

$$= \bar{A}\bar{B} + \bar{A}C + AB + \bar{A}B$$

Semi minimized
expression.

Question Practice Session

Q1. If $x \odot y = \bar{x} + y$ and $z = x \odot y$ Then $z \odot y$ will be:

$$\begin{aligned} z \odot y &= \bar{z} + y = \overline{x \odot y} + y \\ &= \overline{\bar{x} + y} + y = \bar{x} \cdot \bar{y} + y \\ &= \bar{x} \cdot \bar{y} + y \\ &= (\bar{x} + y)(\bar{y} + y) \\ &= \underline{\underline{\bar{x} + y}} \end{aligned}$$

Q2. If $A * B = AB + \bar{A}\bar{B}$ and $C = A * B$. Then which one is correct?

$$A \odot B = AB + \bar{A}\bar{B}$$

$$C = A \odot B$$

$$B \odot C = A$$

$$(A \odot B) \odot (A \odot B)$$

$$x \odot x = 1$$

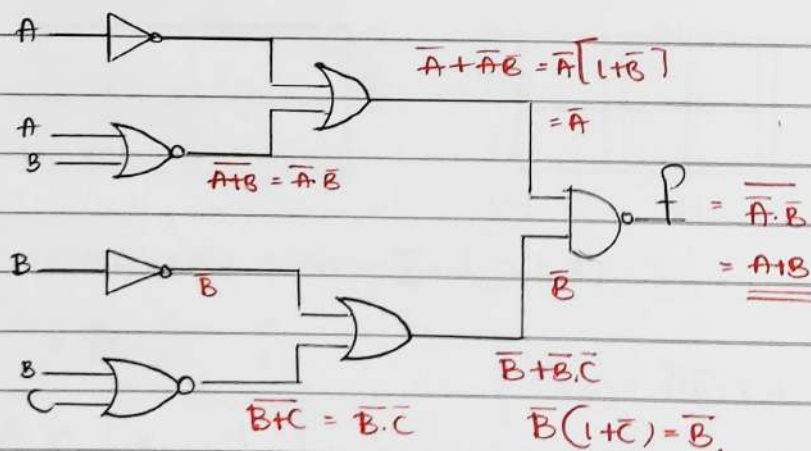
$$(i) A = B * C \checkmark$$

$$(ii) B = A * C \checkmark$$

$$(iii) A \odot B \odot C = 1 \times$$

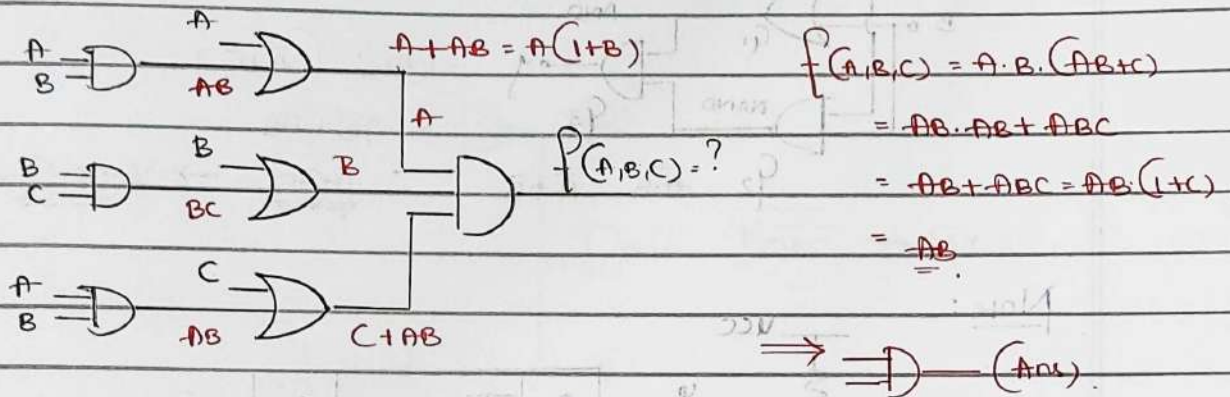
$$(iv) A = B \times$$

Q3. The output for the given logic circuit will be:

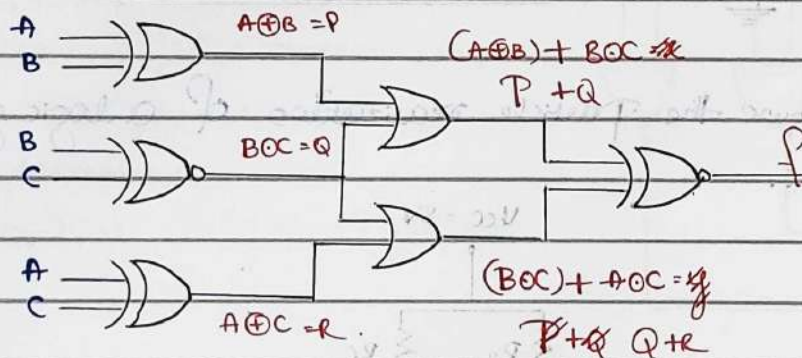


$$\Rightarrow \underline{\underline{F = A + B}}$$

Q4. Consider the given logic circuit with the inputs A, B and C then $f(A, B, C)$ will be.



Q5. The output for the given logic circuit will be:



$$f = \bar{P}\bar{Q}\bar{R} + P\bar{Q}\bar{R} + P\bar{Q}R + P\bar{Q}R + PQR$$

$$f = \bar{P}\bar{Q}\bar{R} + Q[P + \bar{P} + R] + PR$$

$$f = \bar{P}\bar{Q}\bar{R} + Q + PR$$

$$= \bar{P}\bar{Q}\bar{R} + Q + PR$$

$$= (\bar{P}\bar{R} + Q) + PR$$

$$f = \bar{P}\bar{R} + PR + Q \Rightarrow (P \oplus R) + Q$$

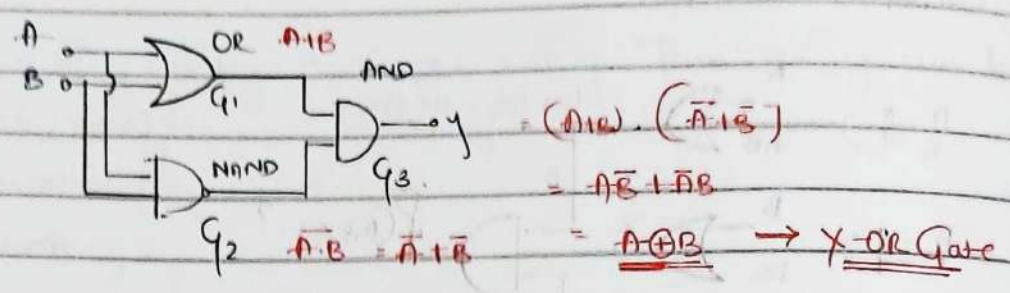
A \ BC	00	01	11	10
0	1		1	
1	1		1	

$$f = \bar{P}\bar{R} + PR + Q$$

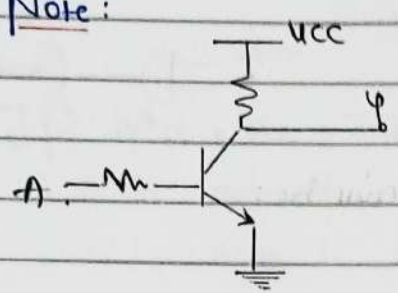
$$= \underline{\underline{B \oplus C}}$$

ABC	P A⊕B	Q B⊕C	R A⊕C	x	y	x⊕y
000	0	1	0	1	1	1
001	0	0	1	0	1	0
010	1	0	0	1	0	0
011	1	1	1	1	1	1
100	1	1	1	1	1	1
101	1	0	0	1	0	0
110	0	0	1	0	1	0
111	0	1	0	1	1	1

Q6. The following logic gate circuit equivalent to



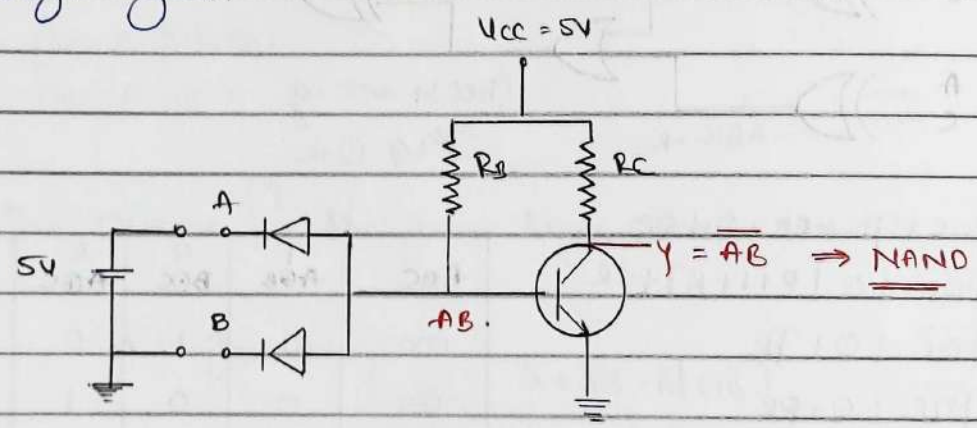
Note:



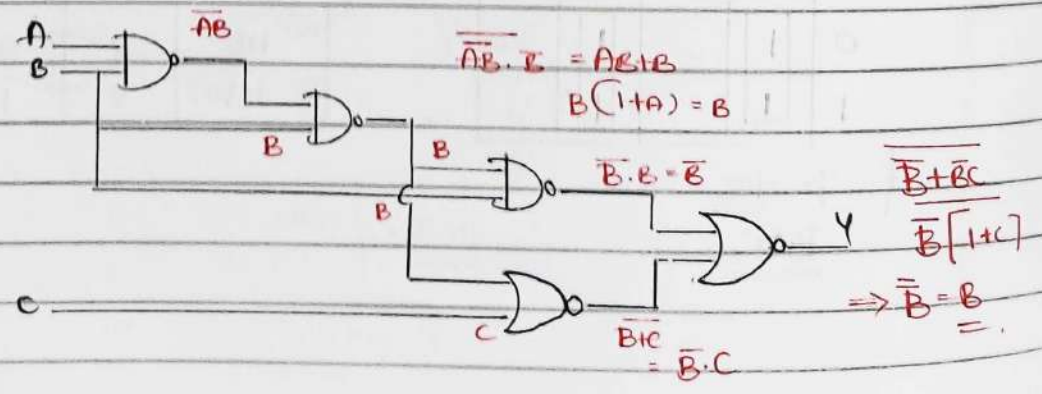
A	Transistor	Y
0	cutoff	1
1	Saturation	0

} NOT Gate!

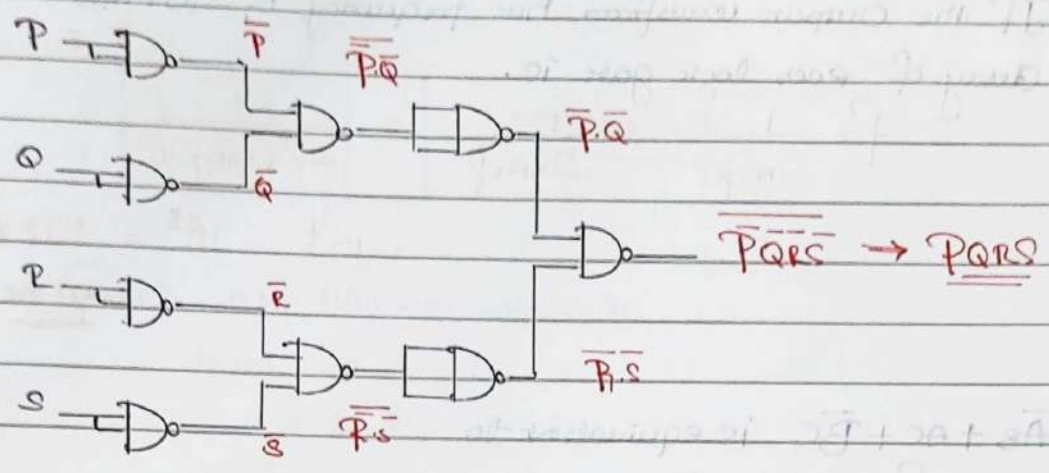
Q7. Figure shows the particle realization of a logic gate. Identify the logic gate.



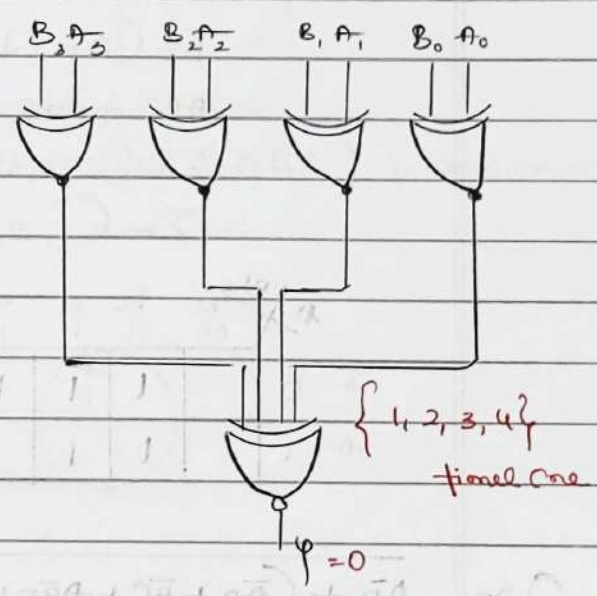
Q8. For the logic circuit shown, the simplified boolean expression for the output y is.



Q13. For the circuit shown in fig. the boolean expression for the output y in terms of inputs P, Q, R and S is.

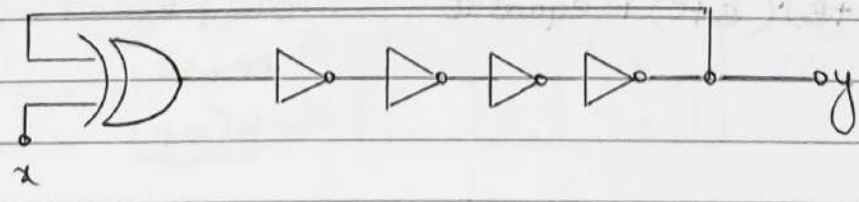


Q14. $A_3, A_2, A_1, A_0, B_3, B_2, B_1, B_0$ is shown in figure. To get output $y=0$, Choose one pair of correct input numbers.
 $\Rightarrow 0010, 1011$



Q15. The minimum number of two input NAND gate required to implement $y = abcd$ is.
 $y = abcd$
 $NAND = (2n-2) + k = (2 \times 4 - 2) + 0 = 6$

Q16. All the logic gates in the circuit shown below, have equal finite propagation delay.



Q17. The Circuit can be used as a clock generator, if $X=1$

Q18. If the output waveform has frequency of 10 MHz the propagation delay of each logic gate is.

$$f = \frac{1}{2\pi \tau p d} = \frac{1}{2 \times \pi \times f} = \frac{1}{2 \times 5 \times 10 \times 10^6}$$
$$= \frac{10^{-7}}{10} = 10^{-8} = 10 \times 10^{-9}$$
$$= \underline{\underline{10 \text{ ns}}}$$

Q 19 $\bar{A}B + AC + \bar{B}C$ is equivalent to

$$\begin{aligned} f &= \overline{A}B + AC + \overline{B}C \\ &= \overline{A}B(\overline{C} + C) + A(\overline{B} + B)C + (\overline{A} + A)\overline{B}C \\ &= \overline{A}B\overline{C} + \overline{A}BC + A\overline{B}\overline{C} + ABC + \overline{A}\overline{B}C + A\overline{B}C \\ &= \sum m(2, 3, 5, 7, 1) \\ &= \sum m(1, 2, 3, 5, 7) \end{aligned}$$

$A \backslash BC$
 $\overline{BC}$ $\overline{BC}$ BC BC
 00 01 11 10

$\overline{A}$ 0		1	1	1
A 1		1	1	

$\Rightarrow C + \overline{A}B$

Q 20 $\overline{A}\overline{B} + (\overline{A}B + \overline{B}C + A\overline{B}D + A\overline{B}\overline{D})$ is equal to

$$\begin{aligned} & \overline{A}B + \overline{A}B + \overline{B}C + \overline{A}B\overline{D} + \overline{A}B\overline{D} \\ & \overline{A}B [1 + D + \overline{D}] + \overline{A}B + \overline{B}C \\ & \overline{A}B + \overline{A}B + \overline{B}C \\ & \overline{A}B \cdot \overline{A}B \cdot \overline{B}C \\ & (\overline{A} + B)(\overline{A} + \overline{B})(B + \overline{C}) \end{aligned}$$

Q21. $\overline{(\bar{A} + \bar{B}) \cdot (\bar{B} + C)}$ is equal to : $\overline{\bar{A} + \bar{B}} + \overline{\bar{B} + C}$

$$\overline{A+B} + \overline{B+C}$$
$$AB + BC$$
$$\Rightarrow \underline{B(A+C)}$$

Q22. $\bar{A}\bar{B} + AC + \bar{B}C$ is equivalent to

$$\bar{A}\bar{B}(\bar{C}+C) + A(\bar{B}+B) + (\bar{A}+A)\bar{B}C$$

$$\Sigma_m(0, 1, 5, 7)$$

A \ BC	00	01	11	10
0	1	1		
1		1	1	

$$\Rightarrow \underline{\underline{\bar{A}\bar{B} + AC}}$$

Q23. $(A+B)(A+C)(A+\bar{C})$ is equivalent to

$$(A+BC)(A+\bar{C})$$

$$= A + BC\bar{C}$$

$$\Rightarrow \underline{\underline{A}}$$

HW

Q24. A logical function is given as

$$f(A, B, C, D) = \bar{B}\bar{C}[A + B\bar{C}D + \bar{B}CD + \bar{A}\bar{B}\bar{C}] \text{ is equivalent to}$$

$$Q25. f(A, B, C) = \Sigma_m(0, 1, 3, 5, 6, 7)$$

A \ BC	00	01	11	10
0	1	1	1	
1		1	1	1

$$= \bar{A}\bar{B} + AB + C$$

$$Q26. f(A, B, C) = \Sigma_m(0, 2, 4, 6)$$

A \ BC	00	01	11	10
0	1			1
1	1			1

$$\bar{C} \Rightarrow \bar{C}$$

$$Q27. f(A, B, C) = \Sigma_m(0, 3, 5, 6)$$

A \ BC	00	01	11	10
0	1		1	
1		1		1

Q28 $f(A, B, C, D) = \sum (0, 2, 4, 6, 10, 11, 13, 15)$

AB \ CD	00	01	11	10
$\bar{A}\bar{B}$ 00	1			1
$\bar{A}B$ 01	1			1
AB 11		1	1	
$A\bar{B}$ 10			1	1

$$\Rightarrow \bar{A}\bar{D} + \bar{A}BD + \bar{A}\bar{B}C$$

$\bar{A}BD$

$\bar{A}\bar{B}C$

Q29 $f(A, B, C, D) = \sum m (0, 1, 2, 4, 6, 9, 10, 11, 12, 13, 15)$

AB \ CD	00	01	11	10
$\bar{A}\bar{B}$ 00	1	1		1
$\bar{A}B$ 01	1			1
AB 11	1	1	1	
$A\bar{B}$ 10		1	1	1

$$\Rightarrow \bar{A}\bar{D} + \bar{A}D + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}B\bar{C}$$

$\bar{A}\bar{B}C$

$\bar{A}B$

$\bar{A}\bar{B}C$

Q30 $f(A, B, C, D) = \sum m (1, 5, 6, 7, 11, 12, 13, 15)$

AB \ CD	00	01	11	10
$\bar{A}\bar{B}$ 00		1		
$\bar{A}B$ 01		1	1	1
AB 11	1	1	1	
$A\bar{B}$ 10			1	

$$\Rightarrow \bar{A}\bar{B}C + \bar{A}\bar{B}CD + \bar{A}B + \bar{A}CD$$

Q31 $f(A, B, C, D) = \sum m (1, 2, 5, 6, 8, 10, 12, 13, 14, 15)$

AB \ CD	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10
$\bar{A}\bar{B}$ 00		1		1
$\bar{A}B$ 01		1		1
AB 11	1	1	1	1
$A\bar{B}$ 10	1			1

$$\Rightarrow AB + \bar{C}\bar{D} + \bar{A}\bar{D} + \bar{A}\bar{C}\bar{D}$$

Quad $\bar{A}\bar{D}$

Q 32. $f(A, B, C, D) = \sum m(1, 5, 6, 7, 11, 12, 13, 15)$

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	1	1	1
10	1	1	1	1
11	1		1	1

$$\Rightarrow \bar{A} + B + C + \bar{D}$$

eg::

AB \ CD	00	01	11	10
00	1	1		1
01		1	1	
11	1	1		1
10	1			1

$$\Rightarrow \bar{B}\bar{D} + \bar{B}C\bar{D} + \bar{A}B\bar{D} + A\bar{C}\bar{B} + \bar{A}\bar{B}C$$

$$\Rightarrow \bar{B}\bar{D} + \bar{B}C\bar{D} + \bar{A}B\bar{D} + A\bar{C}\bar{D} + \bar{A}\bar{C}D$$

Don't Care Conditions:

Combination of inputs on which the output may or maynot depends are called don't care conditions.

$$f(A, B) = \bar{A}B + \textcircled{AB}$$

Q1. $f(A, B, C) = \sum m(0, 1, 6, 7) + \sum d(3)$

A \ BC	00	01	11	10
0	1	1	X	
1			1	1

$\Rightarrow \bar{A}B + \bar{A}\bar{B}$

$f(A, B, C) = \sum m(0, 1, 2) + \sum d(3, 6)$

A \ BC	00	01	11	10
0	1	1	X	1
1				X

eg::

A \ BC	00	01	11	10
0	1	X	X	X
1			X	X

$\Rightarrow \bar{A}$

A \ BC	00	01	11	10
0	X	1	1	
1		X	1	X

$\Rightarrow C$

Q2. $f(A, B, C, D) = \sum m(0, 2, 4, 6, 7, 8, 10, 11, 12, 14, 15) + \sum d(1, 3)$

AB \ CD	00	01	11	10
00	1	x	x	1
01	1		1	1
11	1		1	1
10	1		1	1

$\Rightarrow C + \bar{D}$

eg.:

AB \ CD	00	01	11	10
$\bar{A}\bar{B}$ 00	1	1	x	1
$\bar{A}B$ 01			1	x
AB 11		x	1	
$A\bar{B}$ 10	x	x		1

$$\Rightarrow \bar{A}\bar{B} + \bar{B}\bar{D} + BCD$$

AB \ CD	00	01	11	10
$\bar{A}\bar{B}$ 00	1	1		
$\bar{A}B$ 01	1	x	x	1
AB 11		x	x	1
$A\bar{B}$ 10			1	

$$\Rightarrow \bar{A}\bar{B}\bar{C} + BC + ACD$$

Q3. $F(A, B, C) = \prod m(0, 1, 3, 6, 7) = \sum m(2, 4, 5)$

M1:

A \ BC	00	01	11	10
$\bar{A}$ 0	0	0	0	1
A 1	1	1	0	0

Pos. $\begin{cases} F = (A+B) \cdot (\bar{A}+\bar{B}) \cdot (A+\bar{C}) \\ F = (A+B) \cdot (\bar{A}+\bar{B}) \cdot (\bar{B}+\bar{C}) \end{cases}$

SOP $\begin{cases} f = A\bar{B} + \bar{A}B\bar{C} \end{cases}$

B+C
B $\bar{C}$
00

M2 (Method 2)

A \ BC	00	01	11	10
$\bar{A}$ 0	0	0	0	1
A 1	1	1	0	0

$$\bar{F} = \bar{A}\bar{B} + AB + \bar{A}C$$

$$F = \overline{\bar{A}\bar{B} + AB + \bar{A}C}$$

$$F = (A+B) \cdot (\bar{A}+\bar{B}) \cdot (A+\bar{C})$$

Q4. $f(A, B, C, D) = \sum m(0, 1, 3, 6, 7, 10, 11)$ minimize it in Pos form
 $= \pi M(2, 4, 5, 8, 9, 12, 13)$

$AB \backslash CD$	$C+D$ 00	$C+\bar{D}$ 01	$\bar{C}+D$ 11	$\bar{C}+\bar{D}$ 10
$A+B$ 00	1	1	1	0
$A+\bar{B}$ 01	0	0	1	1
$\bar{A}+B$ 11	0	0	0	0
$\bar{A}+\bar{B}$ 10	0	0	1	1

$$F = (\bar{B}+C) (\bar{A}+B) (\bar{A}+C) (A+B+\bar{C}+D)$$

$AB \backslash CD$	$C+D$ 00	$C+\bar{D}$ 01	$\bar{C}+D$ 11	$\bar{C}+\bar{D}$ 10
$A+B$ 00	0	0		
$A+\bar{B}$ 01	0		0	X
$\bar{A}+B$ 11		X	X	
$\bar{A}+\bar{B}$ 10		0	0	X

$$F = (A+C+D) \cdot (B+C+\bar{D}) (\bar{B}+\bar{C}+\bar{D}) (\bar{A}+\bar{D})$$

Variable inside the K-map.

eg:

$A \backslash B$	$\bar{B}$	B
$\bar{A}$	0	C
A	1	

$$\rightarrow \bar{A}\bar{B}C + \bar{A}BC$$

$$= \bar{A}\bar{B}C + \bar{A}B.C = \bar{A}C(\bar{B}+B) = \bar{A}C$$

eg:

$A \backslash B$	$\bar{B}$	B
$\bar{A}$	0	1
A	1	C

$$f(A, B, C) = \bar{A}\bar{B}.1 + \bar{A}BC$$

$$= \bar{A}[\bar{B}+BC]$$

$$= \bar{A}[(\bar{B}+B) \cdot (\bar{B}+C)]$$

$$= \bar{A}(\bar{B}+C)$$

Method 1

Boolean Algebra

Method 4

Case 1: $C=0$

$A \backslash B$	$\bar{B}$	B
$\bar{A}$	0	1
A	1	0

$$= \bar{A}\bar{B}C$$

Case 2: $C=1$

$A \backslash B$	$\bar{B}$	B
$\bar{A}$	0	1
A	1	0

$$\Rightarrow \bar{A}C$$

$$\bar{A}\bar{B}C + \bar{A}C$$

$$\Rightarrow \bar{A}[C + \bar{B}C]$$

$$\Rightarrow \bar{A}[(\bar{B}+C)]$$

$$\Rightarrow \bar{A}(\bar{B}+C)$$

Method 2

$$f(A, B, C) = \bar{A}\bar{B}(\bar{C}+C) + \bar{A}BC$$

$$= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}BC$$

$$= \sum m(0, 1, 3) = \prod M(2, 4, 5, 6, 7)$$

$A \backslash B$	$\bar{B}$	$\bar{B}$	B	B
	00	01	11	10
$\bar{A}$ 0	1	1	1	0
A 1	0	0	0	0

Sop $f = \bar{A}\bar{B} + \bar{A}C$

$$= \underline{\underline{\bar{A}(\bar{B}+C)}}$$

$A \backslash B$	$\bar{B}$	$\bar{B}$	B	B
	00	01	11	10
$\bar{A}$ 0	1	1	1	0
A 1	0	0	0	0

Pos $= \underline{\underline{\bar{A}(\bar{B}+C)}}$

Method 3

$A \backslash B$	$\bar{B}$	B
	0	1
$\bar{A}$ 0	1	C
A 1		

$\rightarrow \underline{\underline{\bar{A}(\bar{B}+C)}}$

Q1.

$A \backslash B$	$\bar{B}$	B
	0	1
$\bar{A}$ 0	1	C
A 1		C

$$\underline{\underline{BC + \bar{A}(\bar{B}+C)}}$$

$A \backslash B$	0	1
$\bar{A}$ 0		1
A 1		

$$\bar{A}\bar{B}\bar{C}$$

$$\bar{A}\bar{B}\bar{C} + \bar{A}C + BC$$

$$\Rightarrow \bar{A}(\bar{B}\bar{C}+C) + BC$$

$$\Rightarrow \underline{\underline{\bar{A}(\bar{B}+C) + BC}}$$

$A \backslash B$	0	1
$\bar{A}$ 0	1	1
A 1		1

$$(\bar{A}+B)C$$

Q2.

$A \backslash B$	$\bar{B}$	B
	0	1
$\bar{A}$ 0		1
A 1	C	C

$$AC + B(\bar{A}+C)$$

$$\Rightarrow \underline{\underline{AC + \bar{A}B + BC}}$$

$A \backslash B$	$\bar{A}$	A
	0	1
$\bar{A}$ 0		1
A 1	0	0

$$(\bar{A}\bar{B}\bar{C})$$

$A \backslash B$	$\bar{B}$	B
	0	1
$\bar{A}$ 0		1
A 1	1	1

$$(A+B)C$$

$$\bar{A}\bar{B}\bar{C} + AC + BC$$

$$\Rightarrow \underline{\underline{AC + B(\bar{A}\bar{C}+C) = AC + B(\bar{A}+C)}}$$

Implicants and Prime Implicants

1. Implicants: The total number of min-terms in the boolean expression are called implicants. Or in kmap the total no. of 1 is called implicants.
2. Prime Implicants: Total number of possibilities of formation of group are called prime implicants.
3. Essential Prime Implicants / Selective PI: If there are two answers for a boolean expression then the terms common in both the answers are called essential prime implicants.
4. Reduced Prime Implicants: The terms that are not essential prime implicants are called reduced prime implicants.

Q1. $f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}\bar{C} + ABC$
 $= \sum m(0, 1, 3, 6, 7)$

A \ B \ C				
	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$ 0	1	1	1	
A 1			1	1

$$\Rightarrow \bar{A}\bar{B} + \bar{A}B + \bar{A}C$$

Implicants = 5

Prime Implicants = 4

Essential prime implicants =

A \ B \ C				
	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$ 0	1	1	1	
A 1			1	1

$$\Rightarrow \bar{A}\bar{B} + \bar{A}B + BC$$

SPI = 1

RPI = 1

$$PI \{ \bar{A}\bar{B}, \bar{A}B, \bar{A}C, BC \}$$

Q2. $f(A, B, C) = \sum m(2, 3, 4, 5, 7)$
 $PI \{ \bar{A}\bar{B}, \bar{A}B, BC, AC \}$

Q5. $f(A, B, C, D) = \sum m(1, 5, 6, 7, 11, 12, 13, 15)$

$AB \backslash CD$	00	01	11	10
00		1		
01		1	1	1
11	1	1	1	
10			1	

$P_2 \{ \bar{A}\bar{C}D, \bar{A}BC, ACD, \bar{A}BC, BD \}$

Implicants = 8

$P_1 = 5$

$EPI = 4$

$SPI = 0$

$RPI = 1$

$\Rightarrow \bar{A}\bar{C}D + \bar{A}BC + ACD + \bar{A}BC$

$EPI \{ \bar{A}\bar{C}D + \bar{A}BC + ACD + \bar{A}BC \}$

Q6.

$A \backslash BC$	00	01	11	10
0	1	1	1	
1		1	1	1

$P_1 \{ \bar{A}\bar{B} + AB + C \}$

$\Rightarrow \bar{A}\bar{B} + AB + C$

Implicants = 6

$P_1 = 3$

$EPI = 3$

$SPI = 0$ $RPI = 0$

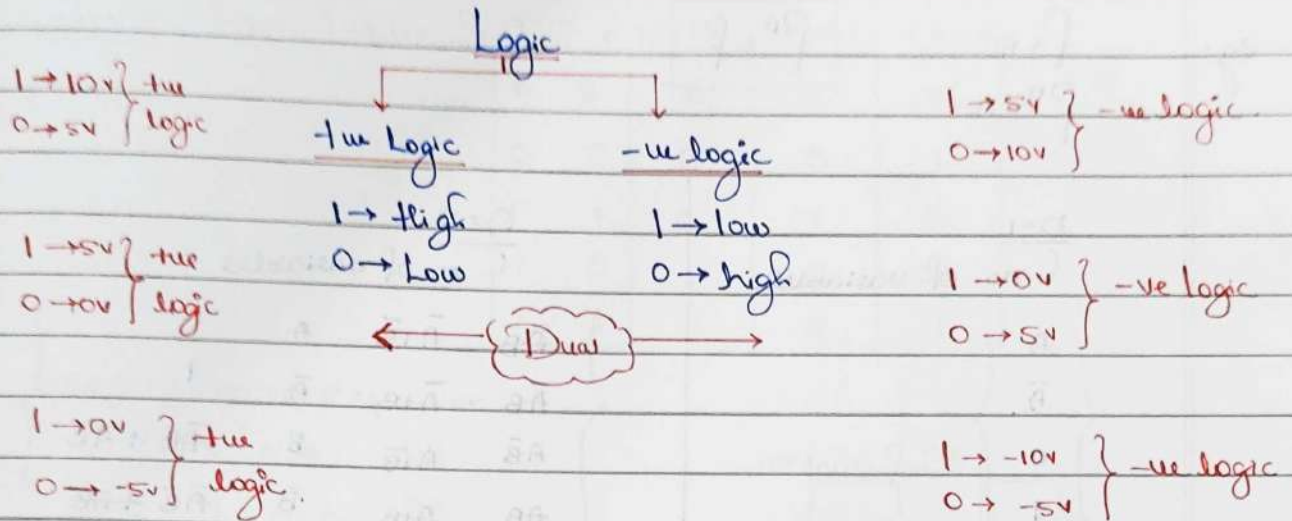
HW

Q7.

$AB \backslash CD$	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$ 00	1	1		
$\bar{A}B$ 01		1	1	
AB 11			1	1
$A\bar{B}$ 10				1

Q8.

$AB \backslash CD$	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$ 00	1	1		
$\bar{A}B$ 01	1	1		1
AB 11		1	1	1
$A\bar{B}$ 10			1	1



AND Case (+ve)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

logic
conversion
↓
Dual

A	B	Y
1	1	1
1	0	1
0	1	1
0	0	0

OR

Dual $\Rightarrow$ +ve logic $\longleftrightarrow$ -ve logic

- | | |
|--|---|
| <p>1. $1 \longleftrightarrow 0$</p> <p>2. AND $\longleftrightarrow$ OR</p> <p>3. $\cdot \longleftrightarrow +$</p> <p>4. NAND $\longleftrightarrow$ NOR</p> <p>5. X-OR $\longleftrightarrow$ X-NOR</p> <p>6. Buffer $\longleftrightarrow$ Buffer</p> <p>7. Inverter $\longleftrightarrow$ Inverter.</p> | <p>$0 \longleftrightarrow 1$</p> <p>$A \longleftrightarrow A$</p> <p>$\bar{A} \longleftrightarrow \bar{A}$</p> <p>$f = A \cdot 1$</p> <p>$f^0 = A + 0 = A$</p> |
|--|---|

eg: $f = AB + CD + \bar{E}F$ $f^0 = (A+B) \cdot (C+D) \cdot (\bar{E}+F)$	$f = \overline{ABC + DEF}$ $f^0 = \overline{(A+B+C) \cdot (D+E+F)}$
--	---

eg: $f = AB + CD$ $f^0 = (A+B) \cdot (C+D)$

$\swarrow$
 $f^{00} = AB + CD$

$f^{00} = f$

Q9.

AB \ CD	CD	CD				
		00	01	11	10	
$\bar{A}\bar{B}$	00	1	1	1	1	Ance 1 $\Rightarrow \bar{B}\bar{D} + \bar{A}\bar{D} + \bar{A}\bar{D}$
$\bar{A}B$	01		1	1		
$A\bar{B}$	11	1			1	Ance 2 $\Rightarrow \bar{A}\bar{B} + \bar{A}\bar{D} + \bar{A}\bar{D}$
AB	10	1			1	

$\bar{A}\bar{D}$ (circled)
 $\bar{B}\bar{D}$ (circled)
 $\bar{A}\bar{D}$ (circled)

Implicants = 10
 PI = 4
 EPI = 2
 SPI = 1
 RPI = 1

Q10. Let a function F which has 3 input variables (x, y, z) . The function F will be high only when atleast two of the input variables are set to high. Draw the K-map for the given function. Let the number of PI in K-map = 'a' and the no. of EPI in K-map = 'b'. Find the quadratic mean of 'a' and 'b'.

Ans:

	x	y	z	f
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	1
7	1	1	1	1

$$f(x, y, z) = \sum m(3, 5, 6, 7)$$

x \ yz	yz	yz			
		00	01	11	10
$\bar{x}$	0			1	
x	1		1	1	1

$$a = \text{PI} = 3$$

$$b = \text{EPI} = 3$$

$$f = xy + yz + xz$$

Quadratic mean:

$$= \sqrt{\frac{a^2 + b^2}{2}} = \sqrt{\frac{3^2 + 3^2}{2}} = \underline{\underline{3}}$$

Q11. Find the number of prime implicants & essential prime implicants in the given K.

PQ \ RS	RS	RS			
		00	01	11	10
00	00	1			
01	01		1		1
11	11	1			
10	10		1		1

$$\text{PI} = 6$$

$$\text{EPI} = 6$$

Q3.

$$f(A, B, C) = \sum m(0, 1, 3, 4, 6, 7)$$

A \ BC	00	01	11	10
0	1	1	1	
1	1		1	1

$$\Rightarrow \bar{B}\bar{C} + \bar{A}C + AB$$

$$\text{Implicants} = 6$$

$$PI = 6$$

$$EPI = 0 \text{ (term common in both)}$$

$$SPI = 3$$

$$RPI = 3$$

$$PI \{ \bar{A}\bar{B} + BC + \bar{A}C + \bar{B}\bar{C} + AC + AB \}$$

A \ BC	00	01	11	10
0	1	1	1	
1	1		1	1

$$\Rightarrow \bar{A}\bar{B} + BC + \bar{A}C$$

Q2
(continued)

A \ BC	00	01	11	10
0			1	1
1	1	1	1	

$$\Rightarrow AB + \bar{A}B + AC$$

$$\text{Implicants} = 5$$

$$PI = 4$$

$$EPI = 2$$

$$SPI = 1$$

$$RPI = 1$$

EPI

A \ BC	00	01	11	10
0			1	1
1	1	1	1	

$$\Rightarrow AB + \bar{A}B + BC$$

$$PI \{ AB + \bar{A}B + AC + BC \}$$

$$Q4. f(A, B, C) = \sum m(0, 2, 3, 4, 5, 7)$$

A \ BC	00	01	11	10
0	1		1	1
1	1	1	1	

$$\Rightarrow \bar{B}\bar{C} + AC + \bar{A}B$$

$$\text{Implicants} = 6$$

$$PI = 6$$

$$EPI = 0$$

$$SPI = 3$$

$$RPI = 3$$

A \ BC	00	01	11	10
0	1		1	1
1	1	1	1	

$$\Rightarrow \bar{A}\bar{C} + BC + \bar{A}\bar{B}$$

$$PI \{ \bar{B}\bar{C} + AC + \bar{A}B + \bar{A}\bar{C} + BC + \bar{A}\bar{B} \}$$

eg. $f = A$
 $f^D = A$ $f^D = f$ $\rightarrow$ Self Dual.

$n=1$ (no. of variables)	$n=2$ (no. of variables)
$\left\{ \begin{array}{c} A \\ \bar{A} \\ 0 \\ 1 \end{array} \right\}$	$\left\{ \begin{array}{cc} \bar{A}\bar{B} & \bar{A}+B \\ \bar{A}B & \bar{A}+B \\ A\bar{B} & A+B \\ AB & A+B \end{array} \right\}$
Self dual = 2	$\left\{ \begin{array}{cc} A & 0 \\ \bar{A} & 1 \\ B & \bar{A}B + A\bar{B} \\ \bar{B} & \bar{A}\bar{B} + A\bar{B} \end{array} \right\}$
	Self dual = 4

* n variables $\rightarrow 2^{2^n}$ Expressions $\rightarrow 2^{2^n-1}$ Self dual expressions.

Combinational Circuit.

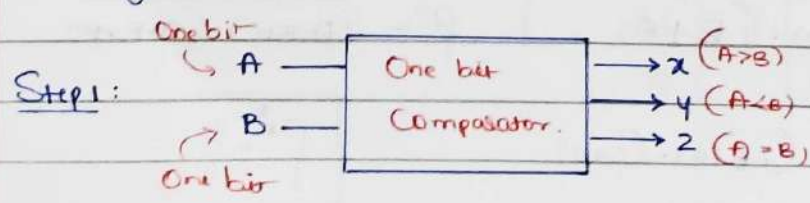
- * A circuit without feedback or memory are called Combinational Circuit.
- * Static Circuit.

Designing of Combinational Circuit:

- Step 1: Find the number of inputs and outputs.
- Step 2: Write the truth table
- Step 3: Write the logical expression
- Step 4: Minimize the logical expression.
- Step 5: Hardware implementation.

Comparator

1. Design a one bit Comparator.



Step 2: Truth Table

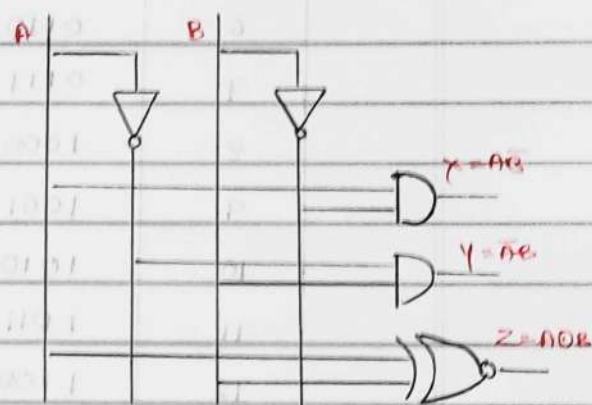
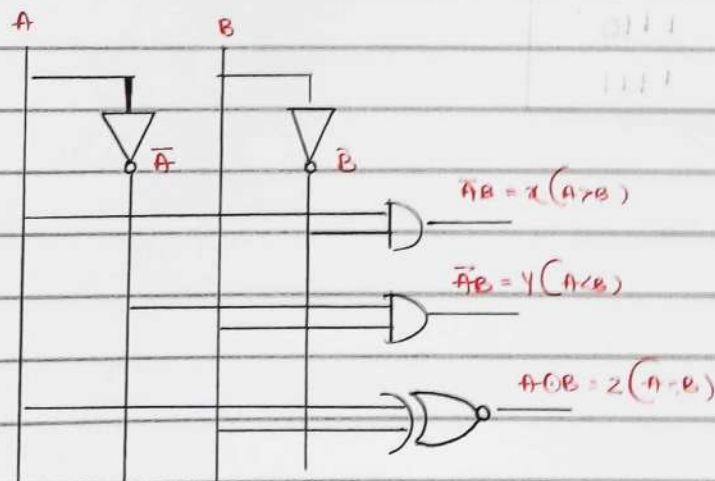
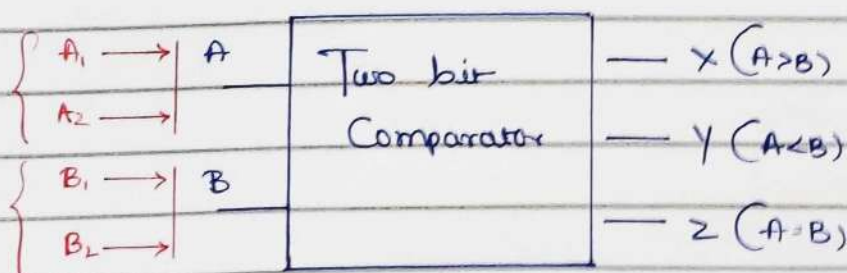
		$(A > B)$	$(A < B)$	$(A = B)$
A	B	X	Y	Z
0	0	0	0	1
0	1	0	1	0
1	0	1	0	0
1	1	0	0	1

Step 3: Logical expression:

$$X (A > B) = \bar{A}B$$

$$Y (A < B) = A\bar{B}$$

$$Z (A = B) = \bar{A}\bar{B} + AB = A \odot B$$

Step 4: Minimization (Already minimized)Steps: Hardware implementation.Two Bit Comparator:

Decimal	Binary	Step 2 Truth Table	A ₁	A ₀	B ₁	B ₀	X	Y	Z
0	0000		0	0	0	0	0	0	1
1	0001		0	0	0	1	0	1	0
2	0010		0	0	1	0	0	1	0
3	0011		0	0	1	1	0	1	0
4	0100		0	1	0	0	1	0	0
5	0101		0	1	0	1	0	0	1
6	0110		0	1	1	0	0	1	0
7	0111		0	1	1	1	0	1	0
8	1000		1	0	0	0	1	0	0
9	1001		1	0	0	1	1	0	0
10	1010		1	0	1	0	0	0	1
11	1011		1	0	1	1	0	1	0
12	1100		1	1	0	0	1	0	0
13	1101		1	1	0	1	1	0	0
14	1110		1	1	1	0	1	0	0
15	1111		1	1	1	1	0	0	1

Step 3 Logical expression:

$$X(A > B) = \sum m(4, 8, 9, 12, 13, 14)$$

$$Y(A < B) = \sum m(1, 2, 3, 6, 7, 11)$$

$$Z(A = B) = \sum m(0, 5, 10, 15)$$

Step 4 Minimization

for X

		B ₁ B ₀			
		00	01	11	10
A ₁ A ₀	00				
	01	1			
	11	1	1		1
	10	1	1		

for Y

		B ₁ B ₀			
		00	01	11	10
A ₁ A ₀	00		1	1	1
	01			1	1
	11				
	10			1	

Semiminimised

$$\begin{aligned}
 X(A > B) &= A_1 \bar{B}_1 + \bar{A}_1 A_0 \bar{B}_1 \bar{B}_0 + A_1 A_0 B_1 \bar{B}_0 \\
 &= A_1 \bar{B}_1 + (\bar{A}_1 \bar{B}_1 + A_1 B_1) A_0 \bar{B}_0 \\
 &= A_1 \bar{B}_1 + (\bar{A}_1 \odot B_1) A_0 \bar{B}_0
 \end{aligned}$$

Minimised

$$\begin{aligned}
 X(A > B) &= A_1 \bar{B}_1 + A_0 \bar{B}_1 \bar{B}_0 + A_1 A_0 \bar{B}_0 \\
 &= A_1 \bar{B}_1 + (A_1 + B_1) A_0 \bar{B}_0
 \end{aligned}$$

Pr. 2'

$A_1 A_0 / B_1 B_0$	00	01	11	10
00	1			
01		1		
11			1	
10				1

$$Z(A=B) = (A_1 \odot B_1) \cdot (A_0 \odot B_0)$$

Semi Minimised

$$\begin{aligned}
 Y(A < B) &= \bar{A}_1 B_1 + \bar{A}_1 \bar{A}_0 \bar{B}_1 B_0 + A_1 \bar{A}_0 B_1 B_0 \\
 &= \bar{A}_1 B_1 + (\bar{A}_1 \bar{B}_1 + A_1 B_1) \bar{A}_0 B_0 \\
 &= \bar{A}_1 B_1 + (\bar{A}_1 \odot B_1) \bar{A}_0 B_0
 \end{aligned}$$

Minimised

$$Y(A < B) = \bar{A}_1 B_1 + (\bar{A}_1 + B_1) \bar{A}_0 B_0$$

H.W.Step 5

Hardware Implementation

ComparatorsOne bit Comparator

Total Conditions = 4

Equal Condition = 2

Unequal Condition = 2

Greater = less Cond = 1

Two bit Comparisons

Total conditions = 16

Equal Condition = 4

Unequal " = 12

Greater = less " = 6

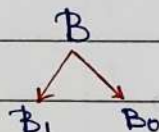
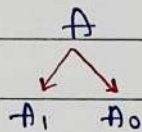
Three bit Comparators

Total combinations = 64

Equal condition = 8

Unequal " = 56

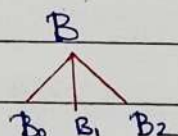
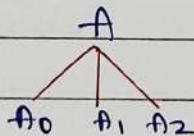
Greater = less " = 28

'N' bit ComparatorsTotal Condition = 2^{2n} Equal Condition = 2^n Unequal Condition = $2^{2n} - 2^n$ Greater = less = $(2^{2n} - 2^n) / 2$ 

$$X (A > B) = A_1 \bar{B}_1 + (A_1 \odot B_1) A_0 \bar{B}_0$$

$$Y (A < B) = \bar{A}_1 B_1 + (A_1 \odot B_1) \bar{A}_0 B_0$$

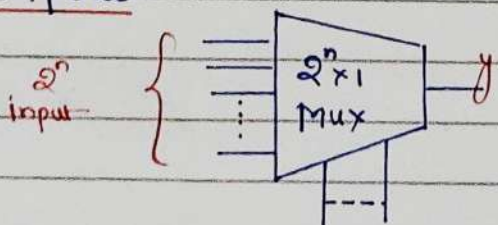
$$Z (A = B) = (A_1 \odot B_1) \cdot (A_0 \odot B_0)$$



$$X (A > B) = A_2 \bar{B}_2 + (A_2 \odot B_2) A_1 \bar{B}_1 + (A_2 \odot B_2) (A_1 \odot B_1) A_0 \bar{B}_0$$

$$Y (A < B) = \bar{A}_2 B_2 + (A_2 \odot B_2) \bar{A}_1 B_1 + (A_2 \odot B_2) (A_1 \odot B_1) \bar{A}_0 B_0$$

$$Z (A = B) = (A_2 \odot B_2) (A_1 \odot B_1) (A_0 \odot B_0)$$

Multiplexer→ 2^n to 1 Converter

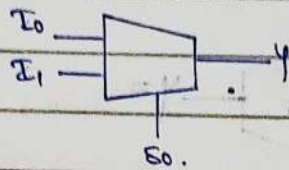
→ Universal Logic

→ AND OR Logic

n-Select line

Q1. Design a 2x1 Mux.

Step 1:



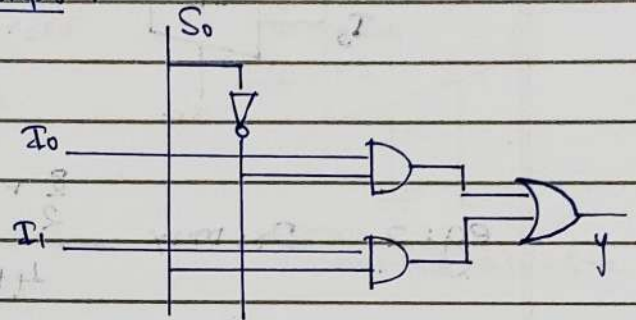
Step 2:

S_0	Y
0	I_0
1	I_1

Step 3: $Y = \bar{S}_0 I_0 + S_0 I_1$

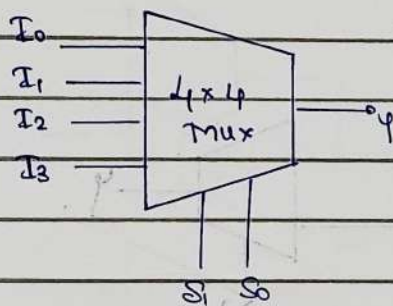
Step 4: Already Minimized

Step 5:



Q2. Design a 4x1 Mux.

Step 1:



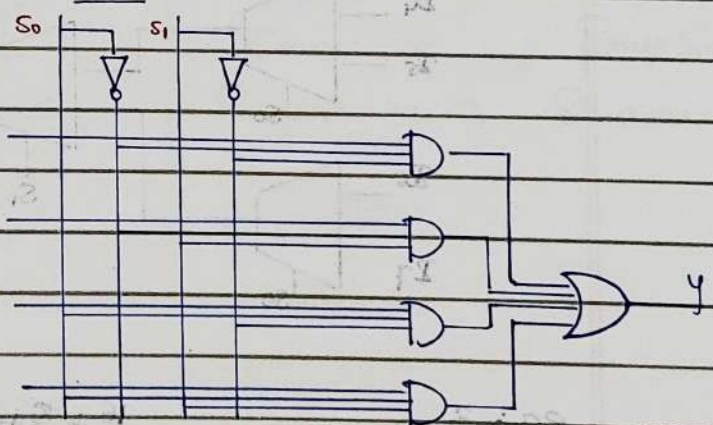
Step 2:

S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Step 3: $Y = \bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3$

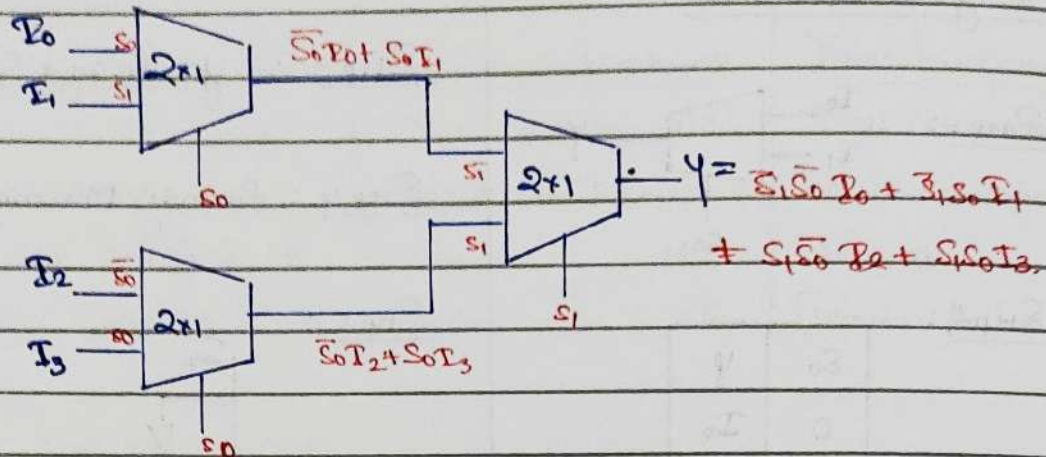
Step 4: No minimization

Step 5:

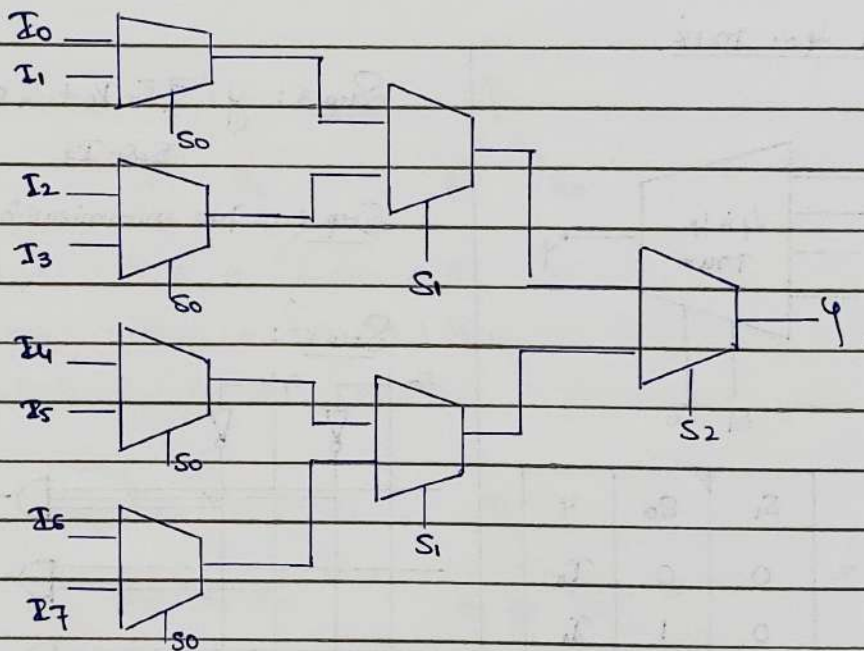


Type: 1 Designing of higher order mux by using lower order mux.

eg. $2 \times 1 \text{ mux} \xrightarrow{\frac{4^2}{2} + \frac{2^1}{2}} 4 \times 1 \text{ mux}$
 $2 + 1 = 3$



eg: 2 2x1 mux $\frac{8}{2} + \frac{4}{2} + \frac{2}{2}$ 8x1 mux
 $4 + 2 + 1 = 7$



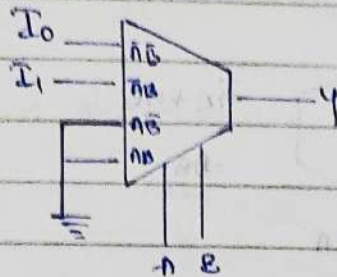
eg: 3 2x1 mux $\frac{16}{2} + \frac{8}{2} + \frac{4}{2} + \frac{2}{2}$ 16x1 mux
 $8 + 4 + 2 + 1 = 15$

2x1 mux 2^3 64x1 mux

2x1 mux $2^n - 1$ $2^n \times 1$ mux

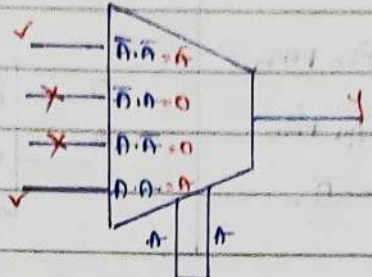
Q1. 4×1 mux $\xrightarrow{\frac{8}{4} + \frac{8}{4}}$ 8×1 mux
2 select line $2+1=3$ 3 select line

Method 1



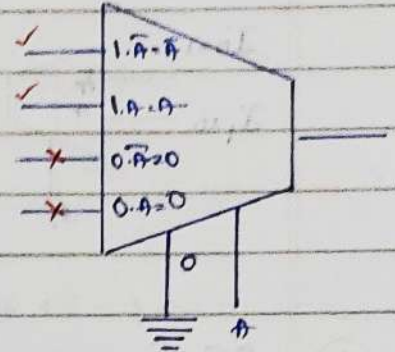
Select-line = 2

Method 2



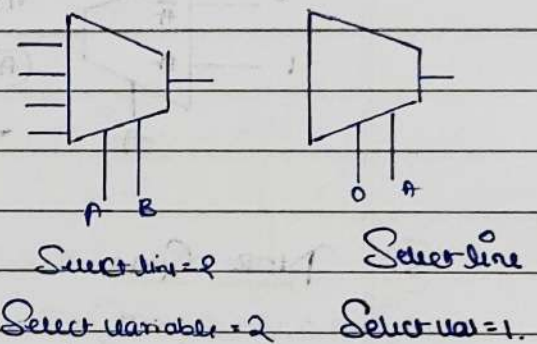
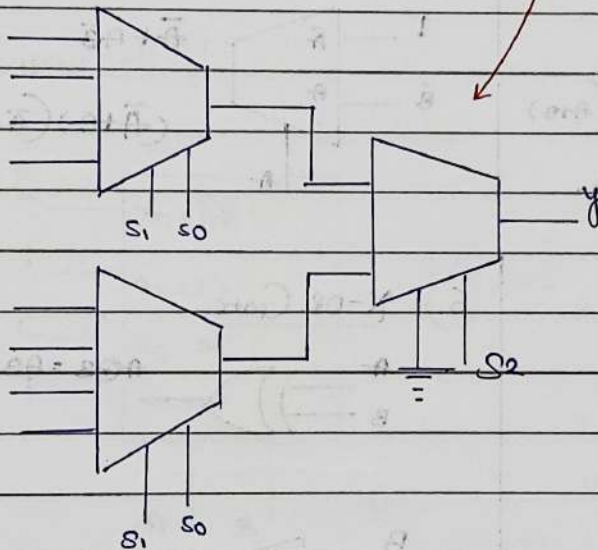
Select-line = 1

Method 3



Select variable = One

4×1 mux $\longrightarrow$ 8×1 mux
3 Select variable



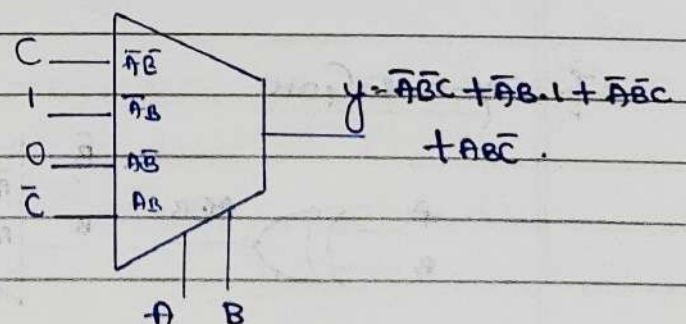
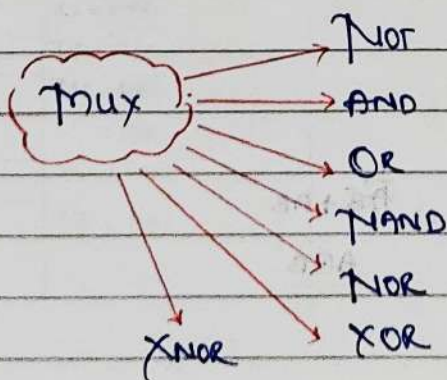
Select-line = 2

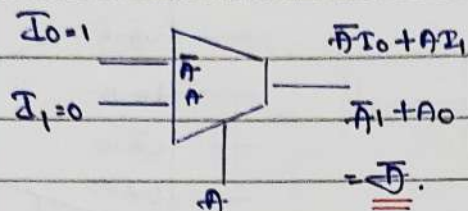
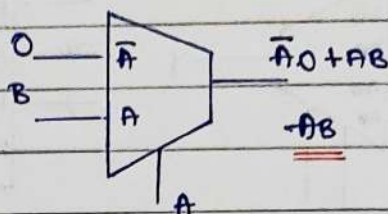
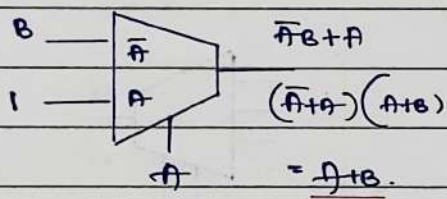
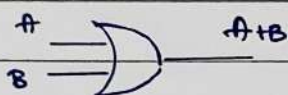
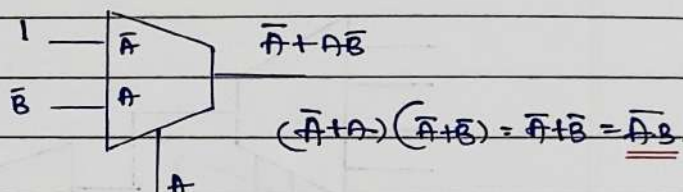
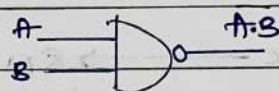
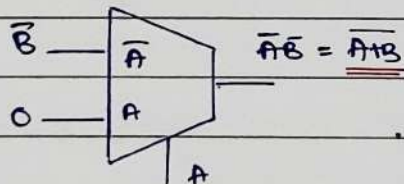
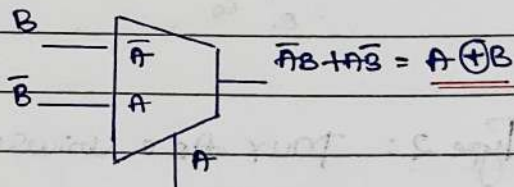
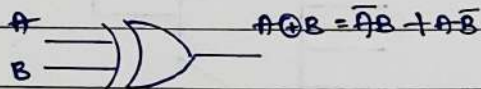
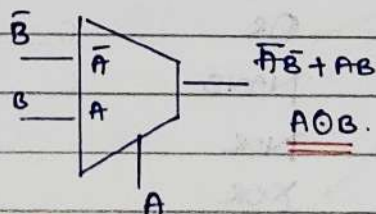
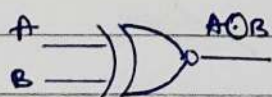
Select-line

Select variable = 2

Select var = 1

Type 2: Mux As a universal logic.

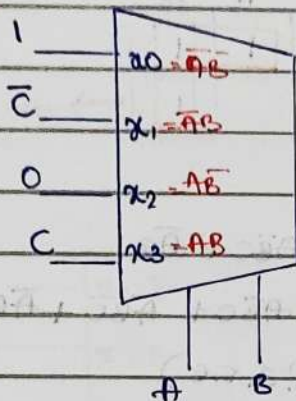


1. Not Gate2. And Gate3. Or Gate4. Nand Gate5. Nor Gate6. X-OR Gate7. X-NOR Gate

Note: To design NOT, AND, OR $\rightarrow$ One 2x1 mux required
To design NAND, NOR, XOR, XNOR, $\rightarrow$ Two 2x1 mux required.

Type 3 Minimization.

Q. Find the output f :

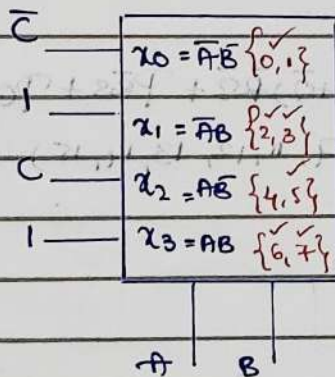


$\begin{matrix} A \backslash B \\ \end{matrix}$	00	01	11	10
0	1	1		1
1			1	

$$\Rightarrow \bar{A}\bar{C} + \bar{A}B + ABC$$

$$\begin{aligned} f(A, B, C) &= \bar{A}\bar{B}.1 + \bar{A}\bar{B}\bar{C} + \bar{A}B.0 + AB.C \\ &= \bar{A}\bar{B} + \bar{A}\bar{B}\bar{C} + ABC \\ &= \bar{A}\bar{B}(\bar{C} + C) + \bar{A}\bar{B}\bar{C} + ABC \\ &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + ABC \\ &= \sum m(0, 1, 2, 7) \end{aligned}$$

Q2. Find output of "f."



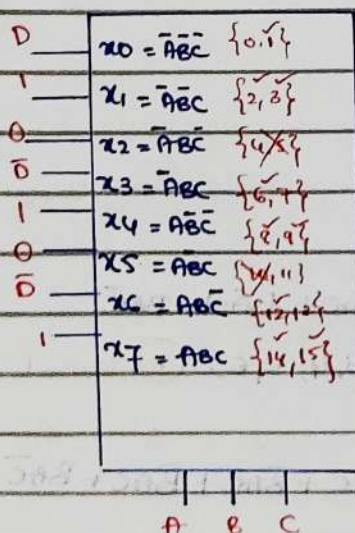
$$\begin{aligned} 010 &= 2 & 011 &= 3 & 110 &= 6 & 111 &= 7 \\ \bar{A}\bar{B}\bar{C} & & \bar{A}\bar{B}C & & A\bar{B}\bar{C} & & ABC & \end{aligned}$$

$$\begin{aligned} f(A, B, C) &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B} + \bar{A}\bar{B}C + AB \\ &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}(\bar{C} + C) + \bar{A}\bar{B}C + AB(\bar{C} + C) \\ &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}\bar{B}C + A\bar{B}\bar{C} + AB\bar{C} + ABC \\ &= \sum m(0, 2, 3, 5, 6, 7) \end{aligned}$$

$\begin{matrix} A \backslash B \\ \end{matrix}$	00	01	11	10
0	1		1	1
1		1	1	1

$$\Rightarrow B + \bar{A}\bar{C} + AC$$

Q3.

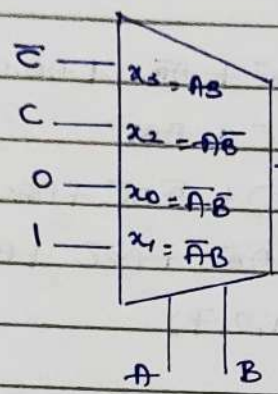


$$\begin{aligned} f(A, B, C, D) &= \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD + A\bar{B}\bar{C}\bar{D} \\ &\quad + ABC \\ &= \sum m(1, 2, 3, 6, 8, 9, 12, 14, 15) \end{aligned}$$

AB \ C	00	01	11	10
00		1	1	1
01				1
11	1		1	1
10	1	1		

(Spain)

Q.4 find F = ?



A \ BC	00	01	11	10
0			1	1
1		1		1

$\Rightarrow \underline{\underline{\bar{A}B + BC + A\bar{B}C}}$

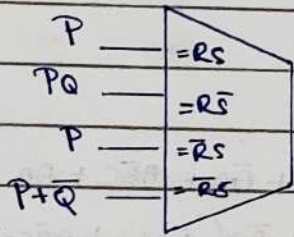
$$\begin{aligned}
 f(A, B, C) &= A\bar{B}\bar{C} + A\bar{B}C + \bar{A}B \\
 &= A\bar{B}\bar{C} + A\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC \\
 &= \sum m(2, 3, 5, 6)
 \end{aligned}$$

$\rightarrow P(\bar{Q} + Q)\bar{R}\bar{S}$

$P\bar{Q}\bar{R}\bar{S} = 1000$

$PQ\bar{R}\bar{S} = 1100$

Q.5

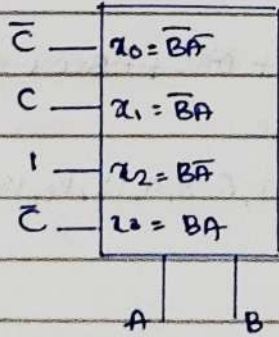


$$\begin{aligned}
 (P, Q, R, S) &= (P + \bar{Q})\bar{R}\bar{S} + \bar{P}RS + P\bar{Q}R\bar{S} + P\bar{R}S \\
 &= \sum m(0, 8, 9, 11, 12, 13, 14, 15)
 \end{aligned}$$

PQ \ RS	00	01	11	10
00	1			
01				
11	1	1	1	1
10	1	1	1	

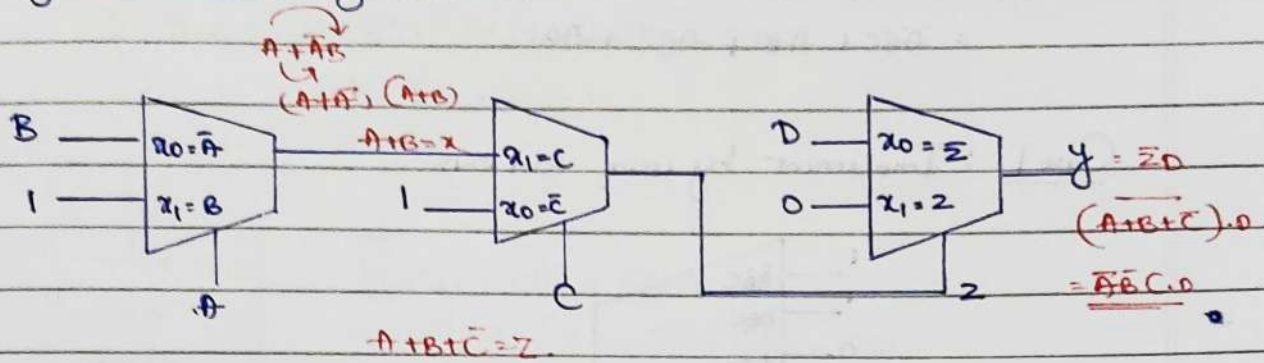
$\Rightarrow PQ + PS + \bar{Q}\bar{R}\bar{S}$

Q.6.

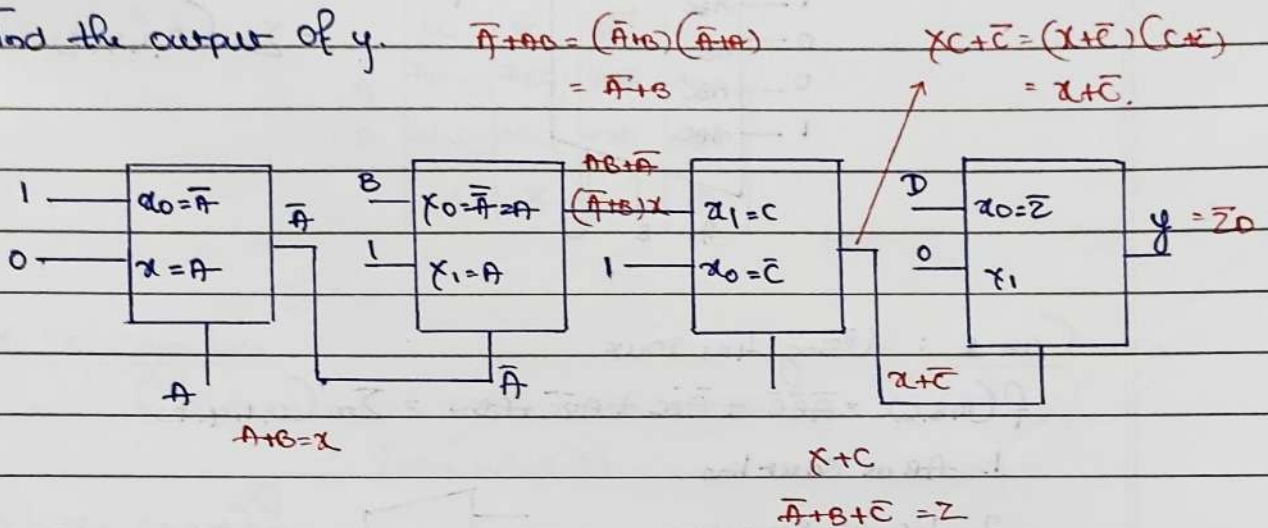
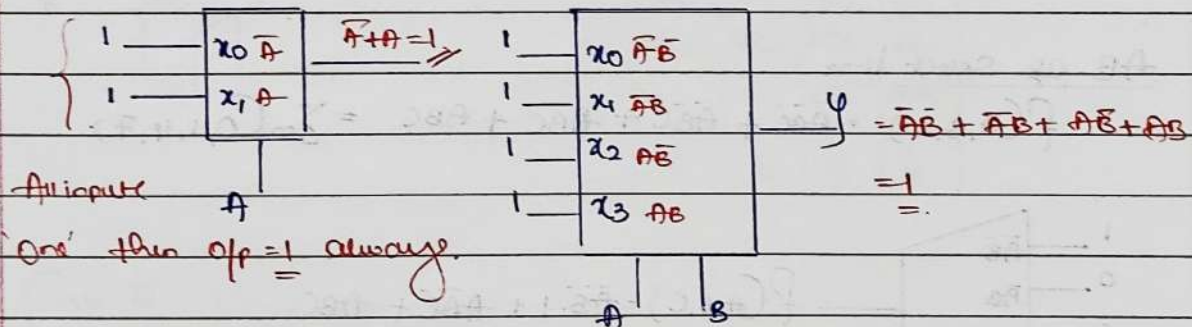
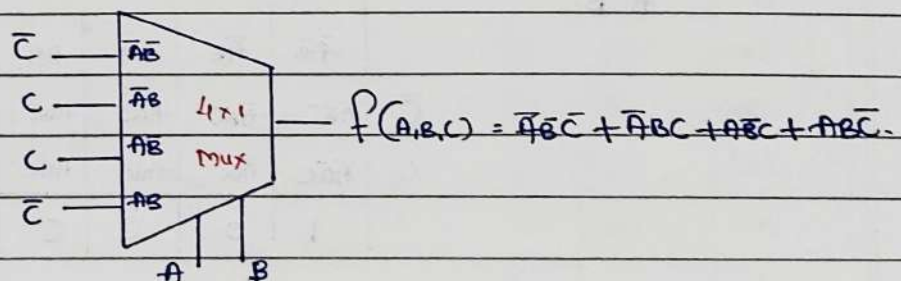


$$\begin{aligned}
 f(A, B, C) &= \bar{B}\bar{A}\bar{C} + \bar{B}\bar{A}C + \bar{B}A + B\bar{A}\bar{C} \\
 &= \sum m(0, 3, 4, 5, 6) \text{ (1) method}
 \end{aligned}$$

$$f(A, B, C) = \bar{B}\bar{A}\bar{C} + \bar{B}\bar{A}C + \bar{B}A\bar{C} + \bar{B}AC + B\bar{A}\bar{C} \text{ (2) method}$$

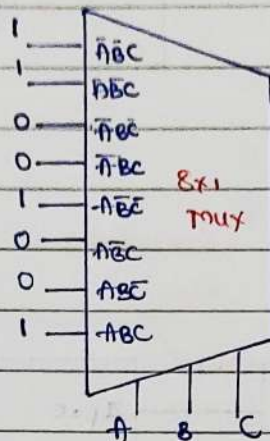
Type 4: Cascading of Mux:

Q1. Find the output of y.

Note :Type 5: Implementation of Functioning :

Ex 1. $f(A,B,C) = \sum m(0,1,4,7)$
 $= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$

Case I : Implement by using 8x1 mux



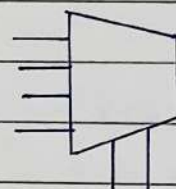
$$f(A,B,C) = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

$$= \sum m(0,1,4,7)$$

Case II : Using 4x1 Mux

$$f(A,B,C) = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC = \sum m(0,1,4,7)$$

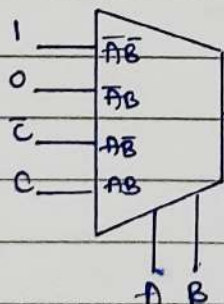
1. AB as select line
2. BC as select line
3. AC as select line



$$f(A,B,C) = \sum m(0,1,4,7)$$

1. AB as select line

$$f(A,B,C) = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC = \sum m(0,1,4,7)$$



$$f(A,B,C) = \bar{A}\bar{B}.1 + A\bar{B}\bar{C} + ABC$$

$$= \bar{A}\bar{B}(\bar{C}+C) + A\bar{B}\bar{C} + ABC$$

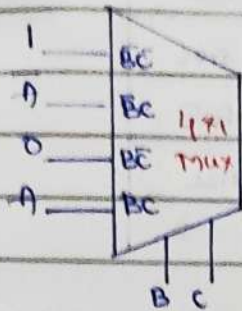
$$= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC$$

	$\bar{A}\bar{B}$	$\bar{A}B$	$A\bar{B}$	AB
$\bar{C}$	$\bar{A}\bar{B}\bar{C}$ 0	$\bar{A}B\bar{C}$ 2	$A\bar{B}\bar{C}$ 4	$AB\bar{C}$ 6
C	$\bar{A}B\bar{C}$ 1	$\bar{A}BC$ 3	$A\bar{B}C$ 5	ABC 7
	1	0	$\bar{C}$	C

2. BC as Select Line

$$f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C$$

$$= \sum m(0, 1, 4, 7)$$



$$f(A, B, C) = \bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}C$$

$$= (\bar{A} + A)\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}C$$

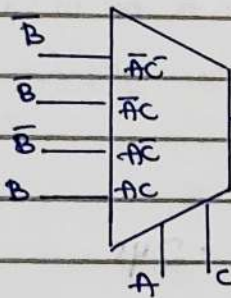
$$= \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}C$$

	$\bar{B}\bar{C}$	$\bar{B}C$	$B\bar{C}$	BC
$\bar{A}$	$\bar{A}\bar{B}\bar{C}$ ①	$\bar{A}\bar{B}C$ ②	$\bar{A}B\bar{C}$ ③	$\bar{A}BC$ ④
A	$A\bar{B}\bar{C}$ ⑤	$A\bar{B}C$ ⑥	$AB\bar{C}$ ⑦	ABC ⑧
	1	$\bar{A}$	0	A

3. AC is Select Line

$$f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

$$= \sum m(0, 1, 4, 7)$$



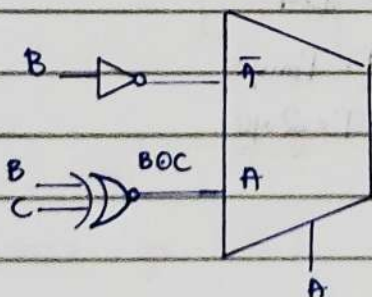
$$f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

	$\bar{A}\bar{C}$	$\bar{A}C$	$A\bar{C}$	AC
$\bar{B}$	$\bar{A}\bar{B}\bar{C}$ ①	$\bar{A}B\bar{C}$ ②	$A\bar{B}\bar{C}$ ③	$AB\bar{C}$ ④
B	$\bar{A}B\bar{C}$ ⑤	$A\bar{B}\bar{C}$ ⑥	$A\bar{B}C$ ⑦	ABC ⑧
	$\bar{B}$	$\bar{B}$	$\bar{B}$	B

Case III: Using 2x1 Mux(i) A as Select line

$$f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

$$= \sum m(0, 1, 4, 7)$$



$$\bar{A}\bar{B} + A(\bar{B}\bar{C} + BC)$$

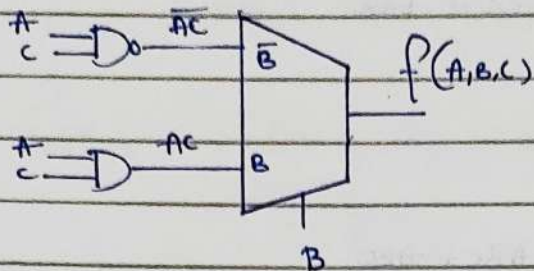
$$\bar{A}\bar{B}(\bar{C} + C) + A\bar{B}C$$

$$+ ABC$$

$$= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC$$

$$= f(A, B, C) + ABC$$

	$\bar{A}$	A
$\bar{B}\bar{C}$	$\bar{A}\bar{B}\bar{C}$ ①	$A\bar{B}\bar{C}$ ④
$\bar{B}C$	$\bar{A}\bar{B}C$ ②	$A\bar{B}C$ ⑤
$B\bar{C}$	$\bar{A}B\bar{C}$ ③	$AB\bar{C}$ ⑥
BC	$\bar{A}BC$ ⑦	ABC ⑧
	$\bar{B}$	$B\bar{C}$

(ii) B as Select line

$$f(A, B, C) = \sum m(0, 1, 4, 7)$$

	$\bar{B}$	B
$\bar{A}\bar{C}$	$\bar{A}\bar{B}\bar{C}$ ②	$\bar{A}B\bar{C}$ 2
$\bar{A}C$	$\bar{A}\bar{B}C$ ①	$\bar{A}BC$ 3
AC	$A\bar{B}C$ ④	ABC 6
$\bar{A}\bar{C}$	$A\bar{B}\bar{C}$ ⑤	$AB\bar{C}$ ⑦
$\bar{A}C + A\bar{B}\bar{C} + A\bar{B}C + ABC$		$\bar{A}C$

$$= \bar{A}(\bar{C} + C) + \bar{A}C$$

$$= \bar{A} + \bar{A}C$$

$$= \bar{A}$$

Ans(iii) C as Select line $f(A, B, C) = \sum m(0, 1, 4, 7)$

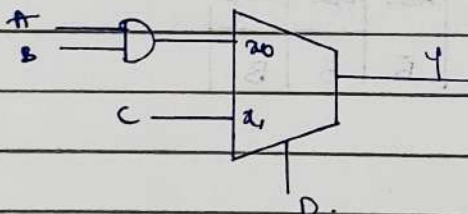
Q1. $f(A, B, C, D) = \sum m(0, 1, 3, 5, 7, 9, 12, 15)$

1. ABD is Select line2. ACD is Select line

→ 16x1 mux, 8x1 mux, 4x1 mux

2x1 mux

(
 AB, AC, AD as Select line
 A as select line.

Type-6: Delay in Mux

$$T_{AND} = 1\mu s, T_{mux} = 2\mu s$$

Case 1: $D=0$

$$T = T_{AND} + T_{mux}$$

$$= 1\mu s + 2\mu s$$

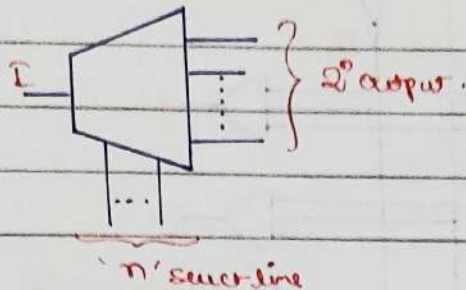
$$T = \underline{\underline{3\mu s}}$$

Case 2: $D=1$

$$T = T_{mux}$$

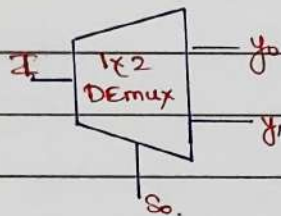
$$T = \underline{\underline{2\mu s}}$$

DE-mux: DE-mux is called And Logic



Q1. Design 1x2 DE-mux

Step 1:



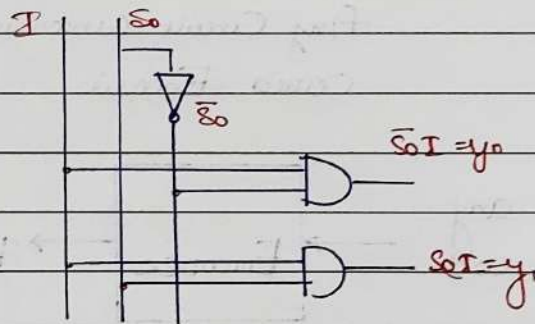
Step 2:

S_0	Y_0	Y_1
0	I	0
1	0	I

Step 3: $Y_0 = \bar{S}_0 I$

$Y_1 = S_0 I$

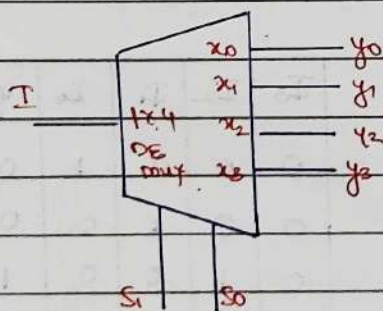
Step 5:



Step 4:

Q2. Design a 1x4 DEMUX?

Step 1:



Step 2:

S_1	S_0	Y_0	Y_1	Y_2	Y_3
0	0	I	0	0	0
0	1	0	I	0	0
1	0	0	0	I	0
1	1	0	0	0	I

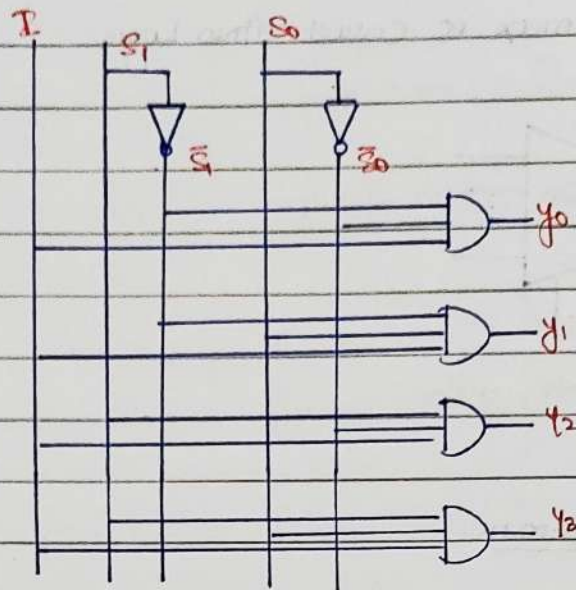
Step 3: $Y_0 = \bar{S}_1 \bar{S}_0 I$

$Y_2 = S_1 \bar{S}_0 I$

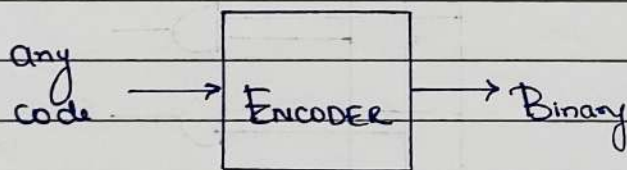
Step 4:

$Y_1 = \bar{S}_1 S_0 I$

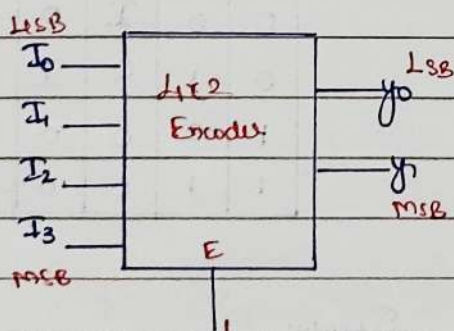
$Y_3 = S_1 S_0 I$

Steps:10wQ. Design a 1×8 DE MUXQ. Design a 1×16 DE MUX.Encoder:

Any Circuit which is used to convert any code into Binary is called Encoder.



- 1) 4×2 Encoder (Quart to Binary)
- 2) 8×3 Encoder (Octa to Binary)
- 3) 16×4 Encoder (Hexa to Binary).

Q. Design a 4×2 Encoder.Step 1:Step 2:

I_3	I_2	I_1	I_0	y_1	y_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

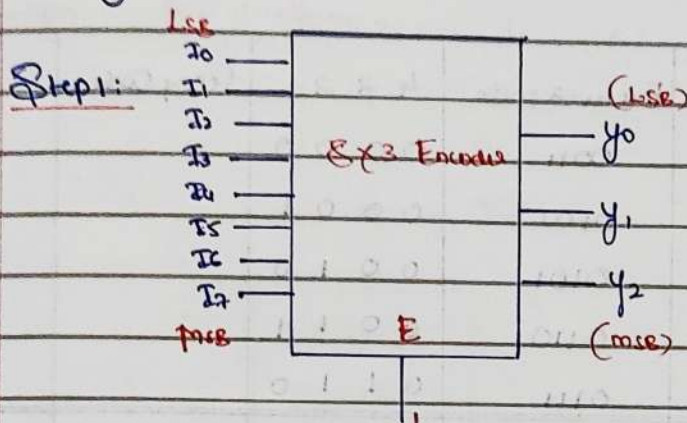
Step 3: $y_0 = I_1 + I_3$
 $y_1 = I_2 + I_3$

Step 5:



Step 4:

Q. Design a 8x3 Encoder.



HW

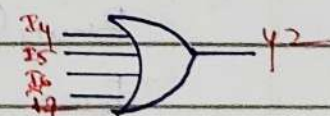
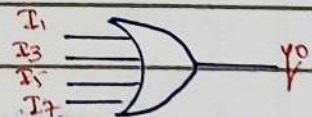
Q. Design a 16x4 Encoder

Step 2:

I_7	I_6	I_5	I_4	I_3	I_2	I_1	I_0	y_2	y_1	y_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

Step 3: $y_0 = I_1 + I_3 + I_5 + I_7$
 $y_1 = I_2 + I_3 + I_6 + I_7$
 $y_2 = I_4 + I_5 + I_6 + I_7$

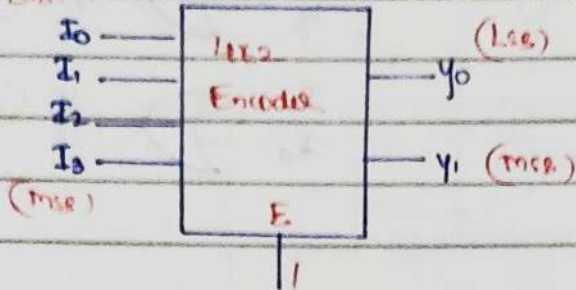
Step 5:



Step 4:

Note: Priority Encoder

(LSE)

LSE Priority Encoder

(MSE)	I_3	I_2	I_1	I_0 (LSE)	Y_1	Y_0
	X	X	X	1	0	0
	X	X	1	0	0	1
	X	1	0	0	1	0
	1	0	0	0	1	1

→ weighted, self-complementing

Decimal	BCD	Excess-3 code	4 2 2 1	Gray Code
0	0000	0011	0 0 0 0	
1	0001	0100	0 0 0 1	
2	0010	0101	0 0 1 0	
3	0011	0110	0 0 1 1	
4	0100	0111	0 1 1 0	
5	0101	1000	1 0 0 1	
6	0110	1001	1 1 0 0	
7	0111	1010	1 1 0 1	
8	1000	1011	1 1 1 0	
9	1001	1100	1 1 1 1	

Decimal → Weighted code
 → Self Complementing

$10^3 \ 10^2 \ 10^1 \ 1$
 $a_3 \ a_2 \ a_1 \ a_0$

 $r \rightarrow$ Base (radix)Complement $(r-1)'s$ $r's$ Complement $r=2$

1's

2's

 $r=8$

7's

8's

 $r=10$

9's

10's

BCD → Weighted Code
 → Not a Self Complementing Code.

→ Binary Coded Decimal

→ Each decimal numbers are represented by 4 bits.

eg.

Decimal - 0
BCD - 0000

Decimal - 9
BCD - 1001

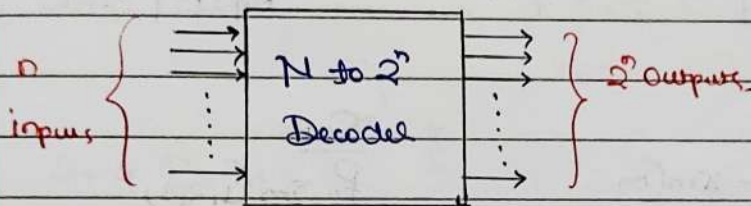
Decimal - 29
BCD - 0010001

Excess - 3 code:

- It is not a weighted code
- Self complemented code

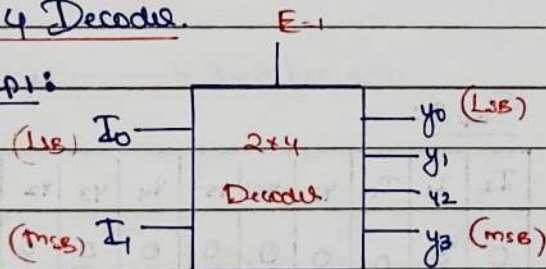
Decoder: Circuit which is used to convert Binary into any other code.

1. 2×4 Decoder
2. 3×8 Decoder
3. 4×16 Decoder.



2×4 Decoder.

Step 1:



Step 2:

I_1	I_0	y_3	y_2	y_1	y_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
ϕ	1	1	0	0	0

Step 3:

$$y_0 = \bar{I}_1 \bar{I}_0$$

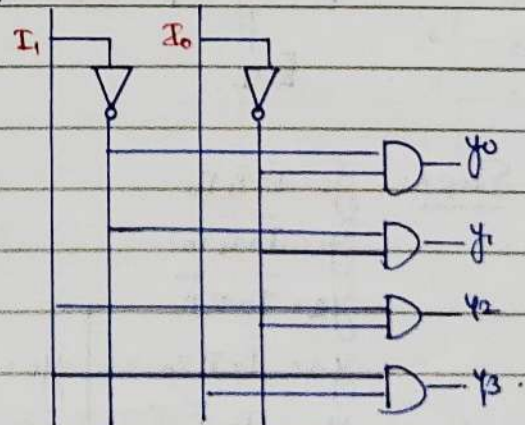
$$y_1 = \bar{I}_1 I_0$$

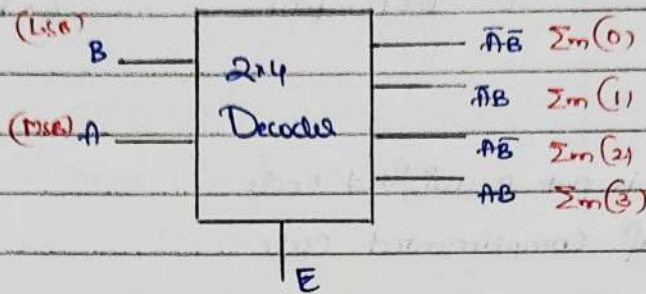
$$y_2 = I_1 \bar{I}_0$$

$$y_3 = I_1 I_0$$

Step 4: minimization

Step 4:



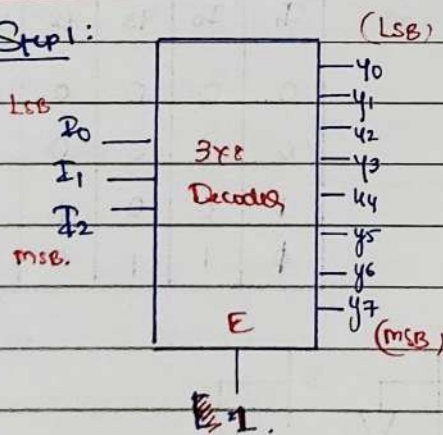
Example:

eg:

(LSB) B	(MSB) A	$\overline{AB}$	$\overline{A}\overline{B}$	$\Sigma m(0)$	A	B	P
0	0	0	1	$\Sigma m(1,2,3)$	0	0	0
0	1	0	0	$\Sigma m(0,2,3)$	0	1	1
1	0	0	1	$\Sigma m(0,1,3)$	1	0	1
1	1	0	0	$\Sigma m(0,1,2)$	1	1	1

$$\underline{\text{Pos}} = P = (A+B) = \pi m(0)$$

$$\underline{\text{Sop}} = P = \Sigma m(1,2,3)$$

3x8 DecoderStep 1:Step 2:

I_2	I_1	P_0	y_2	y_6	y_5	y_4	y_3	y_2	y_1	y_0
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0	0
1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0

Step 3:

$$y_0 = \overline{P_2} \overline{I_1} \overline{P_0}$$

$$y_1 = \overline{P_2} \overline{I_1} P_0$$

$$y_2 = \overline{P_2} P_1 \overline{P_0}$$

$$y_3 = \overline{P_2} P_1 P_0$$

$$y_4 = P_2 \overline{I_1} \overline{P_0}$$

$$y_5 = P_2 \overline{I_1} P_0$$

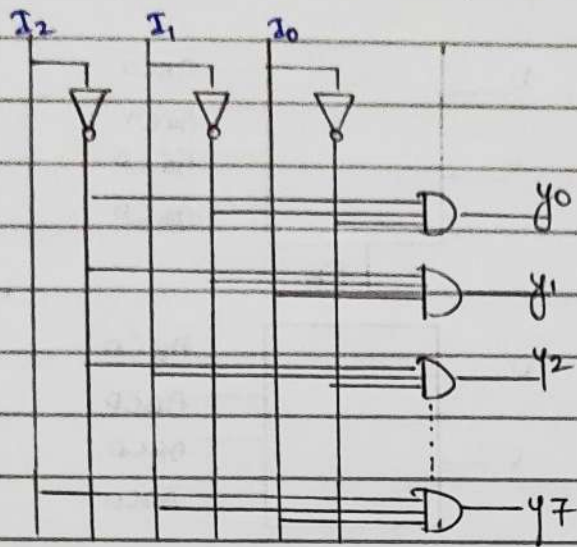
$$y_6 = P_2 P_1 \overline{P_0}$$

$$y_7 = P_2 P_1 P_0$$

Step 4: Minimization

Step 5: Hardware ImplementationHW

Design 4x16 Decoder

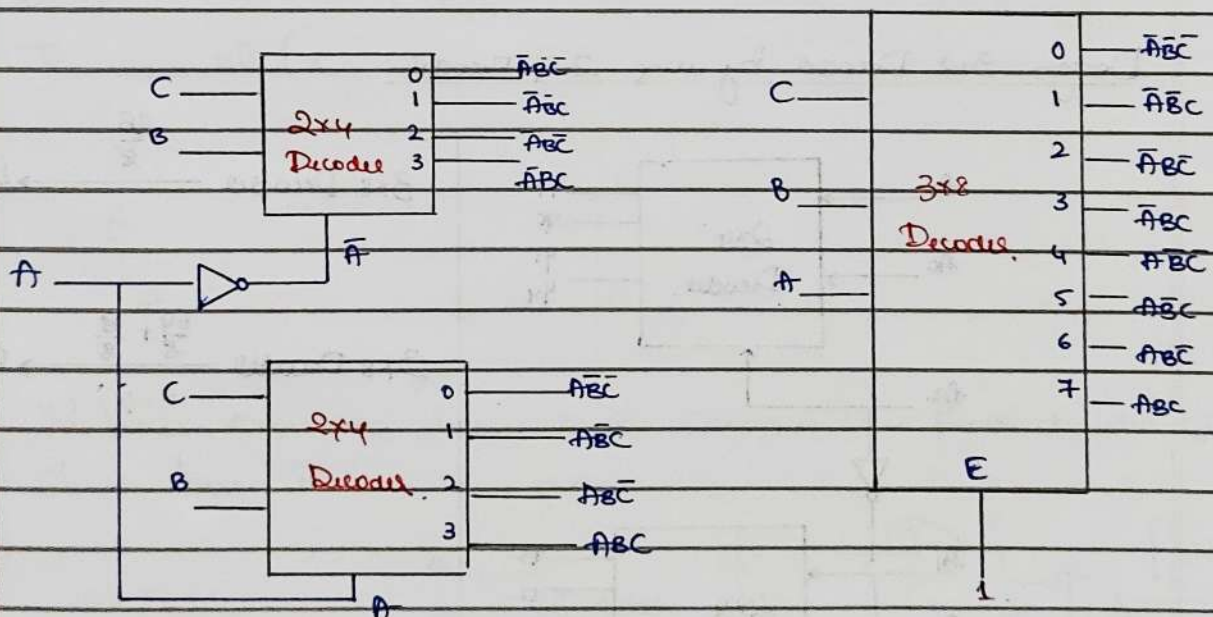


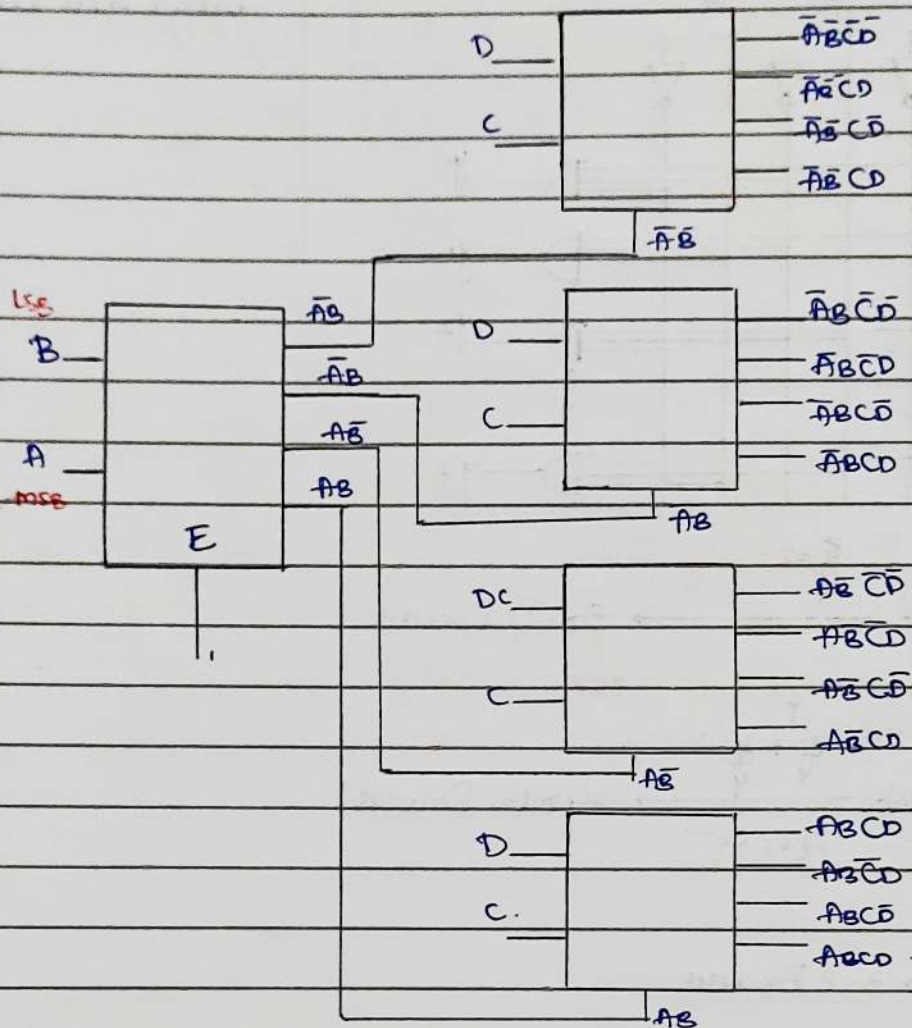
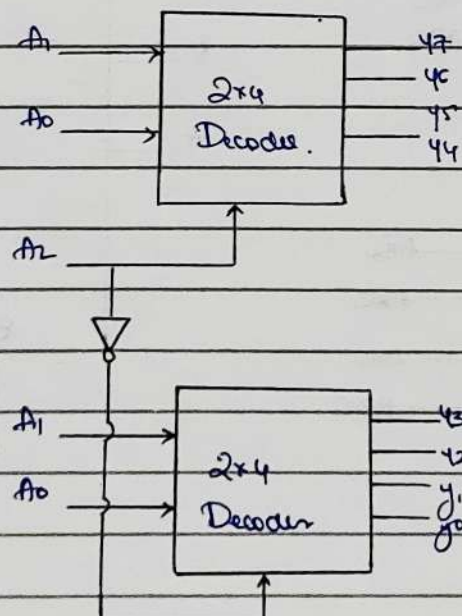
Q.1 2×4 Decoder $\xrightarrow[\substack{8=2^3 \\ 4=2^2}]{}$ 3×8 Decoder

$\uparrow$ input $\uparrow$ output

Q.2 2×4 Decoder $\xrightarrow[\substack{16=2^4 \\ 4=2^2}]{}$ 4×16 Decoder

$4+1=5$

 $2 \times 4 \rightarrow 3 \times 8$ Decoder

2x4 Decoder → 4x16 DecoderDesign 3x8 Decoder by using 2x4 Decoder

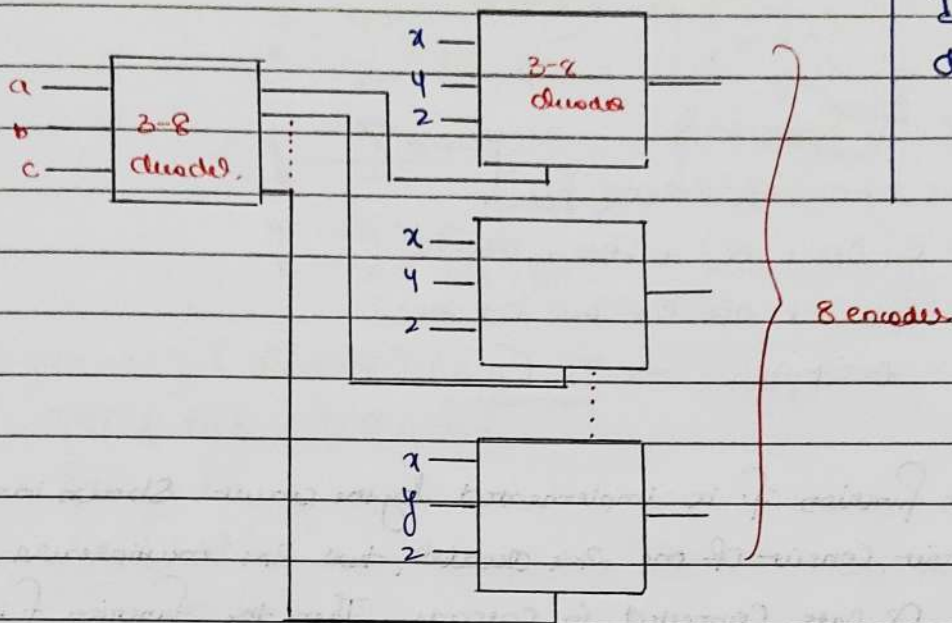
$$3 \times 8 \text{ Decoder} \xrightarrow{\frac{16^2}{8}} 4 \times 16 \text{ Decoder}$$

$$= 2$$

$$3 \times 8 \text{ Decoder} \xrightarrow{\frac{64}{8} + \frac{8}{8}} 6 \times 64 \text{ decoder}$$

$$8 + 8 = 16$$

3x8 Decoder $\rightarrow$ 6x64 Decoder



H.W.
Design Binary to decimal decoder.

Q.1 A logic circuit consist of 2×4 decoders as shown in the figure. The output are as follows:

$D_0 = 1$ when $A_0 = 0, A_1 = 0$

$D_1 = 1$ when $A_0 = 1, A_1 = 0$

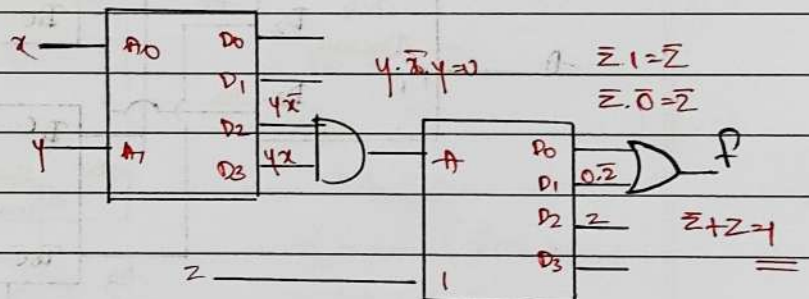
$D_2 = 1$ " $A_0 = 0, A_1 = 1$

$D_3 = 1$ " $A_0 = 1, A_1 = 1$

The name of (x, y, z) is.

Ans

1, 1, 1

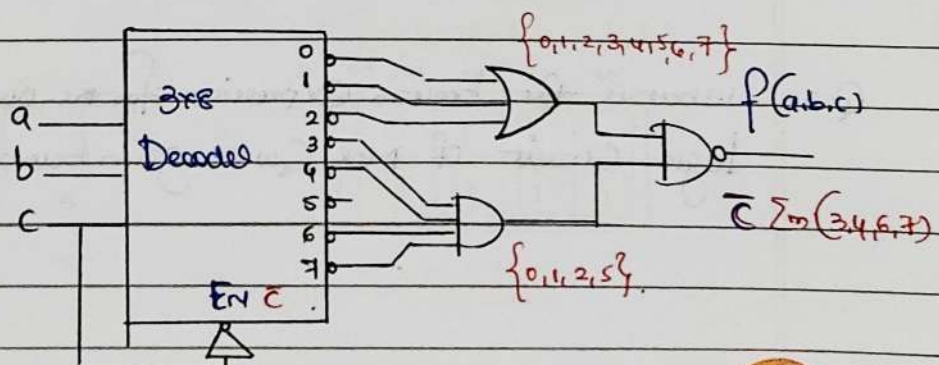


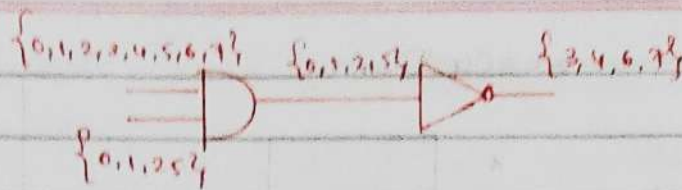
A_1	A_0	y
0	0	$D_0 = 1$
0	1	D_1
1	0	D_2
1	1	D_3

Q2 The boolean expression $f(a, b, c)$ in its canonical form for the decoder circuit shown below is

Ans

$\Rightarrow \Sigma_m(4, 6)$





$$F = \sum m(3, 4, 6, 7)$$

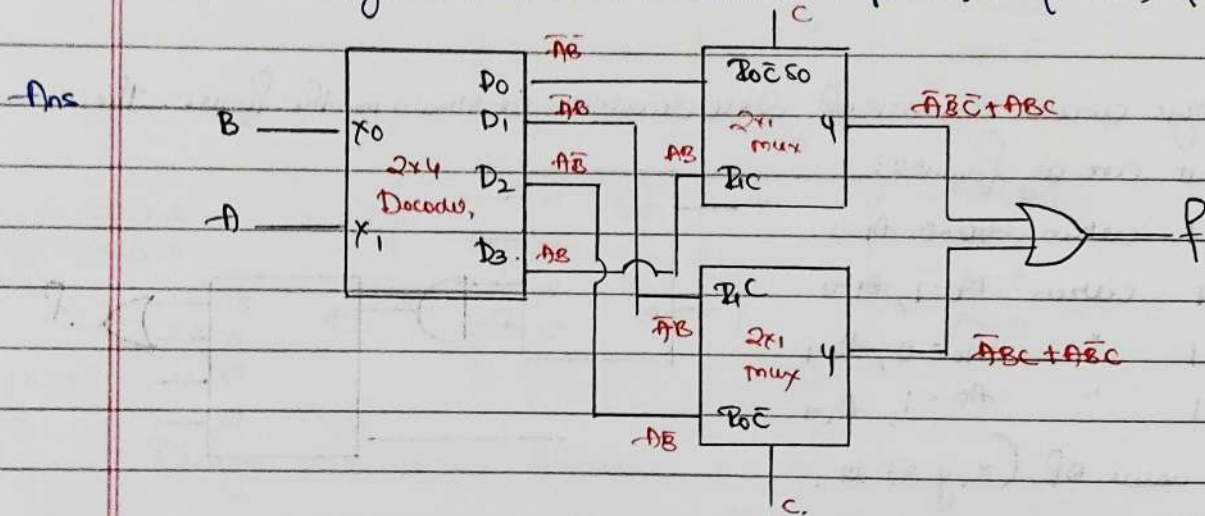
$$= \bar{C} [\bar{A}BC + A\bar{B}C + AB\bar{C} + ABC]$$

$$= \bar{C} [\bar{A}BC + A\bar{B}C + AB\bar{C} + ABC]$$

$$= \bar{A}BC\bar{C} + A\bar{B}C\bar{C} + AB\bar{C}C + ABC\bar{C}$$

$$= A\bar{B}C + ABC \Rightarrow \underline{\underline{\sum m(4, 6)}}$$

Q3. A logic function 'f' is implemented by the circuit shown in the figure below. The circuit consists of one 2x4 decoder, two 2x1 multiplexers and a two input OR gate connected in cascade. Then the function f is equal to



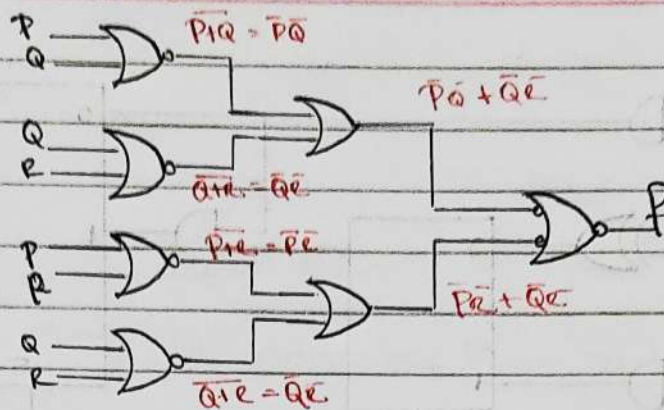
$$f = \bar{A}BC + ABC + \bar{A}BC + ABC$$

$$= \sum m(0, 3, 4, 7)$$

$$\Rightarrow \underline{\underline{\bar{B}C + BC = B \odot C}}$$

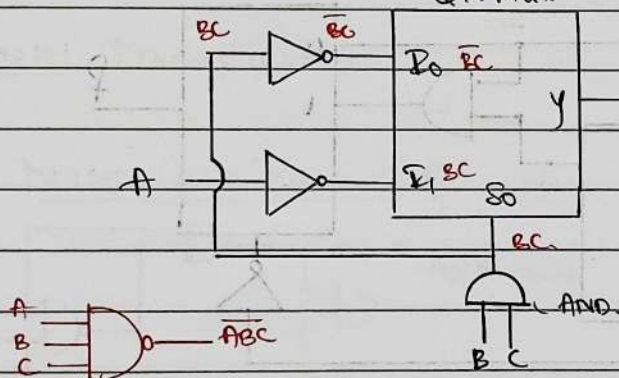
A \ B	00	01	11	10
0	1		1	
1	1		1	

Q2. What is the boolean expression for the output f of the combinational logic circuit of NOR Gates given below.



$$\begin{aligned}
 f &= [\bar{P}Q + \bar{Q}R] \cdot [\bar{P}R + \bar{Q}R] \\
 &= \bar{P}Q\bar{R} + \bar{P}Q\bar{R} + \bar{P}Q\bar{R} + \bar{Q}R \\
 &= \bar{P}Q\bar{R} + \bar{Q}R \\
 &= \bar{Q}R[1 + \bar{P}] \\
 &= \bar{Q}R \Rightarrow \underline{\underline{\bar{Q}R}}
 \end{aligned}$$

Q4. The Combinational Circuit given below implements which of the following 2x1 Mux



$$= \bar{B}C + \bar{A}C$$

$$= \bar{B}C + \bar{A}C$$

$$= \bar{A} + \bar{A}C$$

$$= (\bar{A} + \bar{A}) (\bar{A} + C)$$

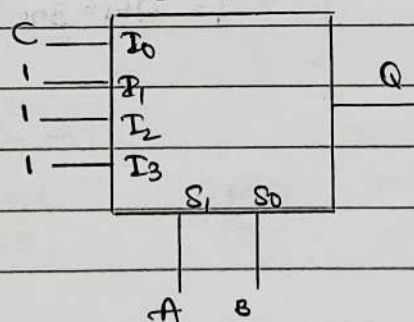
$$= \bar{B}C + \bar{A}$$

$$= \bar{B}C + \bar{A}$$

$$\left. \begin{aligned} &= \bar{B}C + \bar{A} \\ &= \bar{B}C + \bar{A} \end{aligned} \right\} BC = x$$

∴ NAND Gate

Q5. The Combinational logic circuit shown in the given figure has an output Q which is



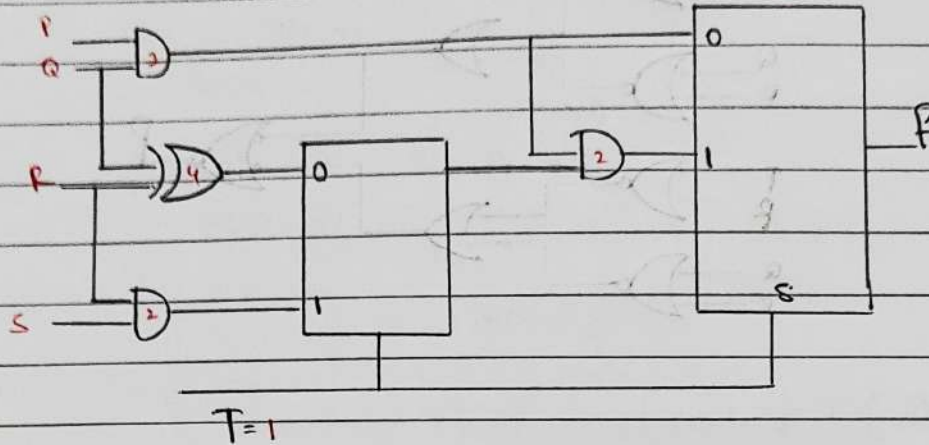
$$= \bar{A}\bar{B}C + \bar{A}B + A\bar{B} + AB$$

$$= \sum m(1, 2, 3, 4, 5, 6, 7)$$

A \ B	00	01	11	10
0	0	1	1	1
1	1	1	1	1

$$\Rightarrow \underline{\underline{A + B + C}}$$

Q6.

Case 1: $T=0$

$$T = T_{AND} + T_{MUX}$$

$$T = 2 + 1 = \underline{\underline{3ns}}$$

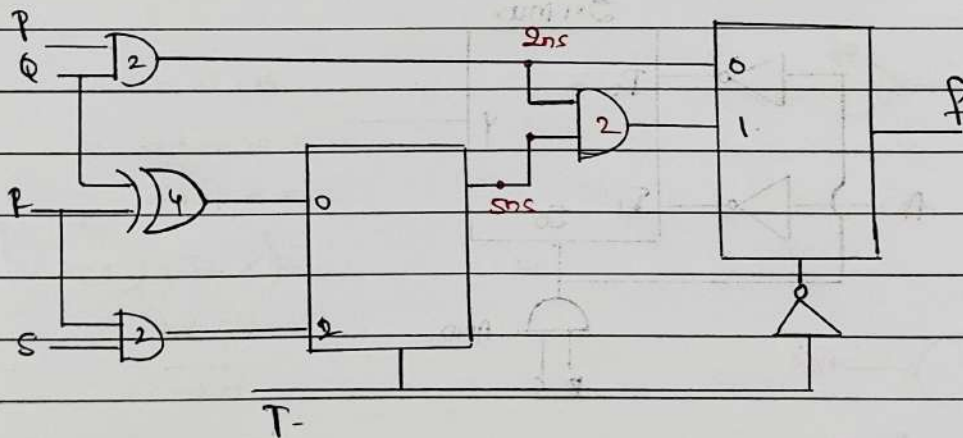
Case 2: $T=1$

$$T = T_{AND} + T_{MUX} + T_{AND} + T_{MUX}$$

$$= 2 + 1 + 2 + 1 = \underline{\underline{6ns}}$$

$$\therefore \text{max} = \underline{\underline{6ns}}$$

Q7.

Case (1): $T=0$

$$T = T_{XOR} + T_{MUX} + T_{AND} + T_{MUX}$$

$$= 4 + 2 + 1 + 1 = \underline{\underline{8ns}}$$

Case (2): $T=1$

$$T = T_{AND} + T_{MUX}$$

$$T = 2 + 1 = \underline{\underline{3ns}}$$

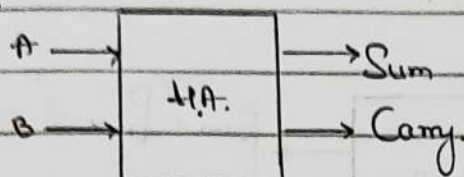
Half Adder

$$\begin{array}{r}
 0011 \\
 +1010 \\
 \hline
 00010110
 \end{array}$$

$\begin{array}{r} 0 \\ -10 \\ \hline 00 \end{array}$
 ↓ Sum
 Carry.

* Two bit adder are known as half adder.

Step 1:



Step 2:

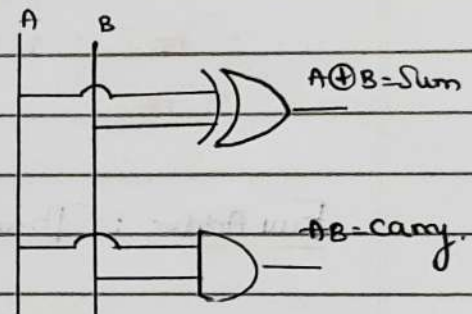
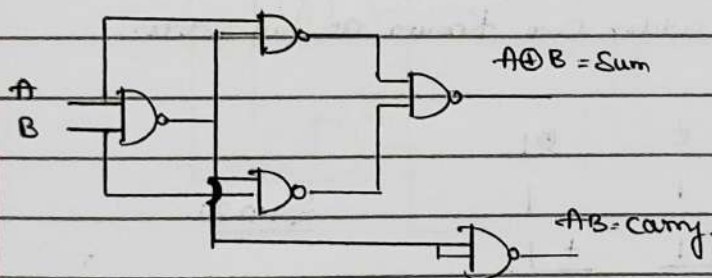
A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Step 3: $Sum = \bar{A}B + A\bar{B} = A \oplus B$
 $Carry = AB$

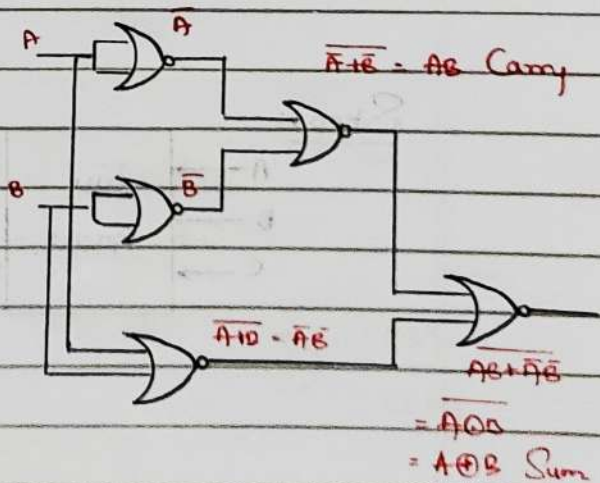
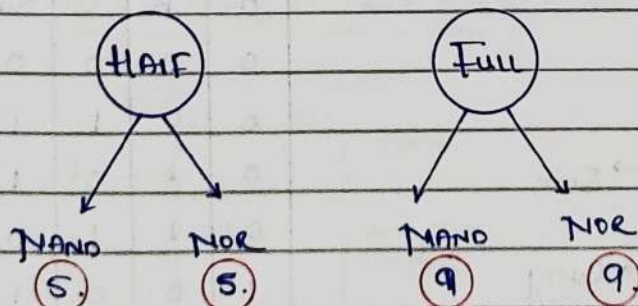
Steps:

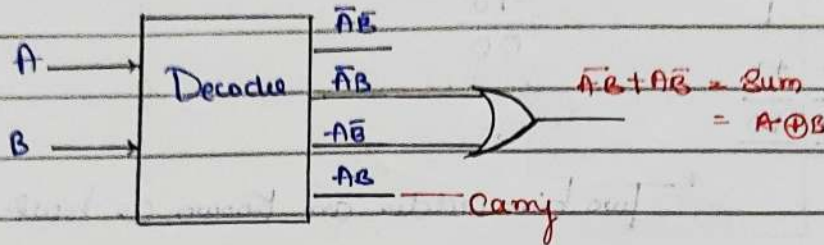
Step 4: Minimization.

NAND Gate Required = 5

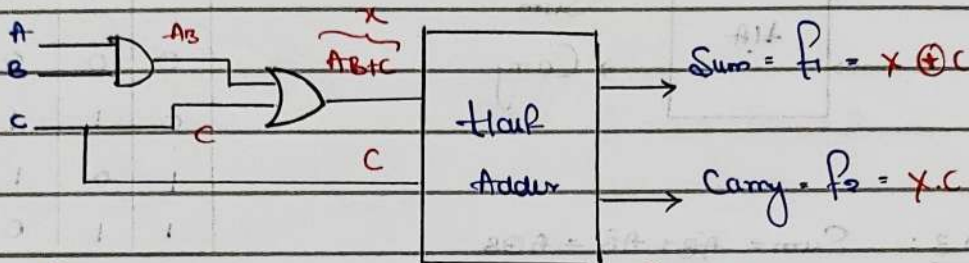


By NOR Gate



Half Adder By Using Decoder.

Q1. What is the value of f_1 and f_2 ?



$$f_1 = (AB + C) \oplus C$$

$$= \overline{AB+C} \cdot C + (AB+C) \cdot \bar{C}$$

$$= \overline{AB} \cdot \bar{C} \cdot C + AB\bar{C} + C \cdot \bar{C}$$

$$= \underline{\underline{AB\bar{C}}}$$

$$f_2 = (AB + C)C$$

$$= ABC + C \cdot C = ABC + C$$

$$= C[AB + 1]$$

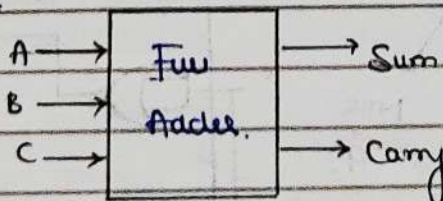
$$= \underline{\underline{C \cdot AB}}$$

Full Adder : Three bit adder are known as full adder.

0	0	0	01
0	0	1	1
+ 0	+ 1	+ 1	+ 1
00	01	10	11

Step 2:

Step 1:

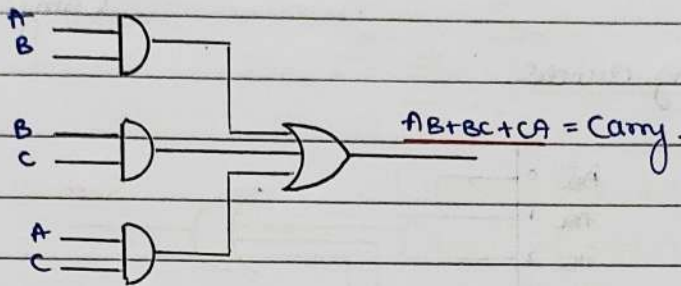
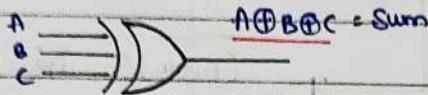


A	B	C	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Step 3: $\bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$
 (Sum) = $\sum m(1, 2, 4, 7)$
 $= A \oplus B \oplus C$

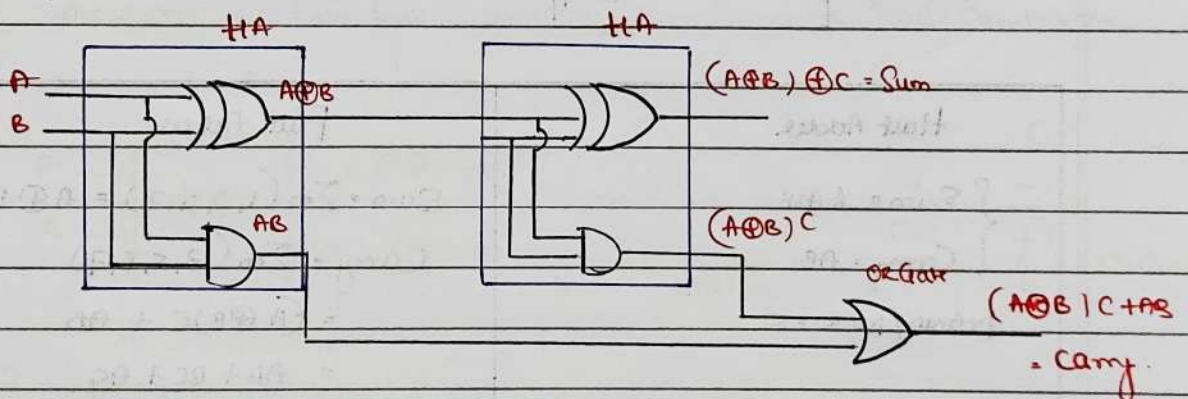
Step 4: $\text{Carry}(A, B, C) = \sum m(3, 5, 6, 7)$
 $= \bar{A}BC + A\bar{B}C + A\bar{B}\bar{C} + ABC$
 $= A\bar{B} + BC + AC$ — minimized exp.
 $= (\bar{A}B + AB)C + AB(\bar{C} + C)$
 $= (A \oplus B)C + AB$ — 2-term minimized.

Steps:

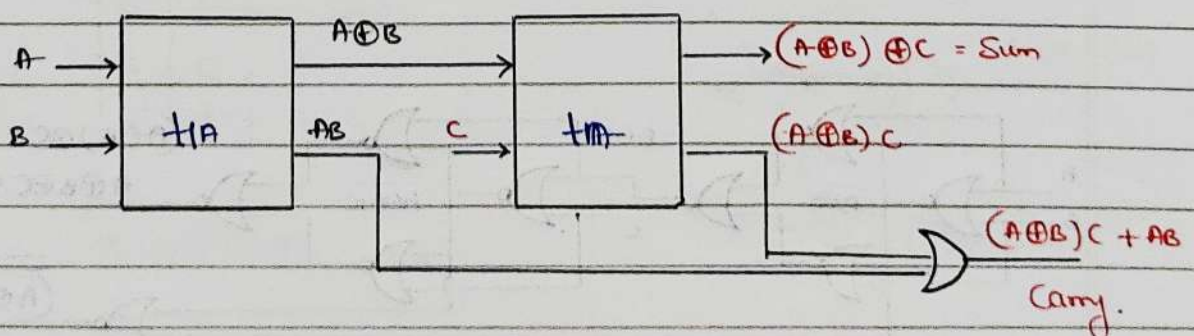


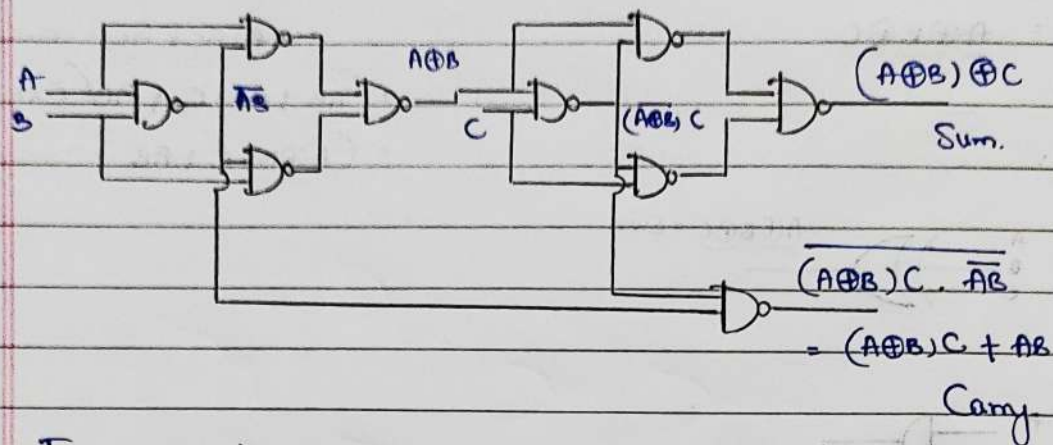
$$\text{Sum} = (A \oplus B) \oplus C$$

$$\text{Carry} = (A \oplus B)C + AB$$

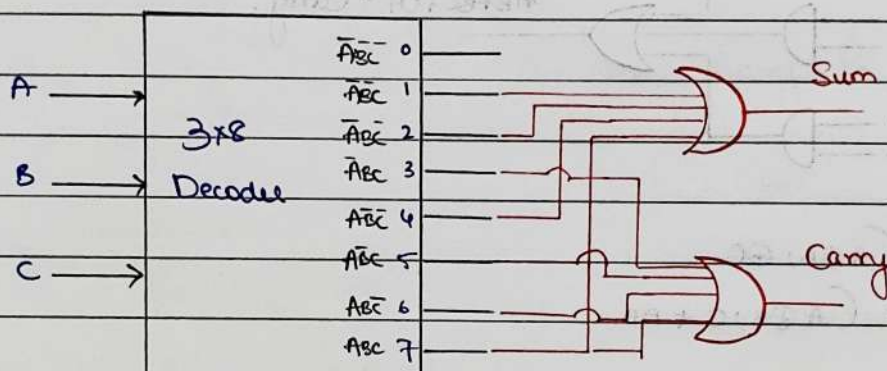


$$1 \text{ Full Adder} = 2 \text{ H.A.} + 1 \text{ OR Gate}$$

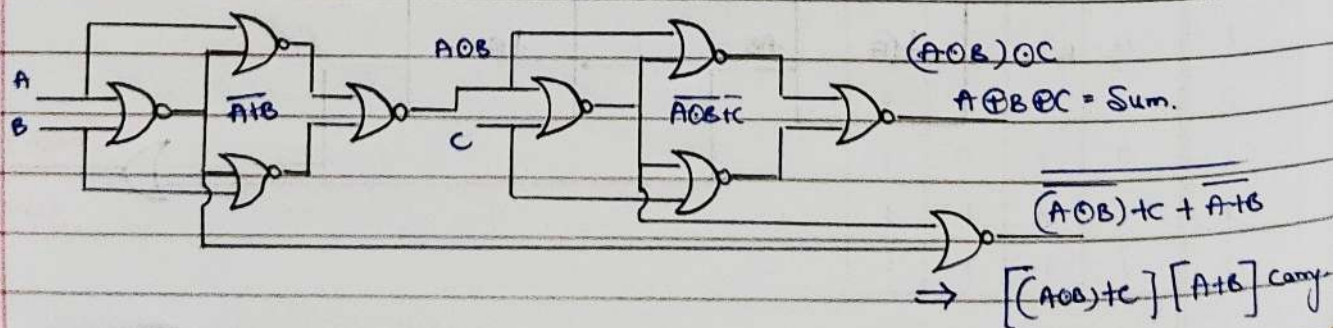


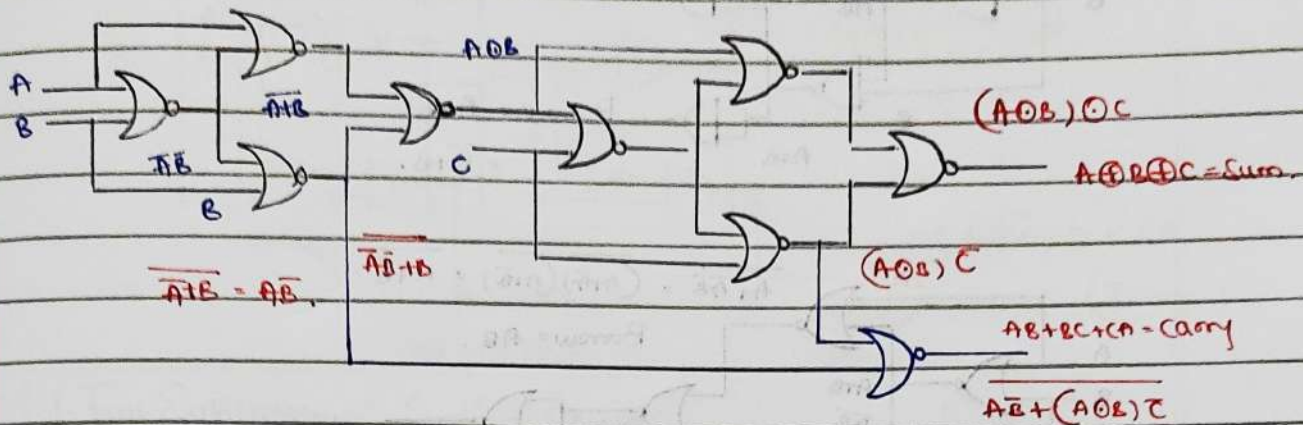
Full Adder by using NAND Gate.

HW
Full Adder by using NOR Gate.

Full Adder by using decoder.

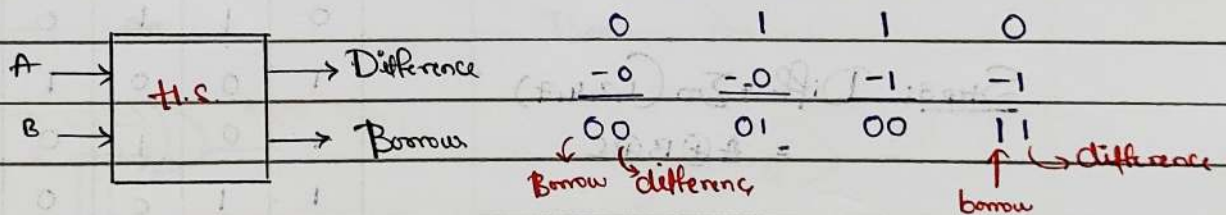
Half Adder	Full Adder
$\begin{cases} \text{Sum} = A \oplus B \\ \text{Carry} = AB \end{cases}$	$\begin{aligned} \text{Sum} &= \sum m(1, 2, 4, 7) = A \oplus B \oplus C \\ \text{Carry} &= \sum m(3, 5, 6, 7) \\ &= (A \oplus B)C + AB \\ &= AB + BC + AC \end{aligned}$
NAND/NOR = 5	NAND/NOR = 9
	1 Full = 2 Half + 1 OR



Full Subtractor

$$\begin{aligned}
 &= \overline{A}B + (A \odot B) C \\
 &= \overline{A}B + (\overline{A}B + AB)C \\
 &= (\overline{A} + B) [(\overline{A}B) + C] \\
 &= (\overline{A} + B) [\overline{A}B + AB + C] \\
 &= \overline{A} + B + \overline{A}C + AB + BC \\
 &\Rightarrow \overline{A}B + \overline{A}C + BC
 \end{aligned}$$

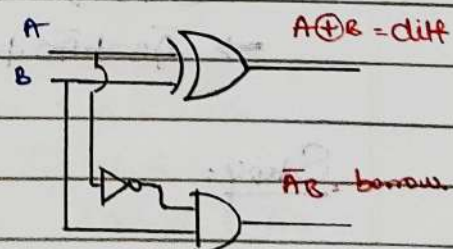
Borrow.

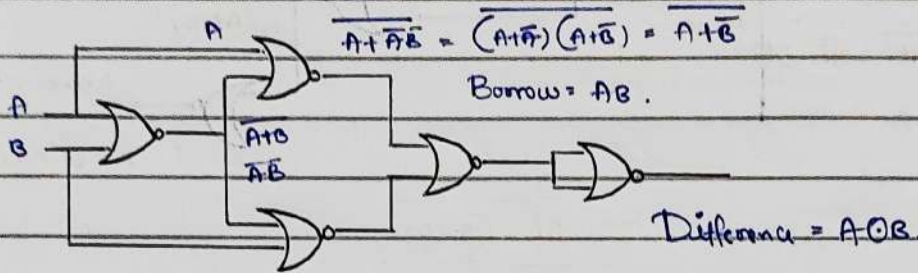
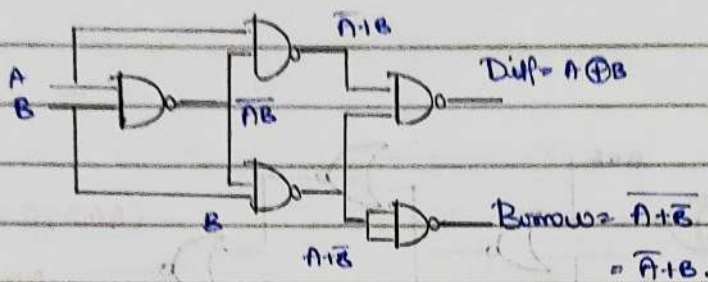
Half Subtractor : Two bit Subtractor are known as half subtractor.Step 1:Step 2:

A	B	Diff.	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Step 4:Step 5:

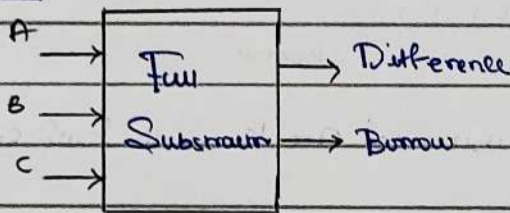
Step 3: Diff = $A \oplus B$
 Borrow = $\overline{A}B$





Full Subtractor

Step 1:



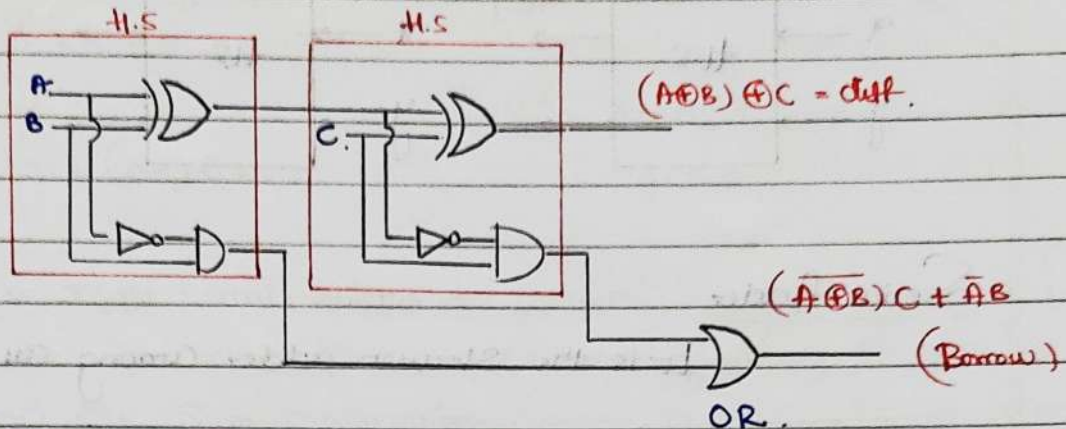
Step 2:

Input			Output	
A	B	C	Diff	Borr.
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

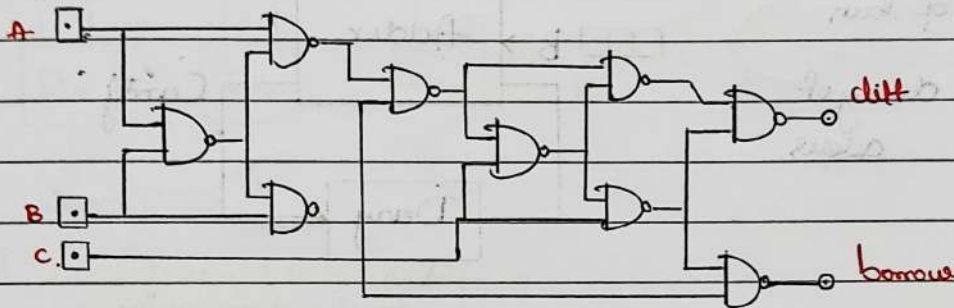
Step 3: $\text{Diff} = \sum m(1, 2, 4, 7)$
 $= A \oplus B \oplus C$

$\text{Borrow} = \sum m(1, 2, 3, 7)$
 $= \overline{A}\overline{B}C + \overline{A}B\overline{C} + \overline{A}BC + A\overline{B}C$
 $= (\overline{A}\overline{B} + \overline{A}B)C + \overline{A}B(\overline{C} + C)$
 $= (\overline{A} \oplus \overline{B})C + \overline{A}B \rightarrow \text{Semi minimised expression}$
 $\rightarrow \overline{A}B + \overline{A}C + BC$

Step 4:

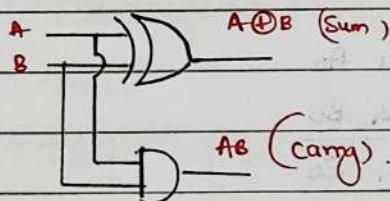
Steps:

1 Full Subtractor = 2 H.S. + Or Gate.

Half adder

$$\text{Sum} = A \oplus B$$

$$\text{Carry} = AB$$

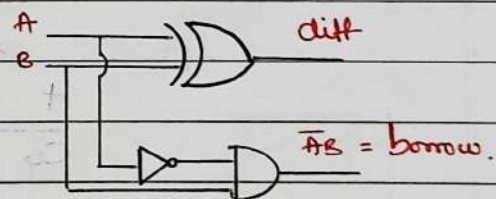


NAND } 5.
NOR

Half Subtractor

$$\text{Diff} = A \oplus B$$

$$\text{Borrow} = \bar{A}B$$



NAND } 5.
NOR

Full Adder

$$\text{Sum} = A \oplus B \oplus C$$

$$\begin{aligned} \text{Carry} &= \sum m(3, 5, 6, 7) \\ &= (A \oplus B)C + AB \\ &= AB + AC + BC \end{aligned}$$

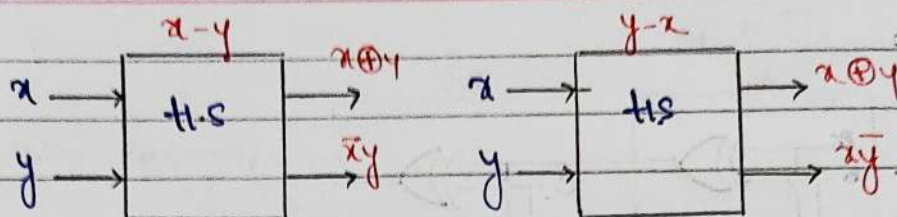
NAND/NOR = 9.

Full Subtractor

$$\text{Difference} = A \oplus B \oplus C$$

$$\begin{aligned} \text{Borrow} &= \sum m(1, 2, 3, 7) \\ &= (\bar{A} \oplus \bar{B})C + \bar{A}B \\ &= \bar{A}B + \bar{A}C + BC \end{aligned}$$

NAND/NOR = 9.



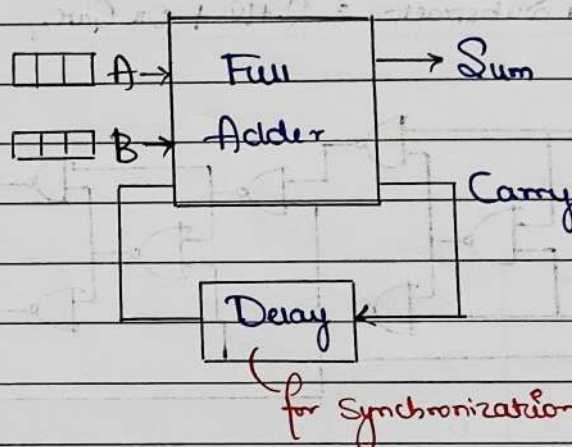
Serial Adder "One by one addition"

It is the slowest adder among all the adders.

$$T = n \cdot T_{\text{delay}}$$

$n \rightarrow$ no. of bits

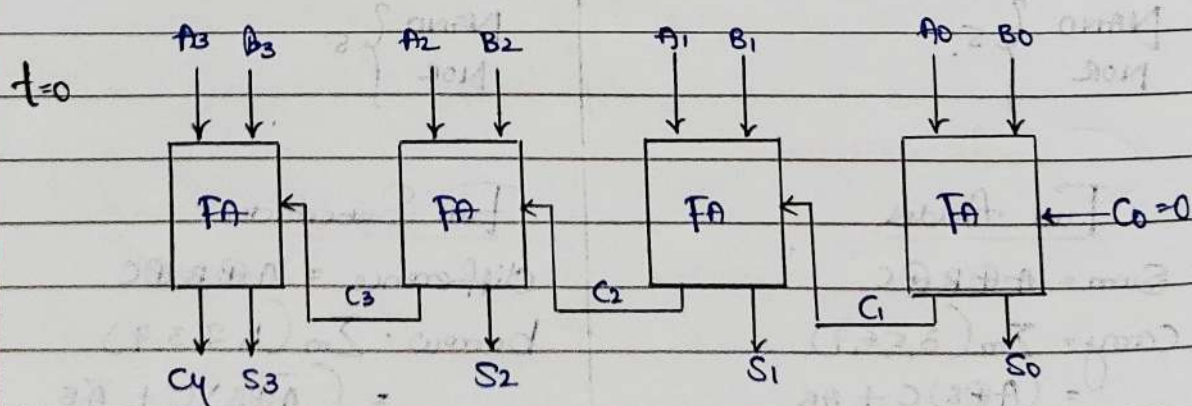
$T_{\text{delay}} \rightarrow$ delay of adder.



Parallel Adder (Ripple Carry Adder)

Inputs are connected simultaneously

$$\begin{array}{r}
 \text{4 Bits} \\
 \begin{array}{cccc}
 C_3 & C_2 & C_1 & \\
 A_3 & A_2 & A_1 & A_0 \\
 + & B_3 & B_2 & B_1 & B_0 \\
 \hline
 C_4 & S_3 & S_2 & S_1 & S_0
 \end{array}
 \end{array}$$

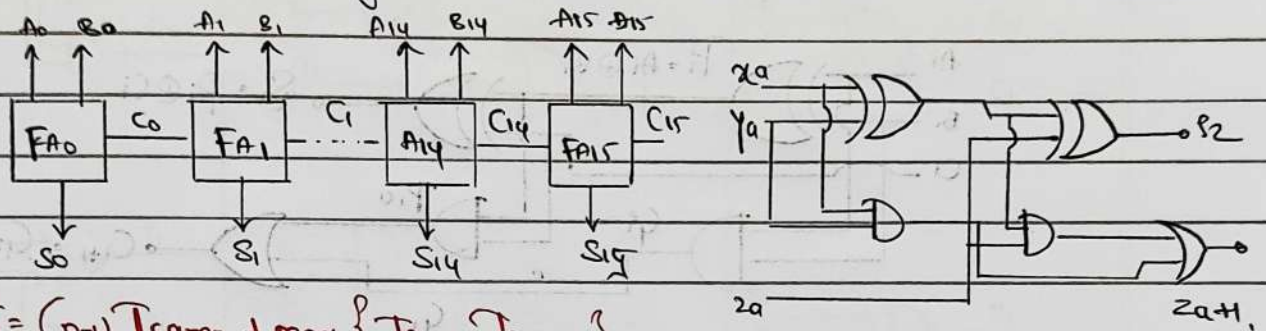


n bit Parallel adder

1. $(n-1)$ FA + 1 HA
2. n-FA
3. $(2n-1)$ HA + $(n-1)$ OR.

$$T = (n-1) T_{\text{carry}} + \max(T_{\text{sum}}, T_{\text{carry}})$$

Q1. A 16-bit ripple carry adder is realised using 16 identical full adders (FA) as shown. The carry propagation delay of each FA is 12 ns and the sum-propagation delay of each FA is 15 ns. The worst case delay (in ns) of the 16-bit adder will be _____

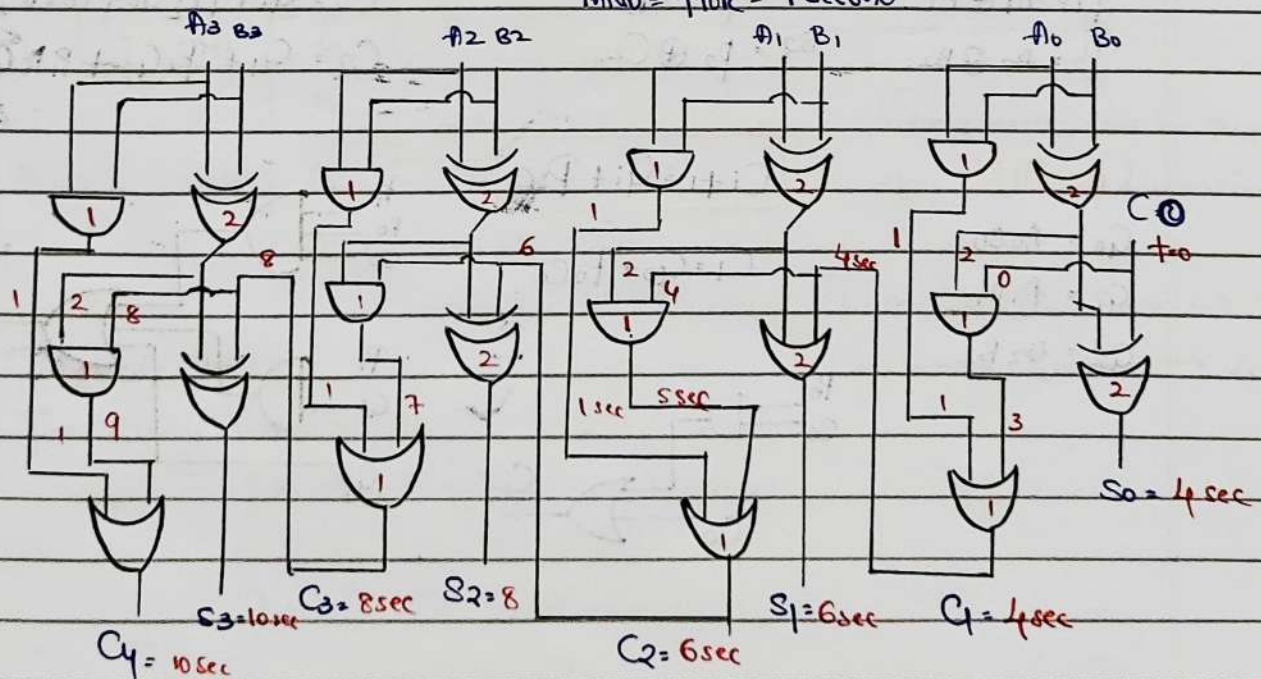


$$\begin{aligned}
 T &= (n-1) T_{\text{carry}} + \max\{T_{\text{sum}}, T_{\text{carry}}\} \\
 &= (16-1) \times 12 \text{ ns} + \max\{15 \text{ ns}, 12 \text{ ns}\} \\
 &= 15 \times 12 + 15 \\
 &= 195 \text{ ns}
 \end{aligned}$$

Parallel Adder (4 bit)

$$T_{\text{xor}} = 2 \text{ second}$$

$$T_{\text{and}} = T_{\text{or}} = 1 \text{ second}$$



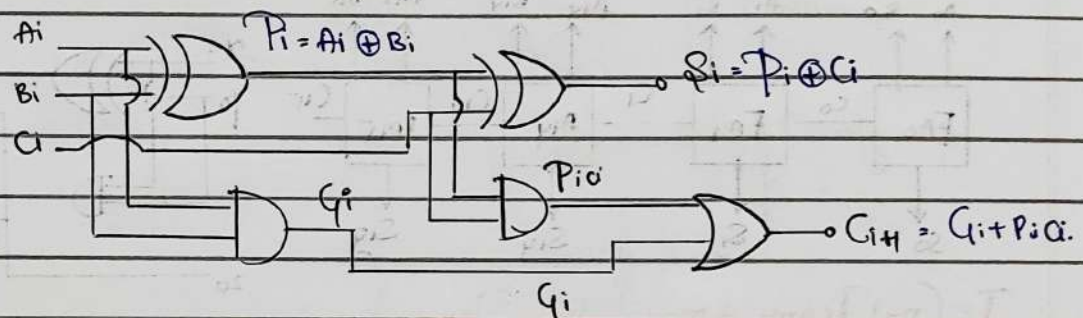
$$T = (n-1) \{ T_{AND} + T_{OR} \} + \max \{ T_{sum}, T_{carry} \}$$

$$\begin{aligned} T &= (4-1) \{ 1+1 \} + \max \{ 4, 4 \} \\ &= 3 \times 2 + 4 \\ &= \underline{10 \text{ ns}} \end{aligned}$$

Parallel Adder, LACA

[Look Ahead Carry Adder]

→ fastest adder among all the adders



$P_i \rightarrow$ Carry propagating term
 $G_i \rightarrow$ Carry generating term

$$P_i = A_i \oplus B_i$$

$$S_i = P_i \oplus C_i$$

$$P_0 = A_0 \oplus B_0$$

$$S_0 = P_0 \oplus C_0$$

$$P_1 = A_1 \oplus B_1$$

$$S_1 = P_1 \oplus C_1$$

$$P_2 = A_2 \oplus B_2$$

$$S_2 = P_2 \oplus C_2$$

$$C_2 = G_1 + P_1 C_1$$

$$C_2 = G_1 + P_1 [G_0 + P_0 C_0]$$

$$C_2 = G_1 + P_1 G_0 + P_1 P_0 C_0$$

$$G_i = A_i B_i$$

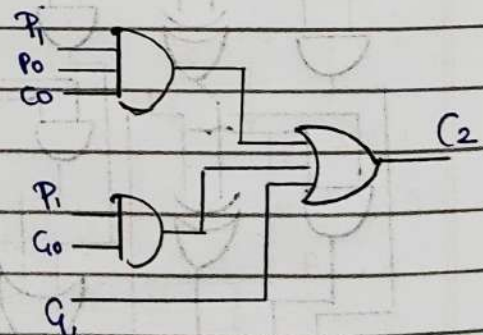
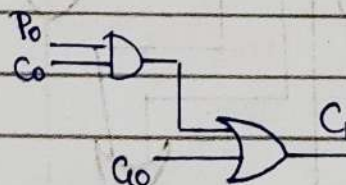
$$C_{i+1} = G_i + P_i C_i$$

$$G_0 = A_0 B_0$$

$$C_1 = G_0 + P_0 C_0$$

$$G_1 = A_1 B_1$$

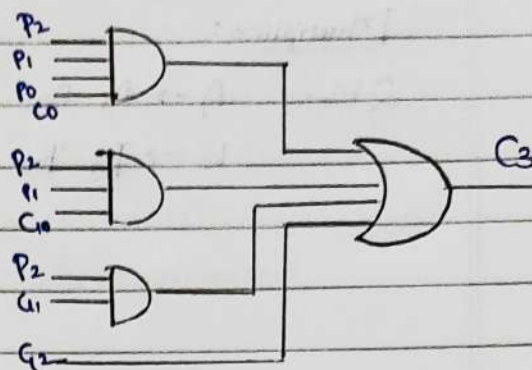
$$G_2 = A_2 B_2$$



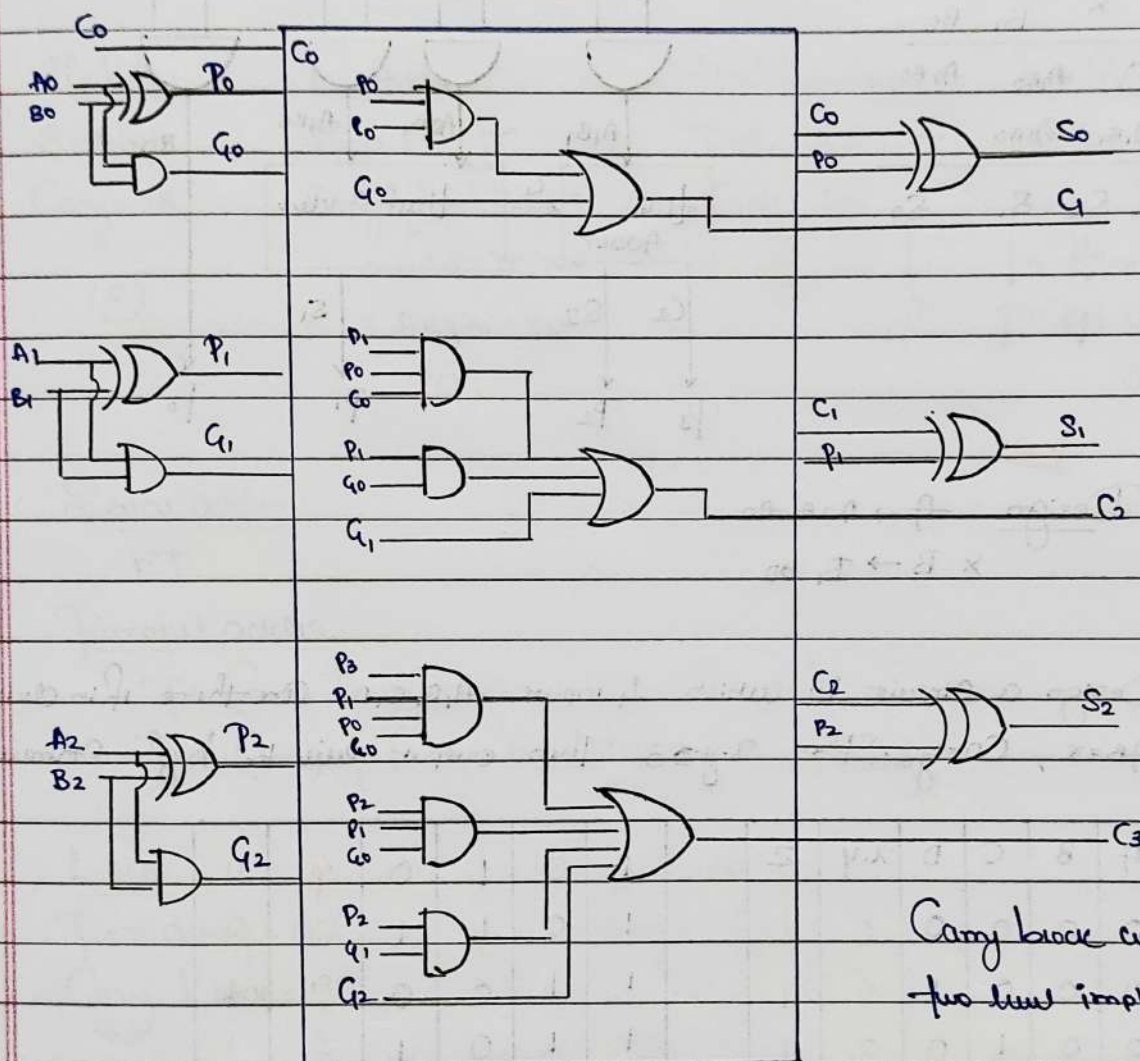
$$C_3 = G_2 + P_2 C_2$$

$$C_3 = G_2 + P_2 [G_1 + P_1 G_0 + P_1 P_0 G_0]$$

$$C_3 = G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 G_0$$



CARRY BLOCK



Carry block can be always two level implementation.

- * Delay for Carry Block $\Rightarrow 27$
- * For entire circuit $\Rightarrow 47$

'n' bit RACA

For carry block:

$$\text{No. of AND Gate} = \frac{n(n+1)}{2}$$

$$\text{No. of OR Gate} = n$$

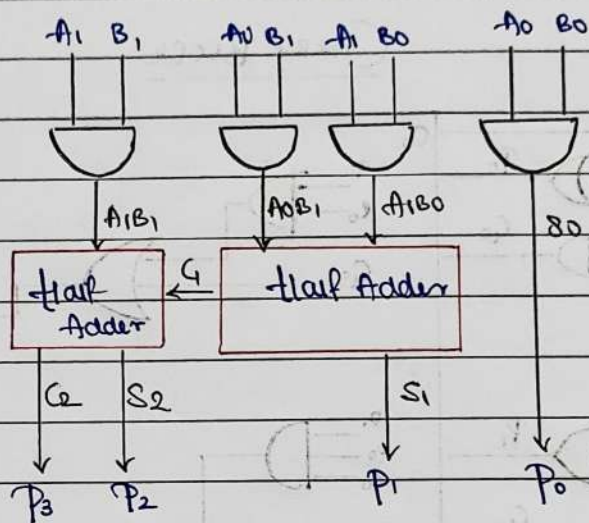
Multiplication:

2 bit. $A \rightarrow A_1 A_0$
 $B \rightarrow B_1 B_0$

$$\begin{array}{r} 52 \\ \times 52 \\ \hline 2704 \\ \hline \end{array}$$

P_3, P_2, P_1, P_0

$$\begin{array}{r} A_1 A_0 \\ \times B_1 B_0 \\ \hline C_1 A_0 B_0 A_0 B_0 \\ \hline D_1 C_1 A_0 B_0 - \\ \hline C_2 S_2 S_1 S_0 \\ \downarrow \downarrow \downarrow \downarrow \\ P_3 P_2 P_1 P_0 \end{array}$$

hw

Design. $A \rightarrow A_2 A_1 A_0$
 $\times B \rightarrow B_1 B_0$

Q1. Design a circuit in which 4 inputs A, B, C, D are there if in decimal $AB=x, CD=y$. Then $x \cdot y \leq 3$ Then output will be high otherwise low.

A	B	C	D	$x \cdot y$	Z
1	0	1	0	4	0
1	0	1	1	6	0
1	1	0	0	0	1
1	1	0	1	3	1
1	1	1	0	6	0
1	1	1	1	9	0
0	1	0	0	0	1
0	1	0	1	1	1
0	1	1	0	2	1
0	1	1	1	3	1
1	0	0	0	0	1
1	0	0	1	2	1

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1		
10	1	1		

$$\bar{A} + \bar{C}$$

$$= \underline{\underline{\bar{AC}}}$$

Combinational Circuit - { static circuit }Quick Summary!Comparator

$$(A > B) \Rightarrow A_1 \bar{B}_1 + (A_1 + \bar{B}_1) A_0 \bar{B}_0 \rightarrow \text{minimised}$$

$$A_1 \bar{B}_1 + (A_1 \odot B_1) A_0 \bar{B}_0 \rightarrow \text{Semi-minimised}$$

Mux, DEMUX

Encoder, Decoder.

Half Adder

$$S = A \oplus B$$

$$\text{Carry} = AB$$

(5)

Full Adder

$$\text{Sum} = A \oplus B \oplus C$$

$$\text{Carry} = \sum m(3, 5, 6, 7)$$

$$= (A \oplus B)C + AB$$

$$= AB + AC + BC$$

(9)

Half Subtractor

$$\text{Diff} = A \oplus B$$

$$\text{Carry} = \bar{A}B$$

(5)

Full Subtractor

$$\text{Diff} = A \oplus B \oplus C$$

$$\text{Carry} = \sum m(1, 2, 3, 7)$$

$$= \bar{A}B + (A \oplus B)C$$

$$= \bar{A}B + \bar{A}C + BC$$

(9)

Serial adder

n.T.

Parallel adder

$$T = (n-1) T_{\text{Carry}} + \max \{ T_{\text{sum}}, T_{\text{Carry}} \}$$

$$T = (n-1) \{ T_{\text{AND}} + T_{\text{OR}} \} + \max \{ T_{\text{sum}}, T_{\text{Carry}} \}$$

LACA

$$\text{Total delay} = 4\tau$$

$$\text{Carry block} = 2\tau$$

Carry Block

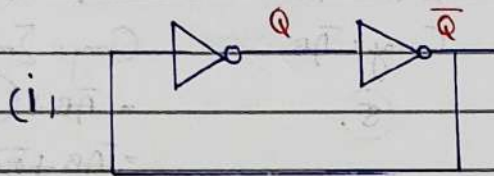
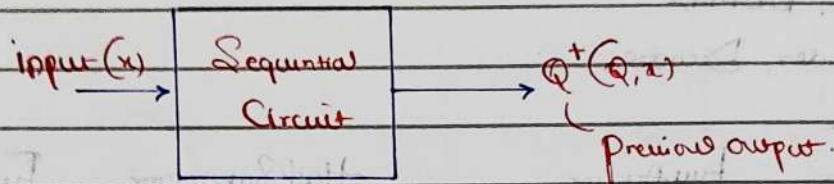
$$\text{AND} = \frac{n(n+1)}{2}$$

$$\text{OR} = n$$

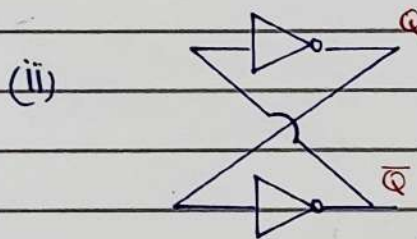
SEQUENTIAL CIRCUIT

Sequential Circuit:

- * A circuit with feedback and memory are called Sequential Circuit.
- * Output of the Sequential Circuit depends on previous output as well as present state of input.



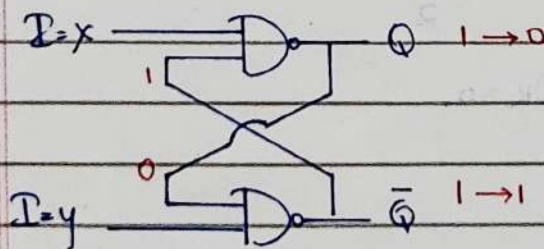
Basic Memory Element
i = ii



Latches : Basic memory element

* Latches are level triggered

* Latches have two output which is complement of each other



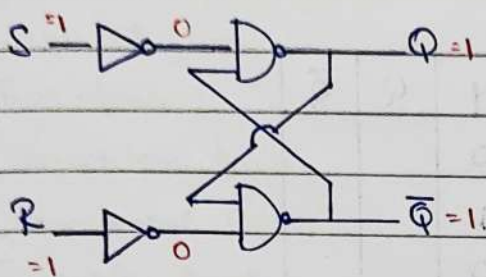
NAND

A	B	$y = A \cdot B$
0	0	1
0	1	1
1	0	1
1	1	0

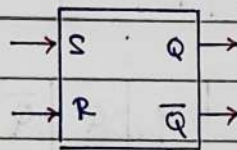
x	y	Q	Q-bar	
0	0	1	1	* Invalid
0	1	1	0	
1	0	0	1	
1	1	Q	Q-bar	→ Hold

Note: Whenever $x=y=1$ is applied and invalid condition occurs than a NAND having lower Propagation Delay first change its Output and other remain on its previous state are called racing Problem or raising Problem.

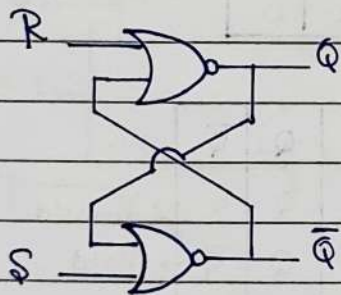
SR: Set-Reset Latch



S	R	Q	$\bar{Q}$	
0	0	Q	$\bar{Q}$	Hold
0	1	0	1	Reset
1	0	1	0	Set
1	1	X	X	(Invalid)



S-R Latch by using NOR Gate:

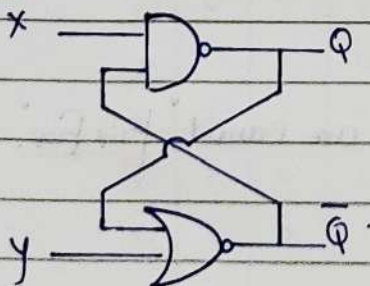


S	R	Q	$\bar{Q}$	
0	0	Q	$\bar{Q}$	Hold
0	1	0	1	Reset
1	0	1	0	Set
1	1	X	X	Invalid.

NOR		
A	B	$y = \overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

Hlw

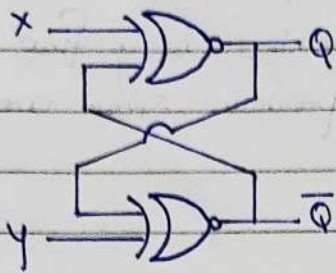
①



x	y	Q	$\bar{Q}$
0	0		
0	1		
1	0		
1	1		

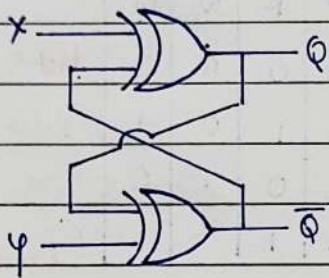
HW

(2)



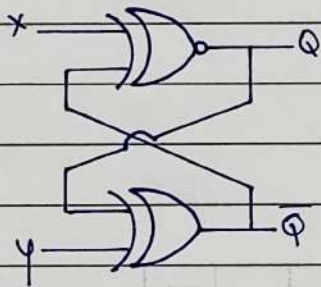
x	y	Q	$\bar{Q}$
0	0		
0	1		
1	0		
1	1		

(3)



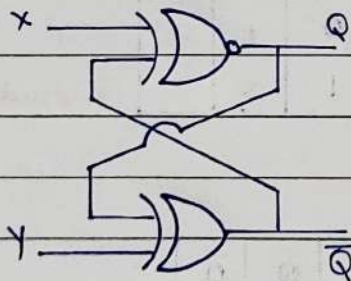
x	y	Q	$\bar{Q}$
0	0		
0	1		
1	0		
1	1		

(4)



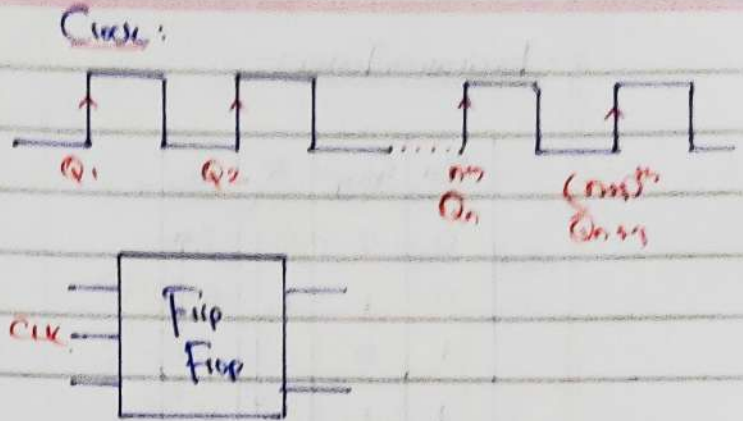
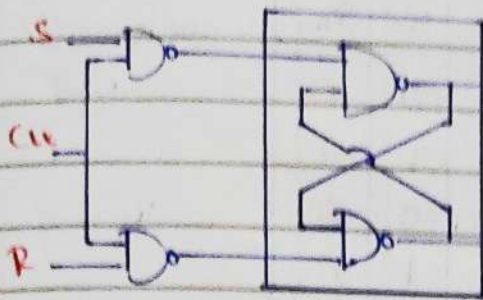
x	y	Q	$\bar{Q}$
0	0	1	0
0	1	1	0
1	0	0	1
1	1	1	0

(5)



x	y	Q	$\bar{Q}$	
0	0	Q	$\bar{Q}$	→ Invalid
0	1	Q	$\bar{Q}$	→ Hold
1	0	$\bar{Q}$	Q	→ Toggle
1	1	$\bar{Q}$	$\bar{Q}$	→ Invalid

→ Latches with control phenomena are called "Flip Flop".

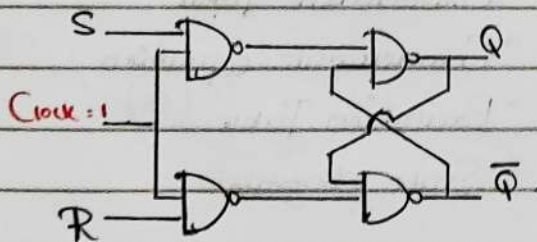


Flip Flop Steps

1. Circuit diagram
2. Truth table
3. Characteristic table
4. Characteristic equation
5. Excitation table
6. State diagram

SR Flip Flop (Ser-Reset Flip Flop)

Circuit diagram:



Truth Table:

S	R	Q_{n+1}	$\bar{Q}_{n+1}$	
0	0	Q_n	$\bar{Q}_n$	Hold/Pres.
0	1	0	1	Reset.
1	0	1	0	Set.
1	1	x	x	Invalid

Characteristic Table:

	S	R	Q_n	Q_{n+1}
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	0
4	1	0	0	1
5	1	0	1	1
6	1	1	0	x
7	1	1	1	x

$$Q_{n+1} = (S, R, Q_n) = \sum m(1, 4, 5) + \sum d(6, 7)$$

Characteristic Equation:

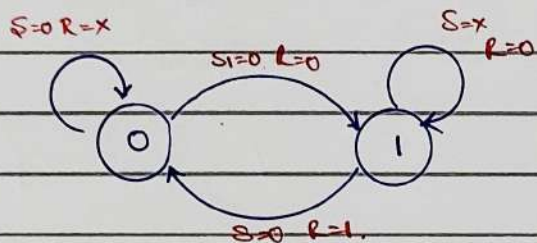
	$\bar{R}Q_n$	$\bar{R}\bar{Q}_n$	RQ_n	$R\bar{Q}_n$
00				
01		1		
10	1	1	x	x
11				

$$Q_{n+1} = S + \bar{R}Q_n$$

Excitation Table:

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

State Diagram:

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

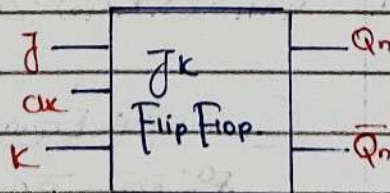
HW

Q Truth Table:

C	J	Q_{n+1}
0	0	$\overline{Q_n}$
0	1	Q_n
1	0	1
1	1	0

Write:

- * Characteristic Table
- * Characteristic equation
- * Excitation Table
- * State diagram

JK Flip FlopSymbol:Truth Table:

J	K	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	$\overline{Q_n}$

Characteristic Table:

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

$$Q_{n+1}(J, K, Q_n) = \sum m(1, 4, 5, 6)$$

Characteristic Equation:

J \ Q_n	$\bar{Q}_n$	$\bar{Q}_n$	Q_n	Q_n
	00	01	11	10
$\bar{J}$ 0		1		
J 1	1	1		1

$$Q_{n+1} = \underline{\underline{J\bar{Q}_n + \bar{K}Q_n}}$$

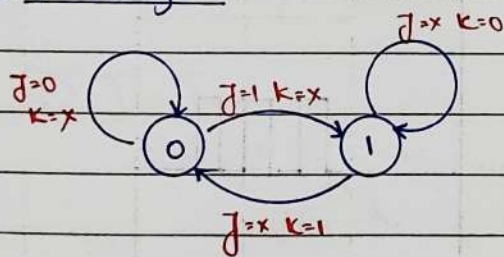
Excitation Table:

Table:

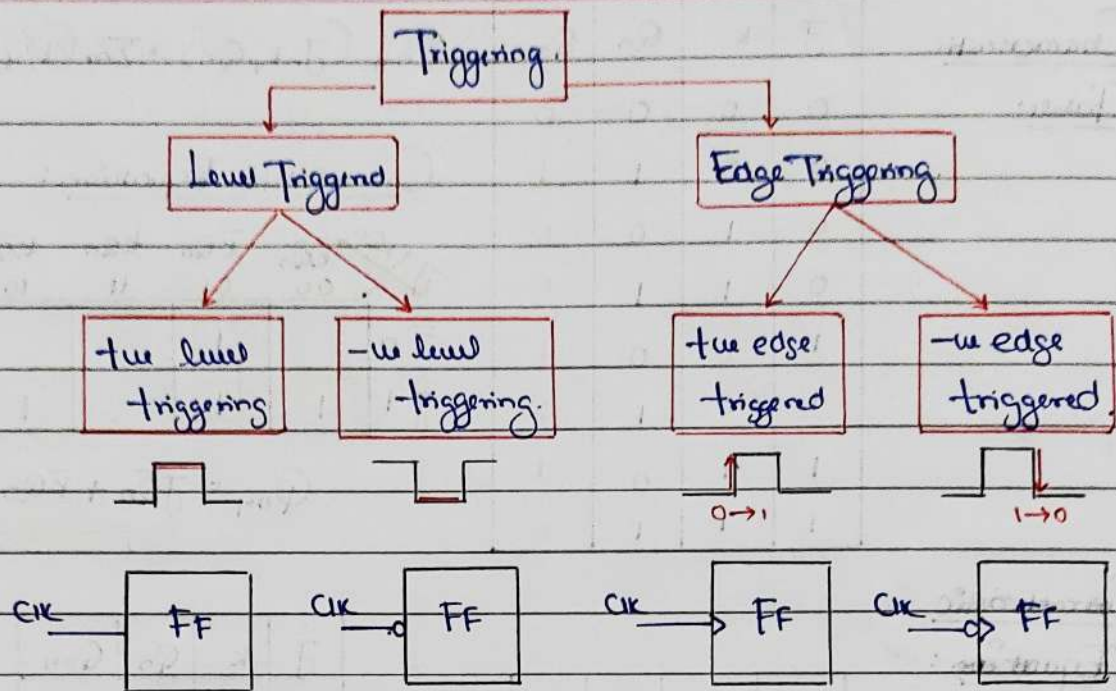
Q_n	Q_{n+1}	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

State Diagram:

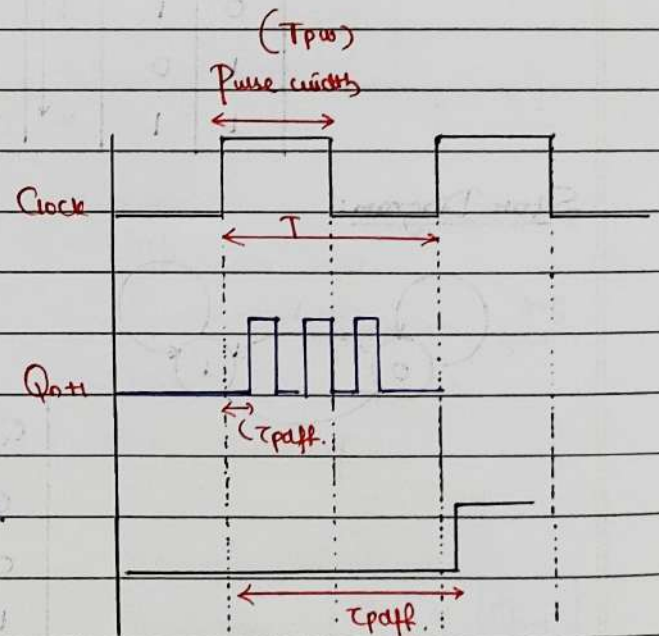
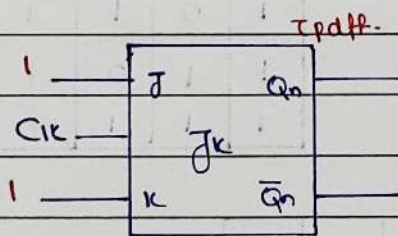


Q_n	Q_{n+1}	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0



Note: level triggered JK flip flop suffers from the problem of "Race Around".

Race-Around Problem:

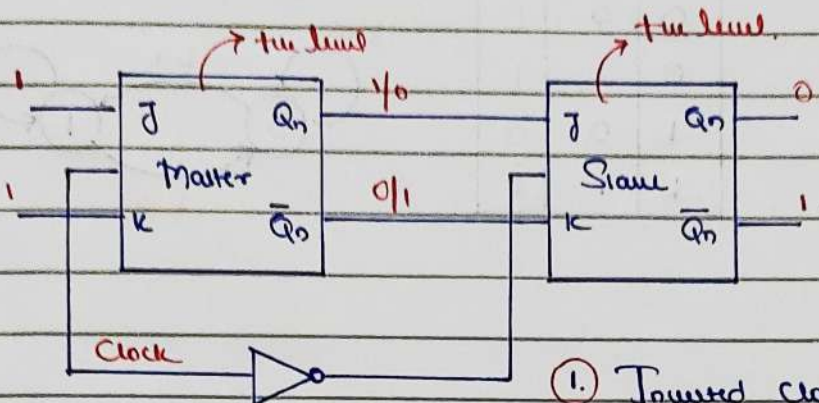


To avoid the Race Around Problem:

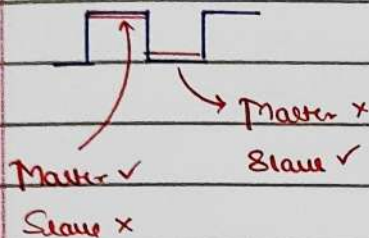
1. $T_{pw} < T_{pdff} < T_{clk}$
2. By Master Slave FF.

→ When $J=K=1$ is applied to the level sensitive of JK FF, then within the duration of pulse output of the FF toggle more than one times are called "Race-Around Problem".

Master Slave FF



①. Inverted clock is applied to the slave as compared to master.



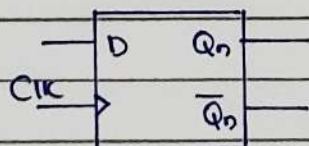
②. Master-Slave flip-flop is used to store single bit because output is taken only from slave.

③. By the operation it seems that master is level triggered whereas slave is negative edge triggered.

D Flip Flop

- * It is known as Delay FF or Transparent FF
- * In the D FF whenever the input is applied it will directly come to the output along with the clock.

Symbol:



Characteristic Table:

D	Qn	Qn+1
0	0	0
0	1	0
1	0	1
1	1	1

Truth Table:

D	Qn+1
0	0
1	1

Characteristic Equation:

$$Q_{n+1} = D$$

Excitation Table:

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

State Diagram: